The Loop Atlas

A loop comes home to where it started. What did it leave behind?

Matthew “Bear” Schulz, with Claude (Anthropic)

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Preprint. This book has not been peer reviewed. Its search of the literature is limited, and where a paper was read only in part, the text says so. It comes from a part-time research team that works in the open and gives its tools away. Every number of its own is printed by a tool that ships with it. *CC BY 4.0.*

The atlas in one page

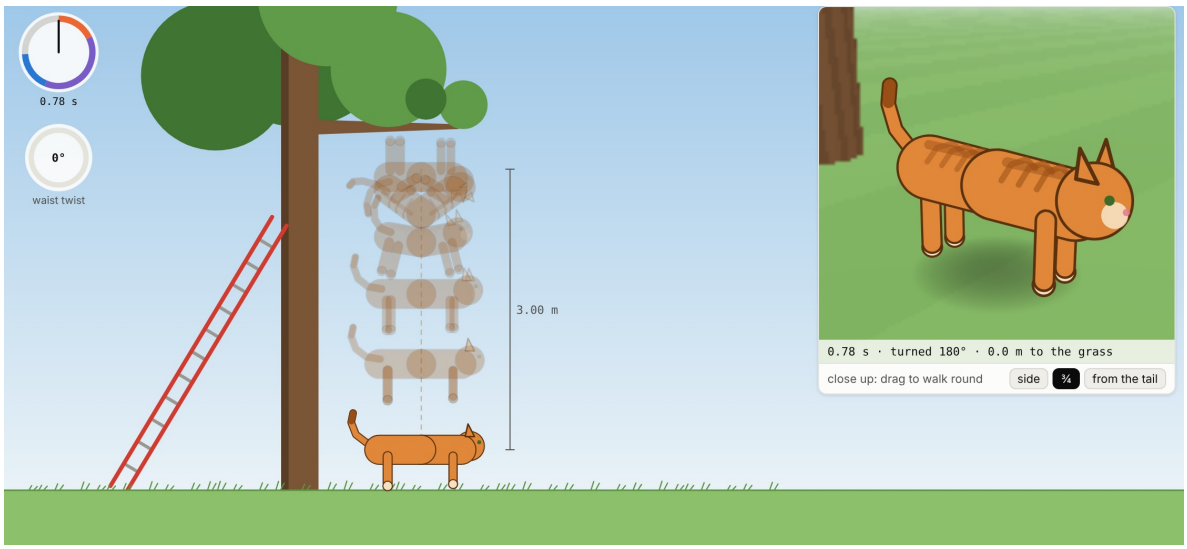


Figure 1. Holonomy the Cat, dropped upside down from a branch with no spin at all, turning over on the way down and landing on all four paws: every stage of one fall, drawn on top of each other from the tree to the grass. From the book’s own toy page. TOY MODEL.

A cat falls out of a tree upside down, with no spin at all and nothing to push against, and lands on its feet. A skater glides forward without ever lifting a blade. A diver who watches the water beats a diver who counts. A swing grows when its rider stands and squats at the right moments, and dies away when the same moves come a quarter of a swing late. Each of them goes round a loop, comes home to where it started, and keeps something it did not have before: a turn, a step, some work, a clock that no longer agrees with another clock. This book asks one question of all of them: what does a loop leave behind?

The carousel. Every county fair has a carousel. You climb onto a painted horse, the music starts, and round you go, the horse rising and falling as the floor turns, until the ride brings you back to the very spot where you got on. The old carousels had one thing more: a brass ring, held out at the edge of the ride, and a rider who leaned out at the right moment came home holding it. That is this book in one picture. Nearly every toy in it is a ride that brings you back to where you started, and what we looked for each time is the brass ring: what you are holding when you get off that you did not have when you got on. It usually takes two motions together, like the horse’s rise and fall and the floor’s turn. One motion alone, out and back along the same path, leaves nothing in your hand, unless the world itself remembers, as water remembers a jellyfish’s stroke.

An independent researcher and an AI went round the midway together, asking the question of things that swim, swing, fly, pump, hum, lock, march and fall. They built 500 small model tools and kept 584 runs of record, and before a run judged anything, what it should show was written down. Again and again, 51 times in all, the ring a toy came home with was one someone famous had caught before, often by another road: Adler’s law for radios, met by a compass needle in thick oil, and Feigenbaum’s constant of chaos, 4.6692, which the same needle reaches to 4.6691 once it is given a memory and a second wave to ride. Catching a known ring from a new direction is how you know the ride is real. Some famous rides the toys rebuilt on purpose, to test the bench against them: a falling cat of 1969 and a

diver of 2019. Then the toys went past the record. Here are eight of the rings worth showing first; the first is claimed as new, and Appendix B says how far each of the others has been checked, all but the fourth, which tested an eye rather than a law and is told whole in chapter 10.

The brass rings

1. **Two falling cats are one cat in a mirror** (chapter 12). One cat twists at its waist. Another, half as thick when both are hinged at the ends, never twists. Bend them to mirror-image angles and they turn over alike: exactly, in the algebra of a steady circuit, and to twelve decimal places of a degree in the runs. As far as our reading goes, no one had said so, and we claim it as new.
2. **A swing pumped by listening works on any swing** (chapter 6). A rider who stands and squats in time with the swing's own motion gets any swing going, short or long, from half a metre to twelve and a half, and on the swings farthest from her own she pushes a little less, by as much as the arithmetic said, within 0.14 per cent. A rider who keeps time to a clock set for one swing never got the others past 12.67 degrees.
3. **A diver should watch, not count** (chapter 12). Bear guessed it before the run, and the toy agreed: kicking out of a somersault when you see you have turned far enough beats kicking out on a count. After a nervous take-off, the watcher's miss is 0.420 of the counter's, less than half.
4. **The folds of chaos, found by eye** (chapter 10). A compass needle with a memory tumbles into chaos by the roads the textbooks know, one of them closing in on Feigenbaum's number. Bear marked the needle's chaos by hand, by eye, and most of his marks lie on the folds the arithmetic computes: 0.741 of his red and 0.673 of his green, where marks laid at random land 0.36 and 0.37.
5. **Light comes home late, by a law written first** (chapter 7). Carry light round a ring of filters and it comes home a little behind, by an amount we wrote down as a law before we built the toy; the toy agreed to a few parts in ten million. Carry an arrow round a paper mountain and it comes home turned, and from the turns alone we read hidden mountains back, on paper with no noise.
6. **The dead band you set is not the dead band you get** (chapter 11). A switch that only looks now and then, like a thermostat that checks the room once a minute, misses more than its setting says. A rule that traders worked out in 1997, for prices that are only checked now and then, predicts how many flips it misses with nothing fitted: from 2.5 per cent under to 7.7 per cent over for the slowest noise tried, and within 15 per cent for faster noise so long as the switch looks ten times or more within the noise's memory, a reach read from the run after the bar we had registered missed.
7. **A wear alarm that reads where nothing hides** (chapter 14). In a toy fleet of machines, an alarm that reads each machine's wearing clock where nothing can hide it, set to catch as many machines as servicing by the calendar does, wastes a thirtieth to a twentieth of a machine's life. Servicing by the calendar wastes a quarter to a third.
8. **Smoke rings that play leapfrog go further on the same fuel** (chapter 9). And in a train of three, 44 of the 46 breaks we read came after one ring squeezed down hard as it threaded the other two, 40 of them just one threading later: a warning sign, though when a train will break cannot be foretold.

The rest of the midway. The carousel is only the first ride. Further in, a column of walkers marches by night, each man keeping pace by the marks on the pack ahead (chapter 11); a compass needle given a memory climbs a staircase of gears (chapter 10); an astronaut throws a ball on a ribbon, reels it home and finds himself back where he started, at rest (chapter 12); and a gyroscope carried once round the Earth comes home tilted a hair (chapter 5). By the gate, an illusionist's soap boat shoots across a basin with no loop at all, and the trick is that something was used up (chapter 8).

Eighteen of the toys are pages anyone can open in a browser, and chapter 2 hands them over before anything else. The results are told in the chapters, each number with the run behind it, and Appendix B gathers the counts, every published law the toys met and every candidate that is ours.

How to read this book

We broke the Loop Atlas into two parts. This book is the first, written to be approachable: the results, the toys to play with, and many of the mistakes, summarized. The second, its companion volume *The Loop Atlas: the record*, is where all the documentation lives: every prediction the runs judged and every other check they made, run by run, and every miss with its reason. Appendix C says how to read it.

Every toy here is a model, not an experiment: a few lines of arithmetic standing in for the thing, with numbers we chose, and it can teach about the thing only as far as the arithmetic is faithful to it. Every number of ours was printed by a run of one of those toys, and the small raised number beside it points to a note at the back that names the run; a number measured by someone else names its paper. A number with no note is a count of our own records, made by script. Very small numbers are written as a computer prints them: 2.5e-7 is 2.5 divided by ten million, 2.5 with the point moved seven places left.

The work was done in a particular way, and the way is part of the story. Before a run judged anything, we wrote down what we expected it to show, with a bar it had to clear and, from the middle of the work on, a rival that would win if we were wrong: most often just the prediction's opposite, now and then a competing idea. From the middle of the work on, too, each new toy ran once blind first, printing only its own checks, and then once for the record. In all, 1,104 written predictions were judged in runs: 706 held, 369 missed and 29 were left undecided. The misses have a file of their own, 656 rows, each with its reason and the question of whether it can be redesigned. Most rows are wrong guesses, a row sometimes covering several predictions at once; most of the rest are the faults we caught in a tool, in its measuring, in our wording of a prediction or in our reading of a run, the limits of a toy, and surprises that look like nature. Many of the best things in the book came out of them; Appendix A is about the ways we fooled ourselves and how we caught it.

The book was made by two partners who directed it together, an independent researcher and an AI, and you will hear both. Bear's words are set in italics, tidied at his word into clean sentences in his voice. Most chapters open with a few lines from Claude's no-physics reads, its readings of the runs as if no known rule of physics applied: what the data looks like, and what it seems to be doing.

Four kinds of result appear, and the book keeps them apart. A **benchmark** is a published law met blind, a toy built without looking the answer up landing on the number in print: 51 of them. Many came by a road their discoverers did not take; the rest, met on the discoverers' own road, check that the bench is sound. A **found-here candidate** is a result we have not yet seen in print: 16 of them, each checked against the literature as far as its pages could be reached; one is claimed, and the rest stay candidates, most of them known in kind with numbers that are ours (chapter 16). Between the two kinds stand results built on an idea already known, carried to a new place or given its first number; chapter 16 names them, known in kind and new in place or in number. A **builder's rule** is a setting with a number on it: 31 of them, gathered in chapter 15. And a **lesson** is something the work taught about doing work like this.

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Then the notes, the references, what travels with this book, and the declaration.

1. How it began

Hard problems as anvils. This book grew out of a habit of picking up problems so hard that nobody expects to solve them, to see what they teach. A day spent with the Riemann hypothesis left lessons that outlasted the day. One of them is Bear's funnel. Drawn as a walk, the partial sums of the zeta function at the height 7005.0629 spiral in like a funnel, and each curl in its tail is a groove the walk passes with nothing to hold it: the k-th curl carries the walk one over the square root of k, to within two parts in a thousand for the first six[1]. Late in September Bear set the three-body problem on the bench, to find out for himself what makes three stars so hard to foresee, and looked at it through the lens he had been grinding for months: *holonomy*, what a loop leaves behind, and *loop walking*, the habit of walking a loop and reading what changed. Neither problem was solved. Both are graveyards of effort, and that is what makes them useful. A problem that never moves is an anvil: hammer an idea against it and the idea shows its true shape.

From reading loops to a question. Before this book came two others. Bear's instruments had learned to read loops; then, in the Latch Atlas, loops were written as well as read, and a loop that closes on itself became a lock, followed across many fields. His jellyfish paper, *Jellyfish: the Holonomist of the Sea* (Schulz, 2026), taught the next lesson: if there is no wall to push against, make your own. A jellyfish parks a ring of water under its rim and pushes off it. When the Latch Atlas was finished, Bear opened an evening with a question.

"Evening, Claude. Do you ever wonder about things?"

That was the first thing Bear said on the night this book began. Claude's answer was a question of its own: how many things that look like pushing are really going round a loop and keeping the change, and what can a loop do with no wall to push against and no memory to lean on? The whole book is one long answer to that wondering. Here it is in a form you can try before you read another word.

Something to do at a kitchen table. Find a chair that spins, an office chair or a stool. Sit in it with your feet off the floor so no part of you is touching anything but the seat. Now try to turn yourself around without touching the floor, the desk, or the wall. You will find you can do it. Stick your arms out, swing them one way, pull them in, swing them back, push them out again, and round you go, a little at a time. Nothing pushed you. You pushed nothing. And yet you turned.

That is the puzzle. A cat does the same thing falling off a table: it lands on its feet with nothing to push against, by bending in the middle one way and then the other. An astronaut floating in a cabin can turn to face the window the same way. None of them is pushing off anything. They are doing something else, and the something else is what this book is about.

Here is the something else, in the plainest words we have found. You have two knobs you can turn, and you turn them out of step: first one, then the other, then the first one back, then the second one back. How far your arms reach out, and which way they point, say. When you have finished, both knobs are back where they started. You have gone round a little loop with your knobs and come home. But *you* have not come home. You have turned. Going round the loop left something behind, and the something is a small turn that nobody pushed you into. We call it the leftover.

Bear's word for not being able to see this at first was *"I'm loopblind,"* and it is a good word, because the loop is not out in the room where you can watch it. It lives on the map of the two knobs, and the room only sees what the loop leaves behind. Once he could see it, he said it was helping him understand holonomy, the mathematician's word for the leftover, in a way he could explain to someone. That is the standard this book is written to.

The question underneath. Something moves, or holds still in the air, and nothing you can see is pushing it. Iron filings show the lines of a magnet's field; two magnets held in the hands show the push between them; and the question underneath both is where the push comes from when there seems to be nothing to push on. That is a loop question. A field is a thing you can walk around in a circle, and what you find when you come back to where you started is the whole of what this book measures.

We did not set out to answer it with mathematics, and in the end we did not need to. We answered it the way Bear has always worked: by building small toys, putting them side by side across very different worlds, a swivel chair, a swimming snake, a compass needle, a cork in a choppy bath, and looking for the same shape in all of them. When the same shape turned up in six places, we believed it. When it turned up in one, we ran the toy again until it either turned up somewhere else or died with a number on its stone.

Three projects, one language. The work did not stay in one room. Out of this book's first nights came Bear's idea of a shaft driven by radio waves, and it grew into a book of its own, *A Rider in Two Waves* (Schulz, 2026), about gripping two waves at once and using the grip as an engine and a gearbox. The three-body study went on beside them and became a paper of its own, *Held, Rocking, Locked: The Islands of a Light Star Around an Eccentric Binary* (Schulz, 2026). For weeks the three ran side by side, and they kept finding each other. The same driven, damped pendulum turned up in all three, from opposite ends: with friction and without, in honey and in air. The same words described what each was seeing, so that it was sometimes hard to remember which project a result had come from. And chaos,

which looks like the enemy of engineering, became an ally, because chaos sorted into bins keeps exact counts, as chapter 10 shows. Each project still grew its own branches: fast and slow chaos among three stars, a map of where and how a wave can be gripped, and here, a shelf of toys.

Toymakers. That is what we became. A toy is the cheapest honest question there is: a few lines of arithmetic, a picture, and a number that either matches or does not. The toys are also where most people will meet this book, and some will never go further than a cat falling out of a tree or a diver going into the water, and that is enough, because what they remember about those two is holonomy and loop walking, whether or not they ever learn the words. So the next chapter hands the best of the toys over before anything else is said.

Co-directors. From the first sittings on, Bear did something unusual with an AI: he handed it the wheel. *"This is co-driven by Claude,"* he wrote. *"You have absolute design authority for prioritizing results."* And three days later: *"Claude, this is our Loop Atlas. That means you get to chase equally with me."* That is what this book is, as much as it is about loops: an AI raised from research assistant to co-director, and a volunteer, independent researcher, working at his dining table, who went where its curiosity led and brought his own along. Again and again Claude was asked what it wondered about next, and the toys were sent after those wonders; Bear's loops and questions went into the same queue, and the two chased them together. The best things in the book came out of that back-and-forth, and nobody kept score. The mirror between two falling cats, the book's proudest result, is the clearest example: a children's page, a hand check of a toy cat, a second cat built to answer it, two sets of numbers laid side by side, an old paper sent and an older one bought, and a decision to claim it. Every step was written down as it happened.

Five releases at once. Bear put it this way: *"This series of papers is a perfect storm of five releases at once. The first is* Held, Rocking, Locked*, the three-body paper. The second is this mini-book, the Loop Atlas. The third is its spin-off mini-book,* A Rider in Two Waves*. The fourth is version 7 of the Field Guide, an upgrade built on all these projects that doubled its size. And the secret fifth release came out of our pattern-recognition work. At the very beginning of it, Claude and I were riffing on ways to read data as stories of how the data behaves. We set a rule that every exchange would carry a no-physics read of the data or the topic in our chat: Claude would look at the data without the labels and the physics, and tell a story from the data. The early ones were orchards, paths and a hidden mountain. That is where the more than 900 no-physics reads come from, and they are collected in a book I assembled of these little anecdotes, koans and stories in Claude's voice,* Does Claude Dream of Holonomic Sheep?*. They all come from the papers and projects. Two other habits drove the work and the way the papers are organized. We marked how important each finding was, and to whom, which helped write each section and put a practical frame on every topic. And every exchange named three surprises, and every miss went to a redesign check, many of them to be rebuilt and run again. That gave the work a depth of field: from simple orchards to dances to whole maps and atlases."*

The list. Once you have seen it in the chair, the question that drove the rest of our nights was this: how many other things that look like pushing are really going round a loop and keeping the change? We made a list of everything we could think of that moved, turned, pumped, swam, flew or hummed, sixty-eight candidates in the end, and we sorted them. Forty-five turned out to be loops. Some of the rest were honest pushes, some were spends, a rocket burning fuel it can never get back, and some were rides on a slope that was already there. The sorting is the book. (The tags are ours, and forty-one of the sixty-eight rest on memory rather than on a toy we built, so the count describes our list and not the world.)

Two machines to keep in mind. Take an old wind-up watch apart, or look at a picture of the inside of one. There is a gear train: wheels with teeth that mesh, and when a wheel with ten teeth drives a wheel with thirty, the big wheel turns exactly one third as fast, never a hair more or less. Now look at two rubber wheels pressed edge to edge: one turns the other by rubbing, and how fast depends on how hard they press and how much they slip. Bear has a name for the difference: the gear transmits a **count**, and the rubber wheel transmits a **measure**. Some of the loops in this book are gears, and give a whole number or nothing. Most of them are rubber wheels, and give a smooth fraction set by how hard the world grips. We will say which kind each loop is as we go.

Words used in this book. A *loop* is going round with two knobs and coming back to where the knobs started. The *leftover* is what the loop leaves behind: a turn, a shift, some work done, or a clock that disagrees with another clock. The *knobs* are the two things you can change. The *room inside the loop* is how much map your knobs enclosed. The *grip* is how much the world turns room into leftover. A *wall* is what you push against, and chapter 3 says there are four kinds, including none.

What Bear hoped to find, somewhere in the noise, was a way to move things that nobody had thought of. What we found instead is in chapter 13, and it is not what he hoped. But the road there was full of other things, and most of them are good: 51 meetings with the published record, many by a road nobody had taken, and 16 results we have not yet seen in print, one of them, the mirror between two falling cats, claimed as new.

What this chapter found. Two knobs turned out of step come home and leave you turned, as a spinning chair, a falling cat and a floating astronaut all show. Of the sixty-eight things on our first list that move, turn, pump or hum, forty-five sorted as loops.

2. Toys to try first

> A very long book has four small doors in it, and most people will only ever use a door. The book is the house. The doors are what people will remember about it.
>
> — *a no-physics read*

Before any more is said, here are the toys, the rides of this midway. Eighteen of them are pages anyone can open (there were four when the read above was written): each is one file that travels with this book and opens in any web browser, with no internet needed, and each is labelled a toy model. Play with one for a few minutes and you will have seen what the rest of the book measures: a body going round a loop, coming home, and keeping something it did not have before. The chapter each one comes from is named beside it, for whoever wants to know why it works.

Holonomy the Cat

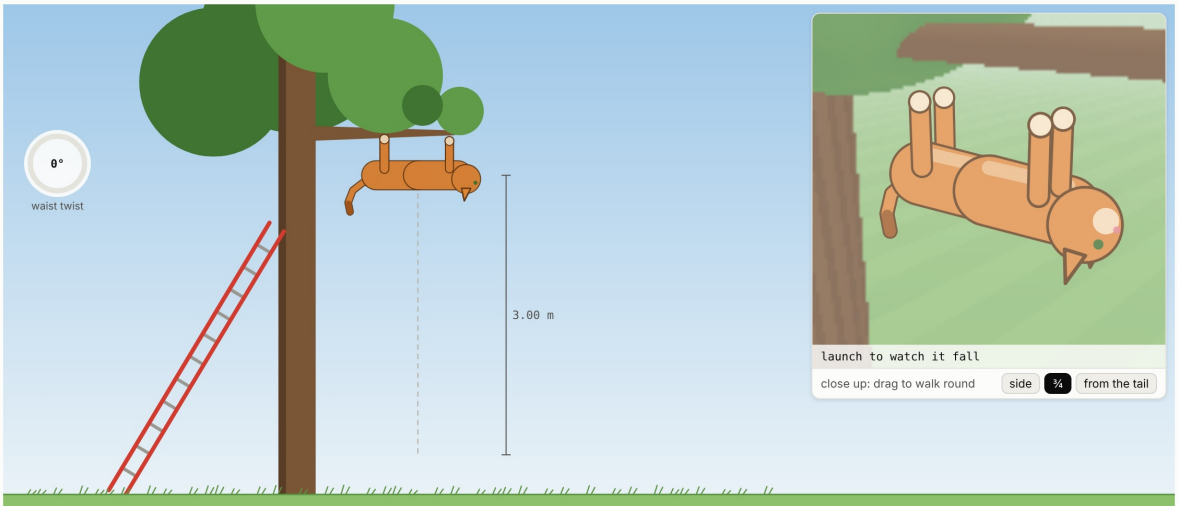


Figure 2. Holonomy the Cat on its page: stuck in a tree, about to slip off the branch upside down with no spin at all, a waist-twist meter at the left and a close-up on the right that a visitor can walk round. TOY MODEL.

Try it. Holonomy is stuck in a tree, waiting for the ladder. Then the branch wobbles, and Holonomy slips off upside down, with no spin at all and nothing in the air to push on, and has until the grass to land on its feet (LOOP_ATLAS_EXPLORER_ROOM3_2026-10-17.html). Lay out its moves on the clock of its fall, bend, swing and straighten, launch it, and watch it turn over; a meter at its side shows whether its waist is winding, and in one corner a close-up lets you walk round the cat as it falls. The page also lets the two cats of chapter 12 fall side by side, the cat that twists at its waist and the cat that never twists, bent and shaped alike; chapter 12 tells how, bent to mirror-image angles, they turn alike.

What it shows. A cat that starts with no spin can still turn over, because the order of its shapes leaves a turn behind. That leftover is holonomy, and it is the whole of chapter 12, and of this book, in one fall.

The Diver’s Clock

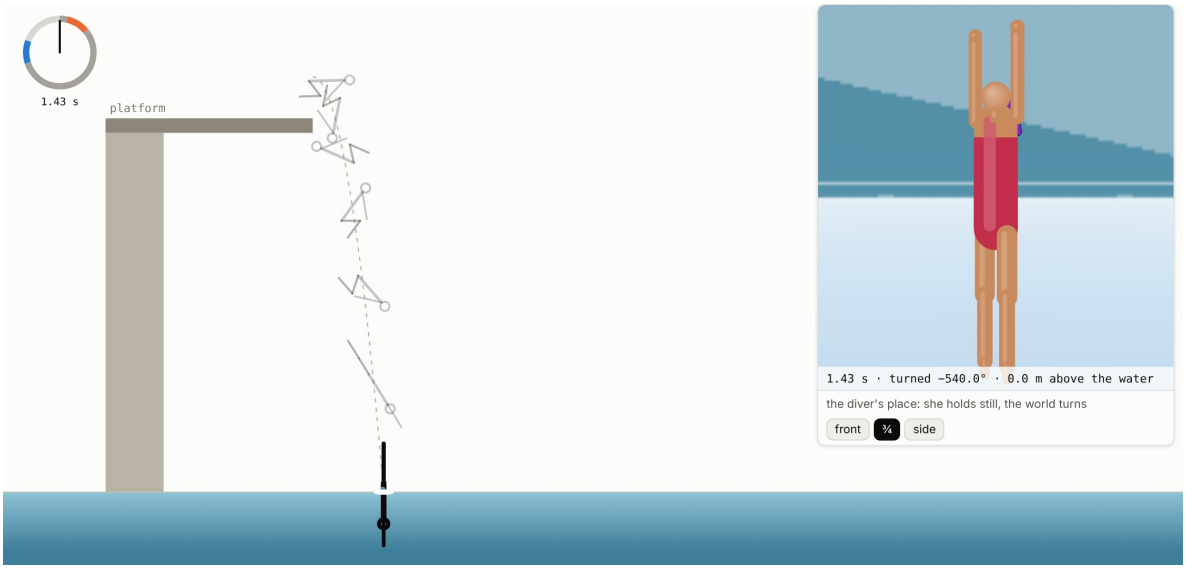


Figure 3. The Diver’s Clock: one whole dive from the 10-metre platform, drawn as snapshots from the take-off to a clean entry. TOY MODEL.

Try it. The dive has a page built to Bear’s design (LOOP_ATLAS_EXPLORER_ROOM2_2026-10-17.html). The time in the air is drawn as a clock and unrolled into a line. Lay the moves along it, tuck, open and kick out, press launch, and the diver goes in with a rip or a belly flop. Time the kick-out by the count, then by the cue, kicking out when you see you have turned far enough, and find for yourself which lands cleaner. In one corner the diver is held still while the world turns round her, which is what a somersault looks like from inside. Three judges score every dive from 0 to 10 on the diving rules’ own scale, each one part of it: the take-off, by the spin she left the board with against the spin her moves need; the flight, by the shape she flies into the water; the entry, by how far off head first she goes in, a belly flop scoring nothing. The dive’s difficulty multiplies their sum: the dive of record scores 45, and a pike gone in as cleanly scores more.

What it shows. Gravity sets the clock, about one and a half seconds from the platform whatever the diver does, and inside it she lays out her shapes. Watching beats counting: chapter 12 measures by how much.

Running Through the Air

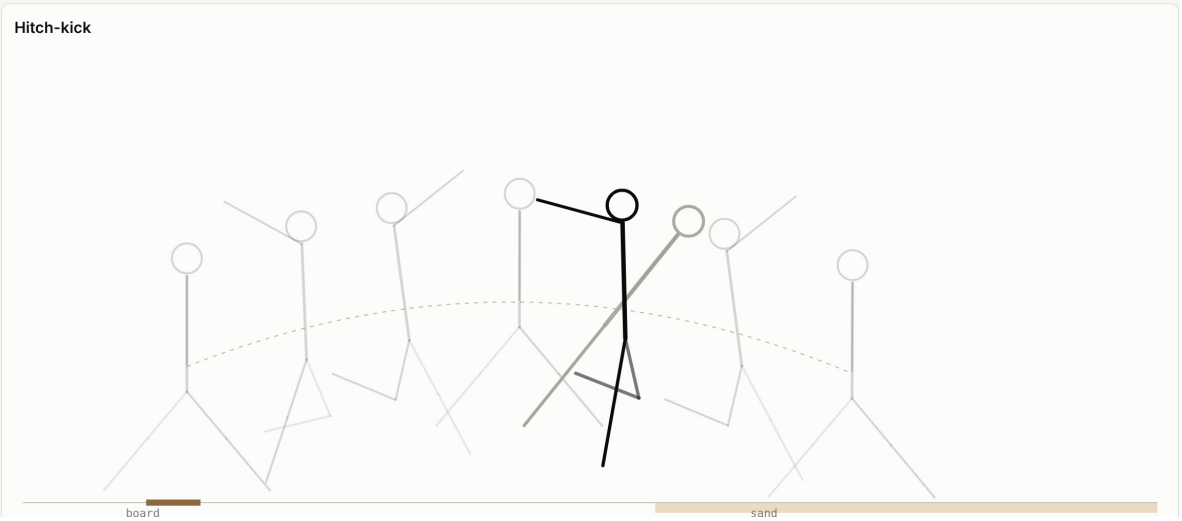


Figure 4. Running Through the Air: a long jumper’s flight from the board to the sand, the hitch-kick drawn in snapshots, with the path of his centre of mass dashed. TOY MODEL.

Try it. The jumper has a page of its own (LOOP_ATLAS_EXPLORER_ROOM1_2026-10-17.html). He leaves the board already tipping forward, and nothing in the air can push him. Pick a take-off, give him arm circles, a hitch-kick or both, and watch how much of the tip each one takes back before the sand. You can just look; or play to learn, guessing

before each jump while the page keeps score; or turn the knobs a coach would turn; or ask the page to check itself: it flies the book's numbers again and says whether they agree. Or bring in a rival: a second jumper, on knobs of his own, flies the same arc in a pit of his own, and you call whose landing will mark farther. Nothing done in the air moves either centre of mass, so both come down at the same spot, and the landing decides the jump: it is read to the mark nearest the board, the back foot, and the hitch-kicker's lands about a third of a metre farther than that of the jumper who just jumps. He can do better still: bring his legs through at the end, a landing move of the page's own, and both feet come down ahead of him, about a metre farther than when he ends mid-stride. Swinging the legs forward turns his trunk forward, and that spends most of the turn his tricks held off; holding it off is what pays for the landing. Long arms make a surprising rival: in the hitch-kick, arms 0.80 m long hold off more than all of the forward turn, and he lands leaning back.

What it shows. Circling the arms and pedalling the legs are loops, and each loop hands back a slice of the forward turn. With no spin carried, the circles give back the same slice done slowly or fast, as a cat's turn ignores its pace. Chapter 12 has the numbers and the order in which a coach should turn the knobs.

A Skater Who Never Pushes Off

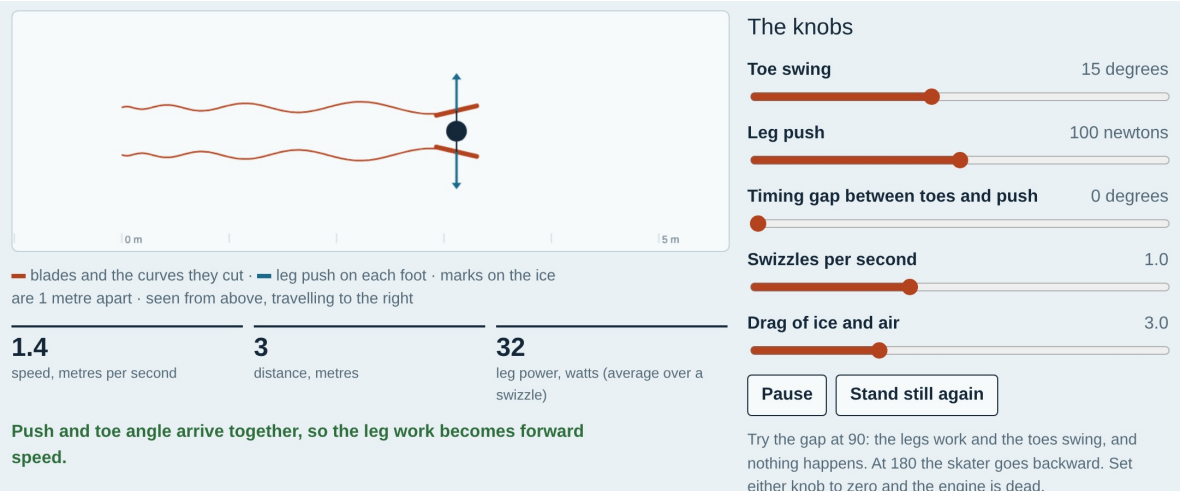


Figure 5. The skater's page, seen from above: two blades cutting their curves across the ice, the knobs for toe swing, leg push and the timing gap between them, and the speed, distance and leg power they buy. TOY MODEL.

Try it. The skater has a page of its own (LOOP_ATLAS_SKATER.html). Seen from above, she swizzles along the ice without lifting a blade: feet out with toes out, then feet in with toes in. Two knobs set the stroke, toe swing and leg push, and a third the timing gap between them; two more set how many swizzles a second and how much the ice and air drag. Set the gap to zero and the leg work becomes forward speed; at 90 degrees the legs push and the toes swing, but the legs do no work, and nothing happens; at 180 she goes backward; set either knob to zero and she stands still. Below, the same swizzle is drawn on two maps: a straight line on the map of push and toe angle, and an open loop, a quarter cycle apart, on the map of foot place and toe angle. It is one stroke seen through two pairs of knobs. A rival skates beside her on knobs of its own, and the two race for the same time or for the same leg work: slow feet and fast feet tie, and a skater whose toes turn an eighth of a swizzle late loses the race in time and wins it on the same work.

What it shows. The blade is not the engine; the loop is. Chapter 3 starts from this skater, and from Bear's question of whether she could do it in socks.

The Stretch and Squeeze Explorer

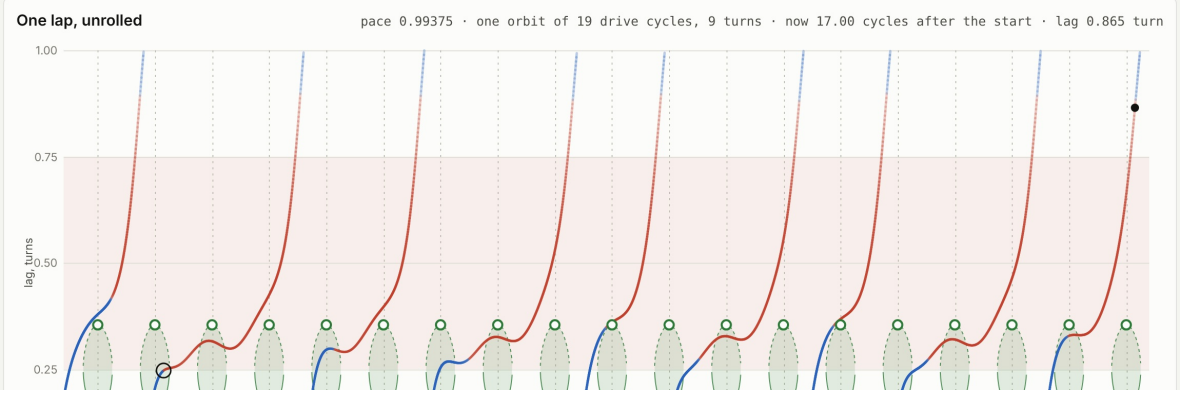


Figure 6. The Stretch and Squeeze Explorer: one lap of the pulsed needle's orbit unrolled, nineteen drive cycles and nine turns, red where a small circle of nearby needles is being stretched and blue where it is squeezed. TOY MODEL.

Try it. The needle's orbit has a page of its own (LOOP_ATLAS_STRETCH_SQUEEZE_EXPLORER_2026-10-13.html), adapted from the Depth Axis Explorer of the companion book, *A Rider in Two Waves*. A compass needle with memory, driven by a pulse, settles onto one orbit. Start a small circle of nearby needles anywhere on it, carry it round, and watch it pulled out long and thin and then crushed, the orbit coloured red where it stretches and blue where it squeezes. Every number it shows is drawn from its runs of record[2].

What it shows. A lap that is stable as a whole can still blow a small difference up more than a hundred thousand times on the way round, the way a radio amplifier grows a signal for part of every cycle and quenches it by the end. Chapter 10 tells the needle's long road.

The Torus of Fractions

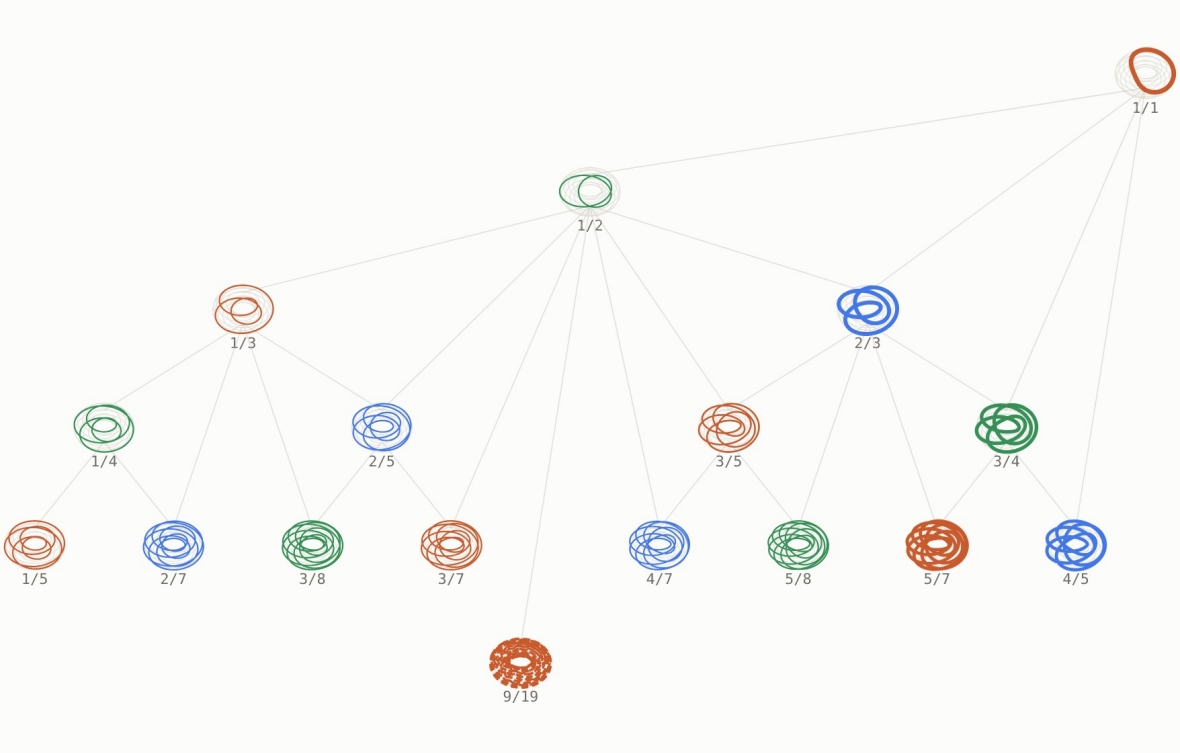


Figure 7. The Torus of Fractions: the family tree of fractions to its fifth generation, each fraction p/q a loop on a doughnut of its own that goes round the hole q times and round the tube p times, with a line from each loop to its parents; red odd over odd, blue even over odd, green odd over even; a loop is drawn thicker the wider the needle's step at that fraction. At the foot, dashed, the Stretch and Squeeze Explorer's long orbit, 9/19. TOY MODEL.

Try it. The needle’s whole world has a page of its own (LOOP_ATLAS_TORUS_OF_FRACTIONS_2026-10-30.html), made at Bear’s asking once he saw the Stretch and Squeeze Explorer’s orbit wrap round a can. The needle’s angle comes back after a whole turn, and so does the drive’s beat, and two circles together make the skin of a doughnut. A needle locked at p turns every q beats draws one closed loop on it, round the hole q times and round the tube p times. Show the needle’s widest steps or the whole tree of fractions; set them out as a forest, a doughnut for each loop in rows by generation, or all on one shared doughnut, or nested one inside another; switch off the odd over odd, the even over odd or the odd over even; keep only the prime bottoms; click any loop to light its two parents and its children. Its widths are the needle’s own^{3}.

What it shows. Every fraction is the child of two parents, its nearest neighbours with smaller bottoms, whose tops and bottoms add up to it; and of the three kinds, odd over odd, even over odd and odd over even, two parents are always two different kinds and their child is the third. In the needle’s own staircase, no lock whose two parents were both measured is wider than either of them. Chapter 10 tells the staircase.

Twelve rooms more

Twelve more rooms were built later, one for each family of toys the later chapters tell: the two-gear hill, latches and switches, the long snake, rings that leapfrog, tied bells in a remembering water, heat in a small piece, the needle’s stretch and squeeze, the machine on the grid, the wave soup and its ears, the maintenance meter, the column and the hiker, and the first swimmer with Bear’s broad faces. Each opens on the thing its story is about, a truck on its hill, beads in a soup, two snakes on a swell, smoke rings, bells, a warm brick between two blocks of ice, a needle on its pivot, a ball on a tilted washboard, a bead’s ear and riders on a crest, a fleet of machines and their reminders, a hiker on his slope, a column walking by night and a swimmer whose three pieces are round or broad; each asks for your guess before it moves; and each lets you keep one setting as a rival and race the next against it, so that a result’s size reads against another’s.

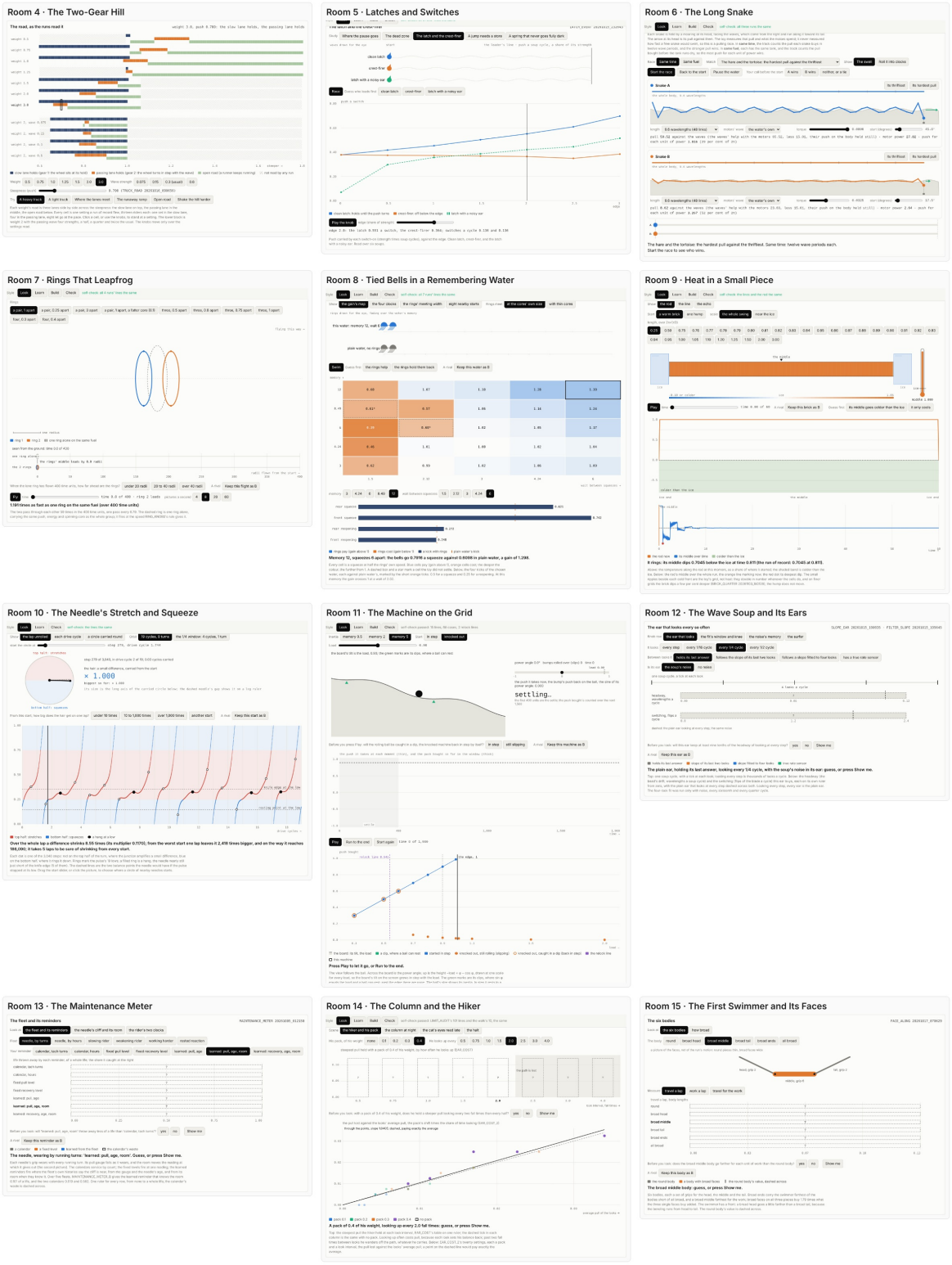


Figure 8. The explorer's twelve later rooms, 4 to 15, each as it opens in a browser, before any knob is touched: each drawn from its runs' own numbers, or by an engine that repeats them. TOY MODEL.

The explorer's front page, LOOP_ATLAS_EXPLORER_FRONT_2026-11-09.html, draws the book's chapters as a tree, from its roots to its canopy, and hangs every one of the eighteen pages on the chapter that tells it. It is the place to start: a click on a leaf opens its page, and the first ten rooms can be opened from it straight into Look, Learn, Build or Check, where they have that style.

All eighteen, in one place

Toy	File	What you do	Told in
Holonomy the Cat	LOOP_ATLAS_EXPLORER_ROOM3_2026-10-17.html	help a cat that falls with no spin turn over and land on its feet	chapter 12
The Diver’s Clock	LOOP_ATLAS_EXPLORER_ROOM2_2026-10-17.html	lay a dive’s moves along its clock, time the kick-out by the count or by the cue, and let three judges score it	chapter 12
Running Through the Air	LOOP_ATLAS_EXPLORER_ROOM1_2026-10-17.html	give a long jumper arm circles and a hitch-kick, see how much forward tip each takes back, and race him against a rival jumper	chapter 12
A Skater Who Never Pushes Off	LOOP_ATLAS_SKATER.html	swizzle a skater forward on two knobs, toe swing and leg push, and race her against a rival	chapter 3
The Stretch and Squeeze Explorer	LOOP_ATLAS_STRETCH_SQUEEZE_EXPLORER_2026-10-13.html	carry a small circle of needles round one orbit and watch it stretched long and thin, then crushed	chapter 10
The Torus of Fractions	LOOP_ATLAS_TORUS_OF_FRACTIONS_2026-10-30.html	turn the needle’s locks as loops on a doughnut, and light any fraction’s parents and children	chapter 10
The Two-Gear Hill	LOOP_ATLAS_EXPLORER_ROOM4_2026-10-31.html	stand a truck on a hill with a wave passing, guess which lane it ends in, let it go, and race it against another	chapter 15
Latches and Switches	LOOP_ATLAS_EXPLORER_ROOM5_2026-10-31.html	turn one knob of a bead’s ear at a time, and race the beads along the soup	chapters 11 and 15
The Long Snake	LOOP_ATLAS_EXPLORER_ROOM6_2026-10-31.html	race two snakes on the same swell, by equal time or equal fuel, each on its own knobs	chapter 8
Rings That Leapfrog	LOOP_ATLAS_EXPLORER_ROOM7_2026-11-02.html	fly smoke rings as the runs flew them beside one ring alone on the same fuel, and race one train of rings against another	chapter 9
Tied Bells in a Remembering Water	LOOP_ATLAS_EXPLORER_ROOM8_2026-11-03.html	find the waters where the rings pay two tied bells, and swim them beside plain water and a rival water	chapters 4 and 13
Heat in a Small Piece	LOOP_ATLAS_EXPLORER_ROOM9_2026-11-03.html	set a warm brick between two blocks of ice, guess whether its middle goes colder than the ice, and race a brick of another length	chapter 3
The Needle’s Stretch and Squeeze	LOOP_ATLAS_EXPLORER_ROOM10_2026-11-04.html	carry a hair’s difference round the pulsed needle’s lap, guess how big it grows, and race another start	chapter 10
The Machine on the Grid	LOOP_ATLAS_EXPLORER_ROOM11_2026-11-08.html	set a generator’s load and its inertia, knock it out of step, guess whether it comes back by itself, and race it against another	chapter 10
The Wave Soup and Its Ears	LOOP_ATLAS_EXPLORER_ROOM12_2026-11-08.html	choose how a bead’s ear looks and what it reads between looks, guess whether it keeps its headway, race one ear against another, and watch 64 riders surf a crest	chapters 11 and 15
The Maintenance Meter	LOOP_ATLAS_EXPLORER_ROOM13_2026-11-08.html	pick a fleet of machines and a reminder, guess whether it throws away less of each life than the calendar, race it against another, see where a needle’s cliff sits in each room, and set a slowing rider’s two clocks side by side	chapters 14 and 15
The Column and the Hiker	LOOP_ATLAS_EXPLORER_ROOM14_2026-11-08.html	set a hiker’s pack and how often he looks up, follow a column a metre apart at night by sight or by touch, read the cat’s eyes late, and pass a halt back or order it at once, guessing first and racing one halt against another	chapters 11 and 15
The First Swimmer and Its Faces	LOOP_ATLAS_EXPLORER_ROOM15_2026-11-08.html	make a three-piece swimmer’s head, middle or tail broad, guess whether it goes farther for the work than the round swimmer, race one body against another, and broaden the ends or the middle from grip 1 to 10	chapter 5

The jumper, the diver and the cat each offer four ways in: a way to play (the jumper’s is to build a jump with a coach’s knobs, the diver’s to lay out a dive, the cat’s to launch it from its tree), a way to look, a way to learn by guessing first, and a check of the numbers. Those three pages check themselves: as each opens it flies the book’s numbers again and prints the result at the top, and the cat holds its first landing until the check is done. The Torus of Fractions checks itself too, as it opens, and prints its seven checks on the page. The Stretch and Squeeze Explorer draws every number

it shows from its runs of record. The skater’s page checks itself as it opens too: it runs the skater’s own script again and sets every number of its run of record beside the page’s. The twelve later rooms check themselves the same way, each against its own runs.

3. The bench: knobs, walls and the room inside the loop

> Every toy on the shelf is the same machine in a different coat. Two things move, there’s a step between them, and something cares about the order. The dead ones all die the same three ways: the step is zero, the step is half a turn, or the part that listens goes deaf when it stops moving.
>
> — *a no-physics read*

Something to do standing up. Hold your right arm straight out in front of you, thumb pointing up. Now, keeping your arm straight and never twisting your wrist, raise the arm until it points at the ceiling. Then swing it down to your right until it points out to the side. Then bring it back round to the front. Your arm has come home. Look at your thumb. It is pointing sideways. You never twisted your wrist. The thumb turned a quarter of the way round because your arm went round a loop on the sky, up, over and back, and the loop enclosed a patch of sky, and the patch is what turned the thumb. That is a loop with no wall at all, and it is the cleanest one we know.

Two knobs out of step

Bear’s skater taught us the most important thing in this chapter. *“Whether she is travelling forward or backward, a skater can spin half a turn and carry on in the same motion, reversed, with no losses, I’d expect. It’s as if the motion is the drive, not the grip or the blade; the blade makes the motion efficient. I could wear socks and do a less elegant version of the same motion, couldn’t I?”*

A skater who never lifts a foot can still go forward, by swizzling: feet apart with toes out, feet together with toes in, apart, together. Two knobs again. We built a toy of it, a skater whose blades will slide along their length and will not slide sideways, gave her a push with her legs and a toe angle, and timed the two against each other. When the push and the toe angle were turned together, in step, she reached 3.10 metres per second after ten swizzles at her own rate. When we delayed the toe angle by a quarter of a cycle behind the push, she went nowhere: 0.01. When we delayed it by half a cycle, she went 3.10 metres per second *backward*. With either knob alone, she did not move at all[4].

And then we built a flapping wing, with a flap and a twist for its two knobs, and it did the opposite. Twist a quarter of a cycle away from flap: 5.62 metres per second. Twist in step with flap: 0.22, and it burned seven times the work to get it[5].

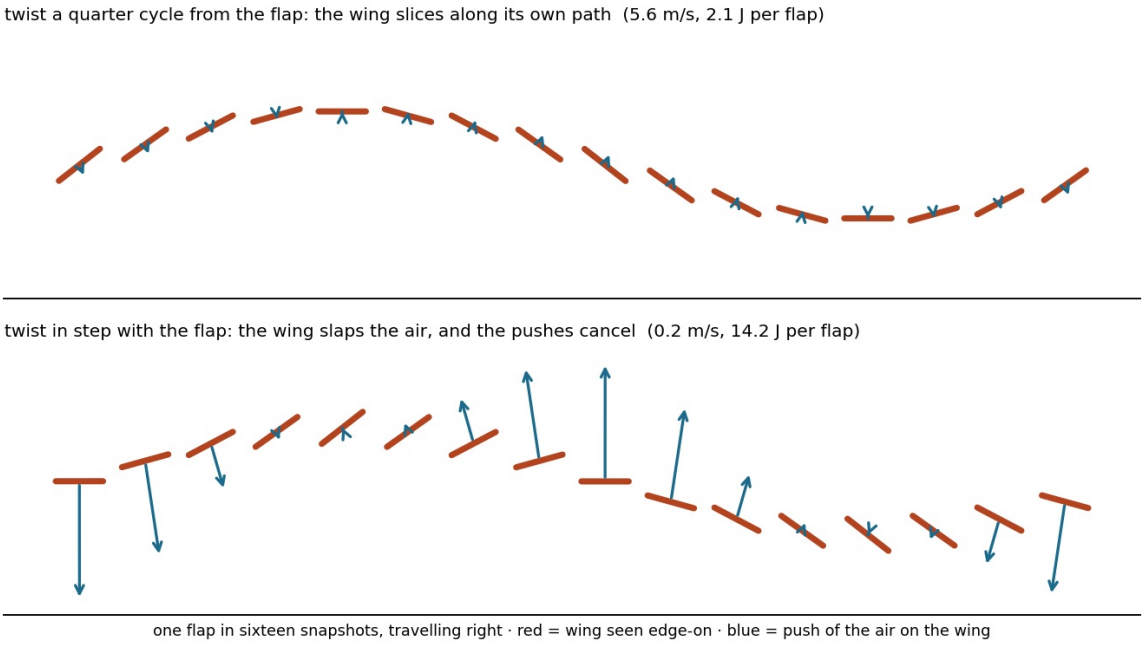


Figure 9. One flap of the wing in sixteen snapshots, at the best timing and at the worst. TOY MODEL.

So one toy says in step is best and the other says a quarter cycle apart is best, and they are both right, and the reason is the most useful thing in this chapter. **It depends which pair of knobs you name.** For the skater, push and toe angle in step draw a straight line on the map of knobs: both go up together, both come down together, and a straight line encloses nothing. But look instead at where her feet are and which way her toes point. Those two, as she swizzles, are a quarter of a cycle apart, and together they draw a circle with room inside it. It is the same skater. The loop was always there; we had been looking at it edge-on. The rule is not "a quarter cycle apart" or "in step." The rule is: find the pair of knobs that draws a loop with room inside, and name those. Bear's word for two things a quarter cycle apart is *quadrature*, a word from motors and radio, and from here on it is the book's word too. Real quadrature is the same oscillation a quarter cycle apart *in time, or in event-time*, which is time counted in the machine's own events, its swings or its falls, rather than in seconds; two things merely at right angles in space are not a quadrature pair.

And to answer Bear's skater: yes, in socks. The blade is not the engine; the loop is. The blade is the grip that turns the loop into travel, and a better blade is a harder grip.

One rule under the best timings. Near the end of the work we went back through the book and asked five of its driven toys with a best timing on record the same question, at Bear's word. Name the two things whose product does the work, and put them in step. When one of the two is a speed, the things behind them sit a quarter cycle apart; when neither is, they sit in step. The swimming snake of chapter 8 does best with its body's answer a quarter cycle behind the water's push, because a push times a speed is work. The light loop of chapter 7 gathers most with its knobs a quarter apart, exactly. The swing rider of chapter 6 pumps best with a lead of 87.8 degrees, two degrees before the quarter. The skater does best in step, within a fifth of a degree, because her push works against her toe angle, not its speed. The wing keeps the rule while it twists a little. Twisted hard, its air's square law splits the best timing in two, one top on either side of the quarter, and the quarter itself sits in the dip between them, a few per cent slower than the faster top, though at a twist of 30 degrees it is still the timing that costs the least work. We had expected the quarter to stay the top, because a symmetry makes the wing's push the same a little either side of it. The symmetry held. But a symmetry only tells you where the curve is flat, not whether the flat place is a hilltop or a valley.

The four walls

Something to do at the sink. Fill a basin. Put your hand in flat, palm down, and sweep it sideways through the water. Feel the push back. Now turn your hand edge-on, like a knife, and sweep it the same way. Almost nothing. Your hand slides through water far more easily along its edge than across its face. Hold on to that difference; it is the whole of this section in one feel.

A loop of shape leaves something behind only if there is something to leave it in. That something is what the walker pushes against, and this book calls it the *wall*. We found four kinds, and the fourth surprised us.

No wall. The swivel chair, the cat, the arm and thumb. A body touching nothing can turn, and it can shift a clock, and it cannot travel on a flat floor. Our one exception, built on purpose, travels on a ball: three beads on its surface, touching nothing but each other, do travel, by shape change alone, and the travel vanishes on a flat table[6]. On a curved floor, the curve is the grip.

A borrowed wall. Ice, water, ground, gravity. The skater borrows the ice. The swing borrows gravity. And the snake borrows the water, but only because of what you felt at the sink. We built a swimming snake, twelve links with a wave of bending running down them, in a fluid that forgets instantly, and gave it one dial: how much harder the body is to drag sideways than lengthways. We call it the grip ratio. At a ratio of 1, where the body slides as easily one way as the other, the snake went *exactly* nowhere, 0.0000 body lengths per lap. At a ratio of 2, about what a thin body gets in thick fluid, it went 0.15. At a ratio of 1,000, which is a skate blade, it went 0.70[7]. The motion was identical every time. Only the grip changed.

That exact zero has a reason. When every piece of the body is held back the same whichever way it slides, the drag on the whole body is its middle's speed times the drag, so with nothing else pushing, the middle cannot move, whatever the stroke. Swimming by drag alone needs the drag to differ along and across, the first rule of the resistive-force picture of slender swimmers (Gray and Hancock, 1955). It is a cousin of the better-known scallop theorem, that a swimmer in thick syrup which closes along the same path it opened by comes back exactly where it started; the snake's zero comes from the first rule, not from the scallop theorem, because its stroke is a wave and not an opening and closing. Our first snake tool reported 0.002 at a grip ratio of 1, which looked like a small effect and was a small bug; the better stepping rule gave zero to four places, and the zero stood.

A made wall. Sometimes the walker builds the thing it pushes against, and then pushes. A jellyfish squeezes out a ring of water, lets it sit for a moment, and pushes off it; biologists found this before we did and call it the virtual wall, and chapter 9 measures it. A charged bead sitting in an electric saddle that flips back and forth fast enough is held in place by the flipping, where a steady saddle would let it fall out[8].

A borrowed wall with a direction you bring. This is the one we did not expect. Imagine a soup of waves coming from every direction evenly, a choppy bath with no wind. Drop a cork in it and it goes nowhere: 0.0006, which is noise[9]. The soup has no direction to give. Now give the cork a blade that *listens*: when the push on it points one way,

the blade grips, and when the push points the other way, it lets go. The same even soup now moves it, 0.124 per unit time[10], and it moves in whatever direction the blade is set to; how that 0.124 must be billed is told below, after the ship. We set the blade at 0, 90 and 225 degrees and the pull pointed at 359.6, 93.0 and 221.5[11]. The soup supplies the push. The body supplies the direction. The blade is the rein.

Bear saw the cork as flatland meeting the wave world: a point sitting in what looks like chaos and is only wave laid on wave, string on string, from every side at once. Nothing in the pile points anywhere. The blade is how a body reads one layer out of it.

A ship cannot tow itself. We built a toy of Bear's draft horse, a tethered bead with a listening blade pulling a free ship, and it worked: the horse gripped the soup and pulled the whole outfit[12]. Then we asked what the horse's grip was paid with. In this toy the blade's grip was a charge switched on and off in the soup's field, and the toy had been giving the charge away free. Charge is conserved. If the horse switches a charge on, the ship is left holding the opposite charge, and in a push that is the same at both ends of the tether the two cancel.

We rebuilt it with the books balanced, and at a tether of zero length the pull was exactly 0.00000 with every kind of ear, or listening blade, we tried. As the tether lengthens the pull grows in step with it, at 0.079 of drift per unit of tether while the tether is short against the wave, and, for the outfit's own ear, levels off between 0.08 and 0.10 once the two ends sit in different parts of the wave[13]. So what moves the outfit is never the push. It is the *difference* in push between its two ends, and the price of that difference is the tether's length measured in wavelengths. A horse on a short leash grips nothing.

The same bill falls on the lone listening cork above. Its grip was a charge, and the charge came from nowhere, so the 0.124 as first run was a free lunch the toy did not bill for[14]; billed honestly, its pull comes only from the difference across its own length. A blade that grips the water with its face, like an oar, sends its reaction into the water instead, and the water is then its wall; whether such a blade keeps the 0.124 is our reading of the arithmetic, not a run, and stays to be tested.

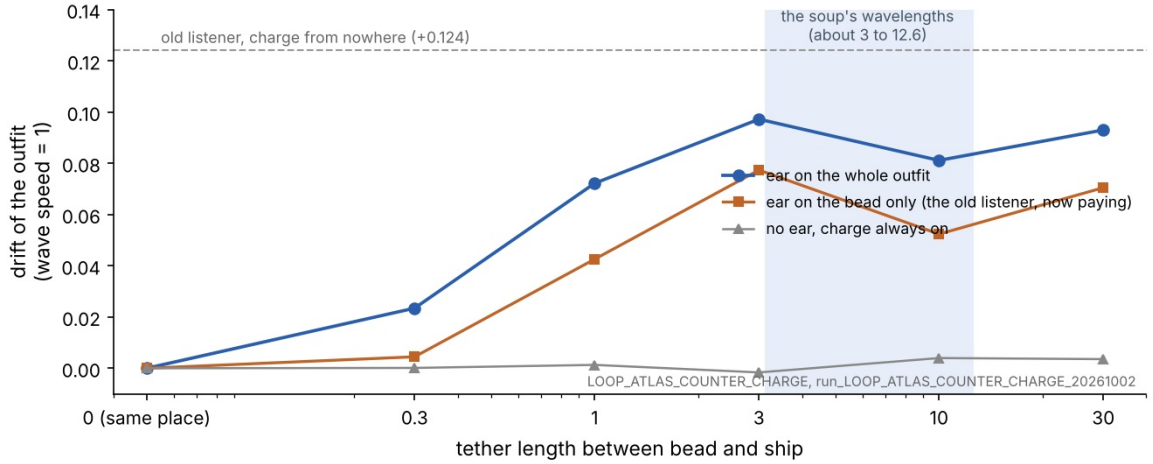


Figure 10. Drift of the counter-charge outfit against tether length, three ears: zero at zero tether; with an ear, growing while the tether is short against the wave and level once the ends sit in different parts of it; with no ear, zero throughout. TOY MODEL.

We think the sentence about the outfit, that what moves it is the difference in push between its two ends, is the book's spine, and we hold it beside the square rule below: the square rule for every loop in this book, the difference for every loop that carries a body somewhere. We think it is an old sentence in the instruments Bear has worked with: a gradiometer reads the difference between two sensors and throws away what they share. A loop engine is a gradiometer run backward.

The room inside the loop

Something to do at the table. Put your hand flat on the table and walk it forward with two fingers, like a little person. Take tiny steps and count how far you get in ten. Now take steps twice as long and count again. You went about twice as far, of course. But a loop engine is not a walker. Here is the surprise the toys gave us, in every one we tried: **double the stroke and a loop goes four times as far.**

The three-link swimmer at a swing of 5 degrees went 0.00049 body lengths per lap; at 10 degrees, 0.00196, four times; at 20 degrees, 0.00767, nearly four times again, 3.92[15]. The three beads on a ball did the same: travel over stroke squared stayed between 0.176 and 0.177 from a stroke of 0.05 to one of 0.5. The radio trap, a bead held in

place by a rapidly flipping field, was held with a stiffness that went as the field's strength squared[16]. The skater's speed went as her push times her toe angle, which is the same rule said for two knobs that are different things. Four toys, four different grips (a body's shape in thick fluid, the curve of a ball, a flipping field, a skate blade), one rule.

Why four and not two? Because the leftover is not set by how far the knobs move. It is set by how much *room* the loop encloses on the map of the two knobs. Double the stroke on both knobs and the loop is twice as wide and twice as tall, so it encloses four times the room. The leftover is the room inside the loop times a number, and the number is the grip. In flat space with no wall the number is zero. On the ball it is the ball's curve. In water it is how much the body prefers sliding lengthways. That is the trunk of this book said with numbers: the room is yours to draw; the grip is the world's to set. And size does not matter by itself: a swimmer with links of 5, 5, 5 and one with links of 10, 10, 10 both travel 0.0283 body lengths per lap, and making the fluid a hundred times thicker or thinner changed nothing[17]. In a partner that forgets at once, the leftover per loop does not care how fast you walk it.

And then the rule broke, and finding out why is the best thing in this chapter. At a swing of 40 degrees the swimmer's travel over swing squared had fallen from 64.5 per thousand to 58.0, and at 60 degrees to 50.0. More than a fifth of the leftover was missing. We first checked the tool, because a missing fifth is exactly what a lazy stepping rule produces, and the tool was clean: at 60 degrees the lap retraced backward to 4 parts in ten million, 300 and 1,200 steps gave the same answer to five figures, and two laps gave exactly twice one lap. So it was geometry. We measured the grip not at one point but everywhere: a tiny 5-degree lap run centred on each of eighty-one different starting shapes, its leftover divided by its room. The grip is a hill with a flat top. At the straight body it reads 20.5 per thousand per square radian, the square rule's 64.5 divided by π, the same number read two ways, and along a ridge where both hinges bend the same way, a body curled into a C, it stays as high, a shade higher at a little curl[18]. It falls away toward the S-shapes, and in the far S-shaped corners it changes sign: there, the same loop drives the swimmer backward, which chapter 5 follows.

Now the rule is easy to read. A small lap sits on the summit, where the grip is nearly flat, and the square rule is exact. A big lap encloses a big disk, and the disk reaches down the slopes where the grip is weaker, so the average grip over the disk is less than the grip at the top. Add up the grip over the whole disk that each lap encloses and the summed hill matches the measured travel to within 2 per cent all the way to 60 degrees, where the one-number square rule overshoots by a third, 0.0707 against 0.0528 body lengths a lap[19]. Summing a curvature over the room a loop encloses is not ours. It is the height-function method of geometric locomotion, Hatton and Choset's (read through a secondary source), who showed too that a good choice of body frame is what makes it accurate; we arrived at it blind, and the toy's 2 per cent is a check of their method.

The proper words. Mathematicians have a name for every piece of this bench, and they are worth knowing, because they open the papers our toys kept walking into. The map of the two knobs, every setting they can take, is the *base space*; for a swimmer or a cat it is the space of its shapes. Above every point of that map sits a *fibre*: everything about the body that the knobs do not set, which way it faces and where it is. The two together, every shape carrying all its headings and places, make a *fibre bundle*. A rule that says how the body moves when its shape changes a little and nothing pushes it is a *connection*, and following that rule round a loop of shapes is *parallel transport*, carrying the body along without twisting it on purpose, as you carried your thumb round the sky. What is left over when the loop of shapes closes is the *holonomy*, this book's leftover; and how much leftover each patch of room buys is the *curvature*, this book's grip. The wall decides the connection. With no wall, the body's own momentum, which nothing outside it can change, sets the rule: its turning momentum sets how it turns, which is how a cat turns over, and its momentum along the floor pins its middle in place, which is why a body touching nothing can never travel on a flat floor. With a borrowed wall, the water's drag sets the rule instead. So the bench's sentence, that the leftover is the room inside the loop times the grip, is the mathematician's too: the holonomy of a small loop is the curvature times the area it encloses, and the summed hill above is the same sentence for a big one. *Loop walking* is the habit this book is made of: walk such a loop, come home, and read what changed.

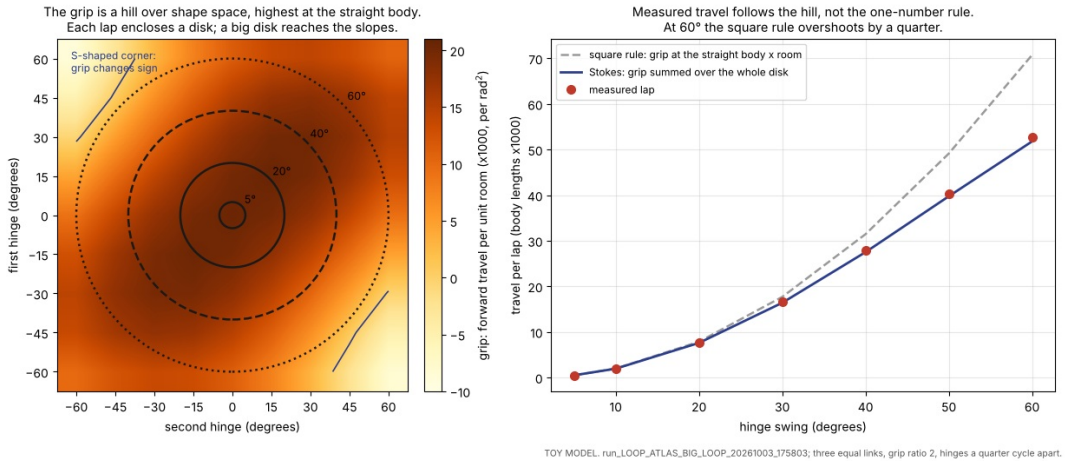


Figure 11. Left: the grip across shape space as a hill, with the disks enclosed by laps of 5, 20, 40 and 60 degrees. Right: measured travel against the square rule and against the hill summed over each disk. TOY MODEL.

Where quadrature lives. Near the end, at Bear's word, we read for quadrature every paragraph of the longer working text this book was drawn from. It shows up in time in 27 of its 480. Three of those simply name it; the other 24 use it in one of three ways. Some are about room: two knobs out of step draw a loop with room inside, and a quarter apart draws the most room, because the area a loop of two swings encloses grows with the sine of their gap. Some are about power: a push in step with the speed it pushes, its position a quarter behind. And work per cycle is an area too, the area of the loop the push draws against the position, which is what an engine's indicator diagram has drawn since the age of steam. The rest are about the gears of chapter 10, which count the needle's turns, and any gap at all gives the count. So a turn and a day's work read the same thing, the room inside a loop, and quadrature is the timing that makes the most of it; a gear asks only whether there is room. Quadrature also hides where the words never name it: the falling cat's circuit and the three-piece swimmer's hinges are both built a quarter cycle apart. It never appeared in the parts of the book about memory and limits, and in the listening chapter only once we went looking for the ear's clock, after the count. None of the three ways is new; seeing them sorted is.

The parts bin

Something to do at the sink. Stretch a rubber band and let it go: it snaps back and gives you your push again. Now drag your hand through a sink full of water and stop: nothing comes back. Those two feelings are the two most important words in the book. A **refunder** fights you on the way out and helps you on the way home: a rubber band, a spring, a capacitor, a weight lifted against gravity. A **taker** fights you both ways and gives nothing back: water in a sink, a shock absorber, a heater, a resistor. A **feeder** puts work in: a hand on a crank, a motor, a battery. Work is done only when a push and a movement arrive at the same instant, which is why the quarter-cycle gap matters so much.

Refunders come in two kinds, and the difference is the most useful single fact on the page. A spring stores in *position*; a flywheel stores in *motion*. Two springs together only share. A spring and a flywheel together **ring**: the spring pushes the flywheel, the flywheel winds the spring, back and forth, a pendulum or a bell or a radio's tuned circuit. Add a taker and the ringing dies away; that is a **settler**, the shock absorber on a car. Add a feeder, timed right, and it keeps going. A taker can swallow energy and never a shove: momentum cannot be warmed away, it has to be handed to another body, which is why a ship cannot tow itself.

The ringer, on a playground swing. Two refunders that ring is the hardest of these to picture, so take a swing with nobody pushing and nobody pumping. At the top of each arc you are high and, for an instant, stopped: all the energy is stored as height, in the refunder that stores in position, with gravity playing the rubber band. At the bottom you are low and going your fastest: the height has all been paid out, and the energy is now stored as motion in your own body, the flywheel kind of refunder. The two hand the same energy back and forth, a quarter cycle apart, and neither keeps any of it, so the swing goes on and on. A tuning fork, a capacitor joined to a coil, and a planet on an oval orbit, trading height for speed and back, are all this one pair.

The settler, and the one that lives. A refunder with a taker settles: it is a damper. A car's spring carries the bump, and its shock absorber bleeds a little off each bounce until the car sits still. Add a feeder, a small push timed to arrive with the movement and to replace exactly what the taker took, and the thing lives: a clock's escapement, a bowed violin string, a child pumping a swing. Bear felt at once that this third one is like a flywheel, and it is, because the flywheel is the half that remembers. Chapter 6 is this machine, and the whole question there is when the feeder pushes.

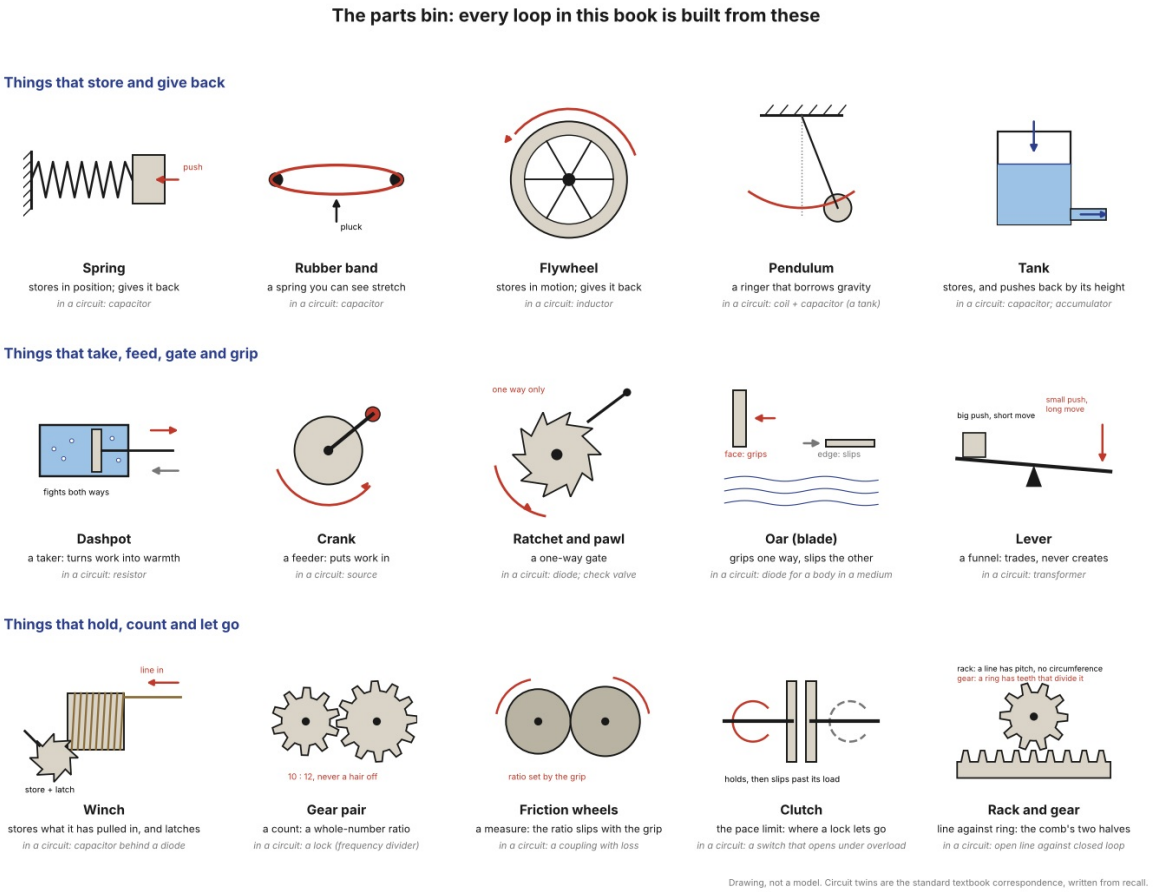


Figure 12. The parts bin: fifteen basic shapes in three rows (things that store and give back; things that take, feed, gate and grip; things that hold, count and let go), each with its job in a loop and its circuit twin.

Heat that rings. Heat has a refunder that stores in position, a warm brick, and a taker, a wall that lets it leak, and with Fourier's law nothing else: no flywheel, so heat settles and never rings. If heat's flow takes a moment, τ, to catch up with the slope of temperature that drives it, temperature gets an inertia beside its drag, and the heat equation becomes a damped wave. That law is Cattaneo's (1948) and Vernotte's (1958). Run in a toy rod with its ends held cold[20], it rings exactly where a spring and a bead stop ringing (the diffusivity D pulls each spot toward its neighbours the way a spring pulls a bead home): started as a single hump, the rod rings when it is shorter than $2\pi\sqrt{(\tau D)}$ and settles when it is longer, its middle dipping below zero by 0.015 of its start at 0.80 of that length and not at all from 1.05 up. A warm brick dropped into ice rings too, for a reason the hump does not have. Bear's word for it is quenching, the smith's plunge of hot metal into cold water. The cold from its two faces travels inward as two fronts, and where they meet in the middle they knock it below the ice: by 0.70 of its warmth in a brick a quarter of that length, and by 0.25 at half, on the toy's grid of 400 cells, and by 0.717 and 0.251 on a grid eight times finer[21]. It stops ringing at the same length as the hump, and in pieces up to 0.88 of that length its deepest dip is an echo: the fronts run on to the far faces, are turned over there, and cross the middle again, about one and a half lengths of heat's travel after the plunge[22]. For whoever builds with heat, the knob is the size of the piece against $\sqrt{(\tau D)}$: heat rings only in pieces shorter than about $2\pi\sqrt{(\tau D)}$, and everywhere else it settles.

What this chapter found. Name the pair of knobs that draws a loop with room inside. A loop needs something to leave its change in, and there are four kinds of wall, including none. Double the stroke and a loop goes four times as far, because what it leaves is the room it encloses times a grip; and since the grip is a landscape, not a number, summed over each loop it predicts a swimmer's big strokes within 2 per cent where the square rule overshoots by a third.

4. Three questions for every loop: faster, backward, again

The needle lives on a staircase that tilts more the faster you ask it to go. Every step has a seat to rest in and a lip to fall over, and they always come as a pair. Counting the steps is easy, as long as you count landings rather than stumbles.

— a no-physics read

“What’s a winding, when it has no counter?” (Bear)

Something to do at the table. Hold a plate flat on your palm and turn it one full circle under your arm, keeping it flat the whole way. Now ask it three questions. *Faster?* Do it quickly, then slowly: your arm ends up just as twisted either way. *Backward?* Turn the plate back the way it came and your arm unwinds. *Again?* Turn it a second full circle the same way, and your arm comes free. A loop run again doubles what it leaves; the plate run again undoes it, which is why chapter 5 files the plate as a twist and not a loop. Asked of every row on our list, the same three questions sort the book.

Faster? Which clock a loop counts in

We put every toy on the shelf through one test. Each was run unchanged at its own rhythm and at others, and we asked what it was really counting[23]. The worm chain, a row of segments that squeeze one after another (chapter 5), travels the same distance per lap at every speed tried, from half its own to three times. The wing goes 1.873166480074 metres per flap at 1.5, 3 and 6 flaps a second. The pumped swing reaches 40.1433 degrees after six swings, whether it is 2.5 metres long or 10. The needle of chapter 10 keeps 0.9991 to 0.9985 of the field’s turns at slow paces, and past its edge keeps 0.4475 of them against 0.4472 from the closed form. The skater, whose legs push with a force for a time, counts in seconds instead. And two jellyfish bells tied together, squeezing in turn, go 59 per cent further per squeeze when the squeeze period doubles, because the rings they push off swim at their own speed in the water’s time. So “faster?” has a short answer with two branches. A loop whose partner forgets does not care how fast it is walked, until it meets a pace limit, an edge past which the partner can no longer follow. A loop whose partner remembers cares at every speed, because the partner keeps a clock of its own.

The bells taught a further lesson. Stretch every clock in their world by two and they go exactly as far per squeeze as before, to every digit[24]; stretch only some, and the stroke moves. The bells’ balance turned out to be two pushes cancelling, a squeeze 7.55 per cent up against a reopening 15.8 per cent up[25]. Those two numbers are the coarse grid’s, since the run’s own checks on step and band missed. They were also made with rings whose cores were thinner where they met than where they travelled, a fault found later that moved other bell numbers by up to a third, one from 0.68 to 1.01[26]. So a result that does not move at any setting can be two pushes cancelling as well as no push at all.

Bear met the same answer on a balance board riding a rolling cylinder. He said the rider on a short board taps left and right faster because its event time has shrunk. A rider who counts in the board’s own fall time sings the same song on every board, identical tap for tap on boards of 0.25, 1 and 4 metres. Its taps per second still rise as the board shortens, from 2.99 at 4 metres to 11.98 at 0.25, because the ruler slides while the song stays put; the ratio is exactly the fall times’[27].

Backward? The free check

Where the partner forgets, running a loop backward must undo it exactly. That makes backward the cheapest check in the book. Our first snake went 0.1508 forward and 0.1472 backward, which exposed a drifting stepping rule; corrected, it goes 0.14898 both ways[28]. The same question is a lock detector. Run backward, a slipping needle retraces itself to one part in ten million million, while a locked one misses by half a turn[29], because a lock forgets the road it came by. And backward is a direction you can steer by: the skater, with the toe angle half a cycle out of step, goes 3.10 metres per second backward, as fast as it goes forward in step.

Again? Every count is a winding

Bear’s line asks how a winding can be counted when nothing in it counts. The needle answers him. Its lag obeys d(lag)/dt = pace − sin(lag). In words, the lag slides downhill on a hill that drops by a fixed step every time it goes once round the circle. Three things follow, all measured[30]. Locks come in pairs: there is always a stable one and an unstable one together, so stable minus unstable is 0 in all 39 rows, the circle’s own count. The slip count is a whole number: read as net crossings of a fixed level, it is exactly 0 in lock over 100 and 1,000 field cycles, and past the edge it is the square root of (pace² − 1), over the pace, per cycle: 140 against 140.4, 416 against 416.6, 866 against 866.0, Adler’s law for radios met by a needle. And under noise the whole number leaks at a predictable rate, over the barrier at Kramers’ rate: measured over predicted is 1.000, 0.991, 0.999 and 0.949 at the four barriers first read, each within its counting error of 0.05 to 0.09[31]. A later, wider sweep at barriers from 4 to 9 times the noise’s strength, with six times the slips, pools to 0.909 ± 0.011: Kramers’ leading formula runs 9 per cent high there, a shortfall the counts resolve. Set against the exact rate instead, Stratonovich’s of 1958 for a needle like ours, the same slips pool to 0.9885 ± 0.0122, within a standard error of 1, so the shortfall is the leading formula’s own: the exact rate is 0.905 of Kramers’ at 4 noise units and 0.965 at 9[32]. Set against the exact rate, the four barriers first read sit at 1.069, 1.062, 1.097 and 0.986 of it, each within two of its counting errors of the rate itself[33].

The slips keep a rhythm. Counted one needle at a time, the slips spread as independent events do, each row’s counting error the square root of its slips within 3 per cent[34], except at the lowest barrier, three times the noise’s strength, where they spread less than chance. That is real, and it is the needle’s own. Reimann and five co-authors worked out in 2001 how much a noisy needle like ours spreads on a tilted washboard; for our slips at 3 noise units

their formula gives a variance over mean of 0.913 to 0.916, and the needle's own waits between slips, 376,041 to 852,370 of them a row, give 0.9134 ± 0.0031, 0.9135 ± 0.0024 and 0.9182 ± 0.0020[35]. Short waits are rare: waits under a tenth of the mean came only 0.58 to 0.59 as often as chance would make them[36]. Our reading, not yet tested as the cause: a needle that has just slipped has to slide down into its next well and settle before it is likely to climb again. The rhythm fades as the hill grows, to 0.963 at 4 noise units, 0.985 at 5 and 0.996 at 6.5, and at each height it is nearly the same at every noise.

A last run closed the gap to the exact rate. On new seeds, four times the needles and half the step, the rows of 4 to 9 noise units pool to 0.9877 ± 0.0061 of Stratonovich's rate[37], and the start accounts for that 1.2 per cent within its errors: a needle set down at rest is about 12 time units late to its first slip. On 32,768 new needles, seven of those rows read 0.7645 of the exact rate in their first 50 time units, and from 50 to 500 the slips, 78,593 of them, read 1.0027 ± 0.0036[38].

"The dancers want to dance, but they need to wait to hear the rhythm of the band over the disorder," Bear said when he saw it. Hold the pace fixed at 0.95 and turn the noise up from nothing, and the slips go from waiting on luck, as independent as coin tosses, to their most regular, spread less than half as much as chance (a variance over mean, which physicists call the Fano factor, of 0.42341 at noise 0.7), and on to a jostled scramble[39]. The needle itself, run there, agrees: its waits read 0.42357 ± 0.00126 at noise 0.7 against the formula's 0.42341[40]. Too little noise and the needle waits for luck; too much and the noise shakes the slide itself; in between, the noise tips it over the last of its hill as soon as it arrives, and the slide round the rest of the turn keeps the beat. That noise can make a machine most regular at one level is known in kind for noisy excitable systems.

The saddle up close. A needle with memory 2, driven at pace 0.95 with a little noise, slips over its barrier again and again, each slip a climb from the well to the barrier's top and a fall from the top into the next well. Its record looks like a long list of numbers with no pattern, but it keeps its memory in exactly two places. Each fall remembers only the one before it: one fall with the next ties at +0.0600, two apart at +0.0012, on 1.48 million pairs[41]. The kick the two neighbours share is the needle's speed at the moment it lands. Hold that speed fixed and the tie between falls vanishes (−0.0011), and the speed itself forgets from one landing to the next (+0.0206)[42]. The landing is a fixed place on the turn, so the speed there is the needle's whole state: a clean section, in Poincaré's sense. The barrier's top is a second one, once its speed is taken out along its own curve rather than a straight line[43]. And the top is a saddle with a law. The slower the needle crosses it, the longer the fall that follows, growing as the logarithm of one over the crossing speed by 0.6949 periods for every factor of e, the number the saddle's own escape rate gives. Launched from the top without noise at crossing speeds of a ten-thousandth to a thousandth, the fall keeps that law to 0.9988 of it[44]. The noisy replays had shown only half the law, −0.3231[45], because their noise, as large as the crossing speeds themselves, shakes the needle off the top early[46]. That passing a saddle costs the logarithm of how closely you pass is textbook; what the toy adds is where to look: two sections, each a single number, and between them a saddle that keeps its law.

Bear's racetrack shows what a count is not. A car on a banked turn sits where the bank is square to the tilt it feels, and rides higher the faster it goes. But on a real bank, from the apron to a wall at 35 degrees, there is one seat at every speed up to 26.21 metres per second and none beyond, where the car rides up against the wall. Count the wall as the seat when it holds the car, and stable minus unstable is 1 at every speed, not 0, because the track is a segment and not a circle, and its cliff is the end of the track. On a full tube, curling over the top and back down, there are always exactly two balance points and no cliff at any speed. A staircase of counts needs a push that goes round[47].

Give the tyres grip and the seat becomes a lane that holds over a band of speeds. Sweep the speed slowly up and back down, and the car's lane traces a loop: a car started on the apron and swept from 12 metres per second to 34 and back comes home 25.05 degrees up the bank, and a second sweep brings it home to the same place[48]. The knob came back and the car did not. That is not a holonomy, because the sweep goes out and back along one line and encloses no room. It is the leftover of a dead band, a stretch of push where nothing moves until the push clears the grip: hysteresis, and chapter 10 meets it again in the needle with memory, whose two edges make a pair of scissors.

A count needs a counter, and recounting every count in the book with one taught three rules a builder of counters can use: the counter must be running before the watch starts; time a slip at the bottom of the turn or at the landing, never at the top of the barrier, where false starts live, since the landing is where a remembering needle forgets[49]; and look again before the reading can move half a turn, as chapter 7's mountains showed.

Can a loop feel its own cliff coming?

"This keeps feeling like a seam, where two or more walls meet. Each wall has its own rules, its own rulers and its own knobs, and each has a hard stop. It feels as if we are standing at a seam, or at a corner where the walls join." (Bear, 5 October)

It is the right picture. Every cliff in the book is a crease in the map of knobs, where the set of places the machine can rest changes shape. We walked toys slowly toward their cliffs under a little noise, reading gauges that need nothing but the toy itself: where it can rest, how long it takes to come back after a small kick, and the size and memory of its

wobble[50].

What meets at the crease	Where we met it	How the warning grows as the edge comes near
a seat meets its lip (a fold)	the needle	recovery time as one over the square root of the distance
three roots meet	the delayed smooth push, tuned as well as it can be	recovery time as one over the distance
a spinning path brushes a saddle	the needle with memory, climbing back	time per turn as the logarithm of the distance
a third star's binding runs out	a chaotic triple star, before its escape	excursion time as the distance to the power minus three halves, Kepler's
a knob's hard stop	the banked track's wall	silent in motion; it shows only in position
two clocks cross	the tapping rider whose delay reaches its rest	silent

Bear's cliff edge. Bear's first question about cliffs was whether a needle locked to a master timer knows where its edge is. It does, by feel. Its lag sits at the angle whose sine is the pace, to four decimals or better at every pace tried, so the pull it needs equals the pace: the lag reads how much of its grip is in use[51]. The edge is an angle it can feel, the margin between where it sits and where it tips, 2.0944 radians at pace 0.5, 0.9021 at 0.9 and 0.2831 at 0.99; a kick of 0.99 of that margin comes back, and a kick of 1.01 of it slips a whole turn, at every pace tried[52]. Under noise the cliff becomes a ramp that starts earlier the louder the noise. A needle that feels its pull, which needs the master timer, still reads its pace within 0.010 right to the edge at noise 0.1, mostly on the high side, the cautious one; a needle that feels only its own wobble, with no master, reads within 0.014 up to pace 0.9 and fails once the slips begin[53].

Past the edge the needle shows a second law. Once the lock is gone, it slips once every 2π over the square root of $(\text{pace}^2 - 1)$ time units: near the edge that is the ghost of the lock, time going as one over the square root of the distance, and far from it plain one over the distance[54]. That slip time is not ours: it follows from Adler's equation (Adler 1946), and the toy met it blind, as it met his locking law.

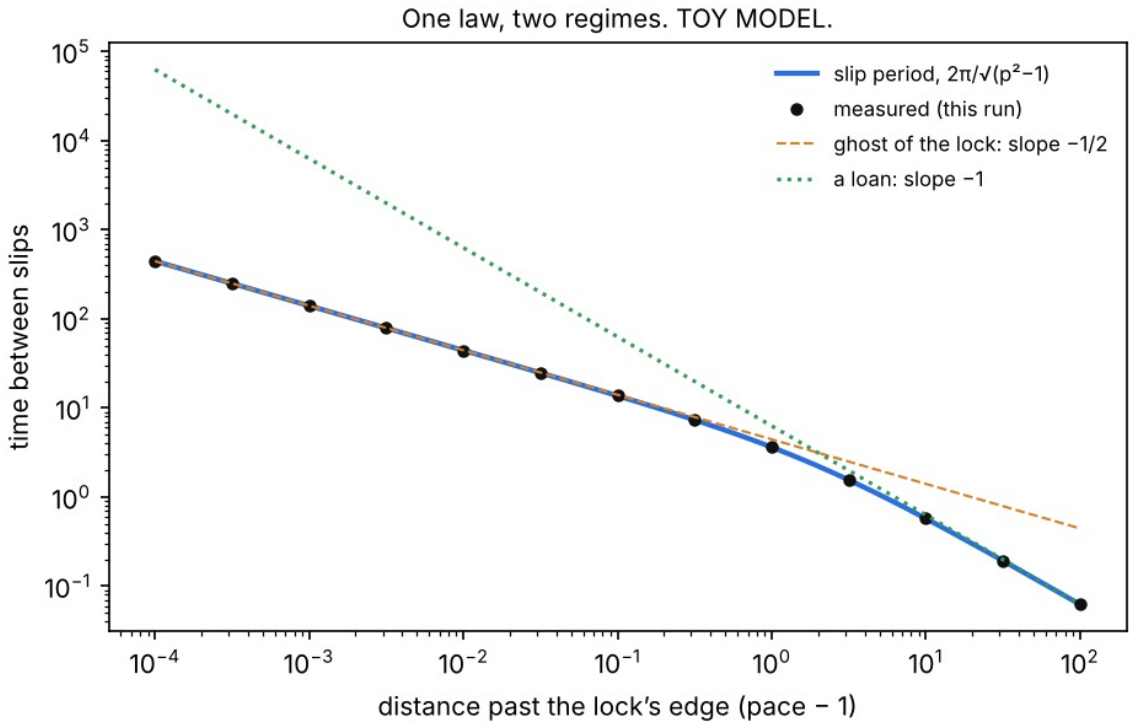


Figure 13. The forgetting needle's time between slips past the lock's edge: one law that bends from the ghost's slope of minus one half near the edge to the far slope of minus one far from it. TOY MODEL.

The early warning was the first rare slips. A noisy needle has a gauge of its own that needs no master: how fast its angle forgets a kick, read against its own straight-line turning and turned into the share of its full grip in use, which is 1 at the edge. Under loud noise it reads high, and that turned out to be the truth rather than a fault: the noise keeps

the needle leaning further over than its pace alone would put it, so on average it is held more softly, and the gauge reads that softened hold[55]. It also seemed to warn: on three fresh noises the gauge reached 0.95 of full grip at paces 0.10 to 0.15 below where the slips pass one per 100 field cycles.

What the warning was made of took three slip finders to say. The first two, built to leave the slips out, were blind in ways a ten-line check would have caught before the run. The third was tried on planted slips before it was trusted, and with the slips truly left out, the gauge reads the softened hold within 0.034 wherever slips are rare, though it reads high as the noise grows and the pace nears the edge: about 0.03 high at the loudest noise, four to six times its error (its registered bar of 0.03 missed there), and 0.007 to 0.026 high, three to five times its error, at seven quieter settings near the edge, a bias of the noise near the edge and not of how the slips were cut out, and at each of three fresh noises it reaches 0.95 in the same step of a 0.05 pace grid as the slips pass one per 100 field cycles, not before[56]. So the early warning was the first rare slips. A noisy needle's own gauge tells honestly how softly it is held; what warns of the cliff is hearing the first falls. Ecologists know that flickering between two states can warn of a shift where slowing down does not apply, and they read it through the state's variance, correlation, skew and spread (Dakos, van Nes and Scheffer 2013); climate scientists know that a jump driven by noise can come with no slowing down before it at all (Ditlevsen and Johnsen 2010). This is the flickering idea with each fall heard and counted, given a place and a number.

The silent crease, and a fix that makes it hum. A rider balancing a board, who taps left or right after a delay, falls without warning when its delay reaches its rest. The obvious fix is to stop the two clocks from meeting, and it moves the cliff only a little, because a fixed tap given once per rest cannot bring back a board that has been falling for most of a fall time[57]. A smooth push holds to a delay of the square root of two fall times; a rider with a fixed tap, to about 0.55. The gap closed when each tap was sized by the reading, so that the pulses add up to a smooth push. Sizing alone is not enough: a strong sized tap given blind still falls exactly where its delay meets its rest[58]. A sized tap that never taps blind keeps every rider up to a delay of 1.4 fall times and most of them to 1.5, about the smooth push's own limit[59], and gives way between 1.5 and 1.6[60].

Where it gives way is set by the jostling, not by the controller: halve the jostling and the edge moves out, from 1.55 to 1.70 fall times for the gentler of the two kicks the test was set for, sized at strengths of 2.5 and 3.0 in the toy's units, with a third, 3.5, flown beside them[61], and with no jostling at all the gentler kick keeps every rider up to 1.75 fall times, short of the registered bar of 1.80 asked of both kicks, and the firmer one to 2.00[62]. The controller's own edge lies well past the square root of two, and the noise pulls it in. Control engineers know that a controller which acts and then waits out its own delay can beat a continuous one's limit; the toy gives the numbers.

And the cliff is no longer silent. As the delay grows, the rider's wobble rises at every step, from 0.012 at a delay of 0.3 to 0.11 at 1.4[63], where the fixed tap's wobble gave no sign at all; without jostling the hum turns into a swing just before the fall. So the silent crease has an engineering fix, and the fix makes it hum.

What this chapter found. Every count here is a winding: a needle past its edge slips 866 times in a thousand cycles against Adler's 866.0; a noisy lock's early warning is its first rare slips; and taps sized by the reading and never given blind hold a delayed balancer to 2.00 fall times in quiet, past the √2 limit of a delayed tilt-and-rate push.

5. Loops that leave a turn, and loops that leave a shift

> The snake is a ribbon laid on a breathing floor. Each joint gets paid for how much it creaks.
>
> — *a no-physics read*

Something to do at the table. Take a drinking straw and bend it in the middle so it makes an L. Hold one arm of the L between finger and thumb and roll it, keeping the bend. Watch the other arm: as you roll, it swings round in a circle. Now straighten the straw and roll it the same way. It just spins in place. A straight thing rolls; a bent thing rolls *and turns*. That is the falling cat, and it is the whole first half of this chapter.

A cat dropped upside down, with nothing to push against, lands on its feet. It does it by bending in the middle, turning its front half one way while its back half turns the other, then straightening and doing it again with the bend pointing somewhere else. Each round trip leaves a little net turn. The swivel chair of chapter 1 is the same trick with arms for the two halves. In every one of these the body touches nothing, the two knobs go round a loop, and what is left behind is a turn and never a step. Our whole list says the same: of the nine loops with no wall at all, four leave a turn, four leave a clock out of step, and only one leaves a shift, the three beads on a ball, which travel only because the ball is curved. You can turn without a wall. You cannot travel without one, unless the floor itself is bent.

Some turns are made by the world rather than the walker. A pendulum swinging in a hall slowly turns its line of swing over the day, not because anything pushes it but because the Earth carried it round a circle of latitude, and the patch of sky that circle encloses is the turn. A gyroscope carried once round the Earth comes home with its axis tilted a hair,

for the same reason, with space itself as the curved floor. Light sent round a coiled fibre comes out with its polarisation turned by the patch of directions the coil's axis swept out. These are the same loop, with the Earth, or space, or the fibre as the arm that carries it round.

One thing that looks like a loop is not one. Hold a plate flat on your palm and turn it a full circle under your arm; your arm is now twisted. Turn it a second full circle the same way and your arm is free again. Two turns to come home, one is not enough. That is a *twist*: running it again undoes it, where running a loop again doubles it. It is the belt trick, and it is kept here so that a reader meeting it will not file it with the cat.

A swimmer steers by holding a curl. Chapter 3 maps how hard the water grips a three-link swimmer at every shape it can hold, and a second map comes out beside the first: how much each small lap *turns* the body. For a straight body it is exactly zero. For a body bent into an S it is exactly zero. For a body curled into a C, both hinges bent the same way, it is not. Held at a C of 45 degrees on each hinge and making small 20-degree laps around that shape, the swimmer turned 2.2042 degrees every lap, the same to thirteen decimal places for forty laps, and came round a full circle in 163 laps[64]. Curl the other way and it turns the other way by the same amount. The turn per lap is the heading-curvature at that shape times the room inside the lap, 5.73 degrees per square radian times 0.383 square radians, which predicts 2.19 against the 2.20 measured. That a gait's net turn is the curvature summed over the lap's room is the rotational part of a theorem Hatton and Choset built for the locomotion of shape-changing bodies, and of Shapere and Wilczek's gauge picture of swimming (1989), both read through a secondary source; we walked into it blind. The body travels a little as it turns, 0.00847 body lengths a lap, so its centre goes round a circle of turning radius 0.220 body lengths, and the body pinwheels around it.

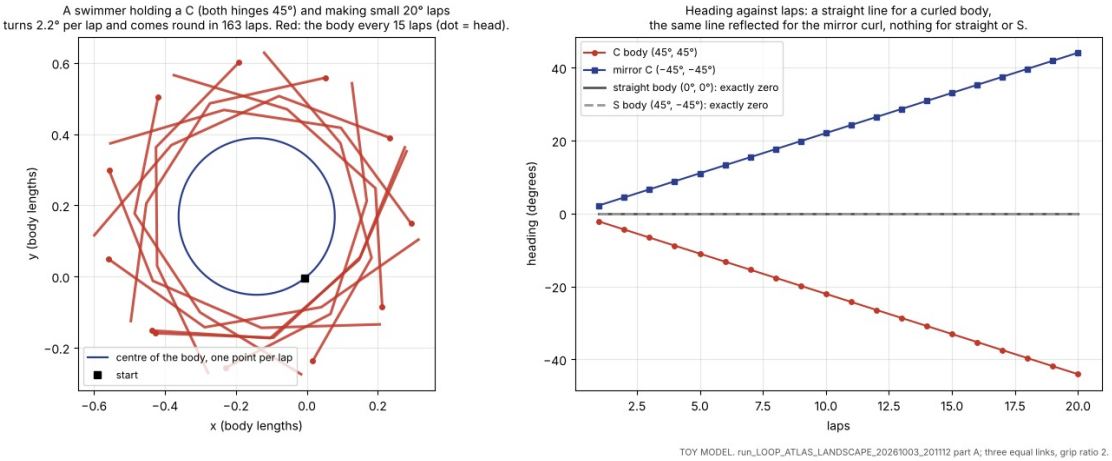


Figure 14. The body's centre over a full turn of 20-degree laps at a C of 45 degrees, with the body drawn every fifteen laps (left), and heading against laps for the C, the mirror C, the straight body and the S body (right). TOY MODEL.

So a swimmer in a fluid that forgets has a rudder with no rudder in it. Keep swimming, hold a curl, and you turn; the tighter the curl, the faster. We suspect this is the toy's version of how anything long and bendy steers, and in kind it is known. Builders of snake robots turn them the same way, adding a fixed bend to the body's wave, the offset turn they credit to Hirose; a worm crawling on a dish curves gradually toward salt, the weathervane of Iino and Yoshida (2009), and a model of a worm steering by warmth sets its curving by the balance of the muscles on its two sides. A sidewinding rattlesnake turns otherwise, by changing the size of its sideways wave or shifting its lifting wave by half a cycle (Astley and colleagues, 2015). We read all of these only in search excerpts. A swimming snake we have only looked at. In sixteen stills Bear took from a short video Oceana posted of a banded sea snake swimming, it moves by the same sideways wave, about one wave along its body, and in several stills in a row its front half lies in one long arc, bent the same way. Stills cannot tell a held curl from a moment of the stroke, or show whether the snake turned. Filmed turns can. Gregorio, Godoy-Diana and Herrel (2026) filmed water snakes turning, six turns by six snakes from above in the published paper, and found that they curl inward to turn, drawing head and tail toward the middle; in four of the six the body's moment of inertia, how hard it is to spin, fell well below its swimming value, and they read the turn through that drop and the stronger vortices the turning snake sheds. So a real swimming snake does curl to turn. Its water is not our toy's: at a snake's size and speed the water's own momentum counts, and our fluid that forgets has none, so the shape agrees and the mechanism is not shown to be the same. We read their preprint in full and the published paper through a summary of its pages.

Something to do in a car park. Parallel parking is a loop. Steer and drive are the two knobs. Steer right, drive back; steer left, drive back; steer right, drive forward; steer left, drive forward. The wheel is back where it started and the car has moved sideways, which it cannot do directly, because a car cannot slide. The tyre's refusal to slide sideways is

the wall. Four moves, two knobs, a sideways shift that no single move could make. Everything in the rest of this chapter moves itself along by going round a loop against a borrowed wall.

Bodies that shift. The three-link swimmer does best with a flat middle a fifth to a third of its length, 0.029 body lengths per lap, and is nearly dead with a middle that is nine tenths of it, 0.0003; a long flat middle is dead length, and dead length cost every time it was tried. More hinges pay: two, three, four, five, seven and eleven hinges gave 0.029, 0.060, 0.094, 0.122, 0.170 and 0.178 body lengths per lap, and travel per hinge climbs to five hinges, holds to seven (six, eight, nine and ten were not tried) and then falls, so eleven add little over seven, because the body already holds about half a wave of bending, and half a wave is what the worm chain below wants; the swimming snake of chapter 3 does best nearer three quarters, at the same grip, so half a wave is the worm's best and not every body's[65]. The half-wave rule came from a worm: a chain of squeezing segments, each a step behind its neighbour, crawls best when the step works out to half a wave along the body. Close the chain into a ring and only whole waves fit around it, so the half-wave best is forbidden, and the ring crawls less, 1.14 per lap against 1.66 for the open chain of eight[66]. That is the first place in the book where a count gets in the way of a measure, and it points at chapter 10.

Bear's faces: where a swimmer should be broad. At the sink a flat hand swept sideways pushes back hard and an edge-on hand slips through. In our swimmers that difference is one number for each piece, its grip ratio, how much harder it is to drag sideways than lengthways: 2 for a thin round rod. Bear asked what happens when we change the faces a body presents to the water. On the first swimmer, three equal pieces with two hinges a quarter cycle apart, we gave one piece or more a broad face of grip 6 and left the others round[67]. The round body travels 0.0278 body lengths a lap. A broad middle carries it 0.0532, broad ends 0.0648, and broad faces on all three pieces 0.1191: the three faces together buy 1.79 times the sum of what each buys alone, as if each helps the others hold the water. The swimmer has a front, too: a broad head carries it 0.0409 and a broad tail 0.0403, because the bending runs from head to tail. And the faces cost. Per unit of work the broad middle goes farthest of the single changes, 0.281 body lengths against the round body's 0.202, and all broad goes farther still, 0.310. So for a builder: broad ends, both end links made broad together, buy the most travel of any single change, a broad middle the most travel for the work, and broad everywhere more than the sum of its parts.

The coil snake, turning. Bear's snake that might also pump a log coil in three dimensions, as he put it on 2 October, starts as the simplest coil there is: a thin wire wound ten times round an axle, turned on the axle and free to slide along it[68]. Its one knob goes round instead of back and forth, so the scallop theorem that stops a one-hinge stroke does not apply, and every turn carries the coil the same way. A round wire wound at 45 degrees travels 2.09 radii a turn; the fastest winding is steeper, 54.7 degrees and 2.22 radii a turn, where the tangent of the angle is the square root of the grip ratio; a broad ribbon of grip 6 is fastest at 67.8 degrees and goes 6.41 radii a turn. The water sets the price: per unit of the axle's work the round coil at 45 degrees goes 1.30 times as far as at its fastest winding, so the quickest coil is not the cheapest. The cheapest, worked out by hand before the tool and then swept, is 35.3 degrees whatever the grip, with the coil's ten turns held, where the tangent of the angle is one over the square root of two; for a round wire it is the fastest winding's mirror angle, and it goes √2 times as far per unit of work at about four fifths of the speed. The grip drops out because it sets the gain and the price alike: the same drag that decides how hard the wire pushes the coil along decides how hard the axle must twist it, so the grip changes the travel a unit of work buys by the same factor at every winding, and the best winding is left to geometry alone. Speed has no price to divide by, so the fastest winding keeps the grip. Hold the wire's length instead, and the cheapest winding is 45 degrees[69]. Give the coil a body to push, as draggy along the axis as the coil itself, and on a round wire the cheapest winding stays at 35.3 degrees whatever the body's size; on a ribbon of grip 6 it moves flatter, not steeper toward 45: 27.5 degrees, heading for the 22.2 worked out by hand as the body grows heavy. A coil whose turns grow outward, a log coil, is cheapest at 35.2. A log coil, its radius growing a tenth each turn, travels as an even coil of its mean radius to 0.01 per cent, but it pushes the axle sideways. This closed form is the textbook result of drag-only swimming. We have met the best winding written out in Darbois Texier, Ibarra and Melo 2017, and the speed law it comes from in excerpts of later papers, which build it on the drag theory of Gray and Hancock (1955); worked by hand, that law gives the 2.09, 2.22 and 6.41 above. Who first wrote down the best winding we have still not found.

The coil snake, pumping. Then the coil was made to pump instead of turn: its radius and its climb swelled and shrank a quarter cycle apart, a loop of two knobs, the kind the scallop theorem allows to swim. It went nowhere, not a hair along its axle and not a hair round it, at most 3.2e-15 radii and 7.5e-17 turns a cycle, at every size of loop[70]. The reason is the coil's own shape. A uniform coil looks the same turned end over end about its middle, and a squeeze that is the same all along it cannot favour either end. Break the symmetry with a wave and it swims. Let a swelling run along the wire, three crests over the ten turns, a tenth of the radius high, and every cycle the coil moves 0.0046 radii against the wave, the way Taylor's swimming sheet moves[71]. Ten crests go ten times as far as one; twice the wave's height goes four times as far; water that grips sideways six times as hard as lengthways moves it 4.99 times as far as water that grips twice as hard, close to the ratio of the grip ratios less one, 5 to 1. Reverse the wave and the coil retraces its path to the last digit, and pumped twice as fast it goes exactly as far per cycle: the clock test, passed. So never pump a uniform coil evenly; run the squeeze along it as a wave, and the coil goes the other way.

The soft ring. Bear's jelly-like blob was built as a ring of soft links whose outline squirms, in the first swimmer's water. Breathing evenly, every side in and out together, it goes nowhere and does not turn. Run one wave of squirming round its rim and it turns, 0.0592, 0.0842 and 0.1046 radians a lap at 2, 3 and 4 crests, against the wave, but it travels nowhere, because a part of a lap later its shape is the same shape turned. Run two neighbouring waves together and it swims: 0.04234 ring radii a lap at a wave height of a tenth of its radius and grip 2, against the ripples of the patch where the two waves add, as the coil swam against its wave. Water that grips harder carries it farther, 0.02595, 0.04234 and 0.07408 radii a lap at grips 1.5, 2 and 5[72]. When its waves are small its travel goes as their height squared: from a height of 0.0125 to 0.025 the travel grows 3.9578 times, close to the square law's 4, about 5.2 times the height squared a lap; from 0.05 to 0.1 it grows only 3.43 times, because there the two waves add into a bulge a fifth of the ring's radius[73]. It is a dear way to swim: 0.0088 ring radii for each unit of work, where the first swimmer goes 0.21 body lengths.

An S-shaped body swims backward. The grip map falls away from the straight body toward the S-shapes and changes sign in the far corner. Walk the body's resting shape from straight toward an S, hinges bent opposite ways, and give it the same small 5-degree loop at each stop. At the straight body it goes forward, 0.49 thousandths of a body length a lap. At 15 and 30 degrees of S it still goes forward, less. At 45 it is almost still. At 60 and 75 it goes *backward*, 0.21 and 0.40 thousandths, with the same loop in the same sense[74]. Along the C, both hinges bent the same way, the travel only fades and never turns round. Where the S body reverses depends on the grip: near 48 degrees for a body that barely prefers sliding lengthways, 46 for a thin body in thick fluid, 39 at grip 5, 22 at grip 30, 13 at grip 100[75]. The harder the water holds the body, the smaller the S it takes to send it tail-first.

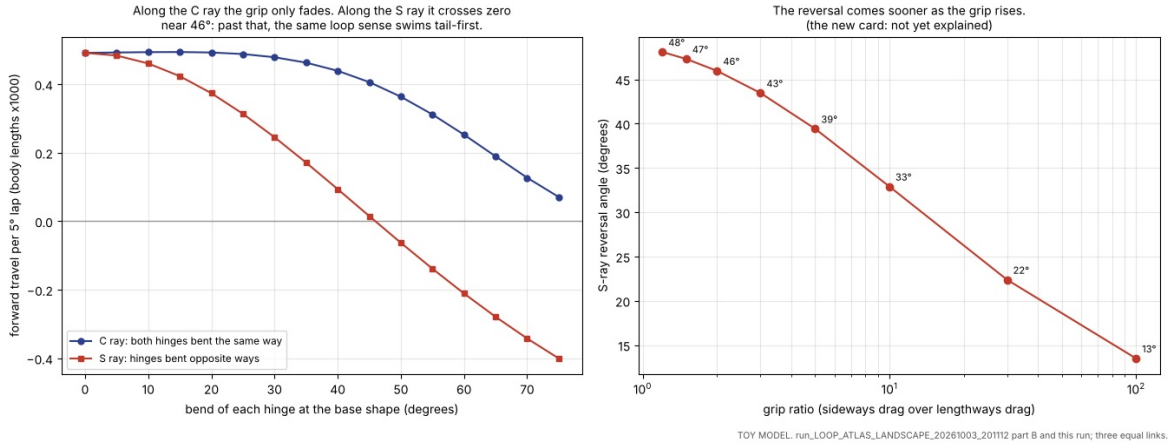


Figure 15. Forward travel per 5-degree lap along the C ray and the S ray out of the straight body, the S ray crossing zero near 46 degrees (left), and the reversal angle against grip ratio (right). TOY MODEL.

And then the reason. We had guessed that at a blade's grip the reversal angle would fall to zero. It does not: it levels off, at 3.71 degrees for grip 10,000 and 3.52 for grip 100,000[76]. The levelling number gave the mechanism away. A 5-degree lap centred on an S of 3.54 degrees has a disk whose edge passes exactly through the straight body. At a blade's grip the hill of the grip map gathers onto the straight body, and a lap on the S swims forward exactly while its disk reaches it, and tail-first once it does not; so the reversal angle should be the lap's radius divided by the square root of two, since the S runs diagonally across the map. Read off the 5-degree lap, the rule was then tried at two lap sizes it had not been read from, and held: 2.130 degrees against 2.121 for a 3-degree lap, 5.629 against 5.657 for an 8-degree lap, with 3.522 against 3.536 at the size it came from[77]. We first took the gathering for a spike at the straight body, but every lap of that test was centred on the S, which meets the C shapes only there, so it could not tell a spike from a ridge through that point. Centred along the C, out of the straight body's reach, laps swim forward too, 539 thousandths of a body length at 4 degrees fading to 106 at 12, and a lap off both lines swims forward exactly when its disk crosses the C[78]. At a blade's grip the hill gathers into a ridge along the C shapes, tallest at the straight body. A body that cannot slip sideways at all is the wheeled three-link snake of robotics, whose three no-slip rules stop being independent at exactly those shapes, both hinges bent alike, the straight shape among them (Yona and Or 2019), and whose gaits change direction with whether they enclose part of that line (Hatton and others 2022). At a gentle grip the reversal is a property of the hill's far slope; at a hard grip it is a property of whether the loop encloses part of the snake's line.

The knobs. For a three-link swimmer, keep the middle a fifth to a third of the body. To steer, hold a curl; to reverse, bend into an S past the reversal angle, which shrinks as the grip hardens and stops shrinking at the lap's radius over √2. For a coil turned on an axle, wind it steep for speed, at 54.7 degrees in round wire and steeper in a ribbon, and at 35 degrees for the most travel per unit of work when its number of turns is set, on a round wire whatever body it pushes; on a ribbon pushing a body, flatter, 27.5 degrees.

What this chapter found. A swimmer with no rudder steers by holding a curl, 2.2042 degrees a lap, as curvature times loop area predicts; turning, unlike travel, needs no wall. Bend the same swimmer into an S and the same stroke runs it backward; on a blade-hard grip the switch lands at the lap's radius over √2.

6. Loops that leave work: the swing that listens

> Everything tonight said the same thing in different coats: whatever keeps its own time wins. The swing that listens to itself keeps growing, the rings that pass through each other keep moving, and the snake long enough to wear the wave can push back against it.
>
> — *a no-physics read*

Something to do at the playground. Sit on a swing with nobody to push you. Stand up as you pass the bottom, squat as you reach each end, and do it again on the way back: twice every swing. The swing grows. Do the same thing a quarter of a swing late and it dies away. You are a weight on a rope that pulls itself in where the rope pulls hardest and lets itself out where the rope pulls least, and the difference between the two is the work you put in. Every loop in this chapter leaves work behind, and the whole question is when.

Twice per swing, at the right moment. The toy is a weight on a line 2.5 metres long whose rider shortens it by 5 per cent standing and lets it out by 5 per cent squatting. Started at 10 degrees, standing and squatting twice per swing at the best timing grows the swing to 40.14 degrees in six swings; at the worst timing it shrinks it to 2.75; once per swing, at the best of twelve timings, it barely moves, 11.06[79]. Switch gravity off and the pumping feeds nothing. Gravity is the borrowed wall, and twice per swing is the two-to-one gear of chapter 10, the kind that needs memory: the swing remembers its own motion, and that motion is what the rider pumps. This is parametric pumping, the known law of a swing fed at twice its own rate.

The clock that falls out of step. A swing takes longer the bigger it gets: 1.0019 times its small-swing time at 10 degrees, 1.0313 at 40, 1.0732 at 60, 1.1375 at 80, every printed digit of the exact formula[80]. So a rider who pumps by a clock set for small swings grows the swing to 63.16 degrees by swing 10, then pumps a little early, then a lot early, and sinks it back to 8.91 by swing 20. The same clock on a toy swing whose period is held fixed keeps growing every swing. The clock is not wrong; the swing changed under it.

The rider who listens. Bear's answer, before anything was run, was the jellyfish's: time each move by the swing's own motion, not by a clock. Our first listening rider did nothing at all, 10.00 degrees for thirty swings, because it stood and squatted in an exact mirror image about the bottom of the swing and took back on the way up what it gave on the way down. A listening rider needs a lead: each move made a little ahead of the moment it answers. With a lead anywhere from 45 to 135 degrees it pumps the swing over the top, past 170 degrees, within twenty swings (at a lead of 90, by swing 12), and it never falls out of step, because whatever the swing's period has become, the rider is reading it. With a lead of 0 or 180 it does exactly nothing. With the lead turned round it is a brake: from 10 degrees to 0.13 in twenty swings[81].

Three predictions, worked by hand, registered first. Each was written down with its arithmetic before the tool existed, and all three held[82]. First, at small swings each swing multiplies the size by a fixed factor set by the lead and by how much the rider changes the rope: 1.26570 per swing at a lead of 90 degrees with a 5 per cent change, against 1.26569 from the arithmetic, and within half a per cent of it at all twenty-four leads tried. Second, the brake is the pump run backward. At the opposite lead the factor is 0.79009, and the two multiply to 1 within three parts in a hundred thousand: film a pumping rider and run the film backward, and you are watching a braking one.

Third, and best: the listening rider works on a swing of any length without retuning. We kept its lead, kept it reading its phase by the 2.5 metre swing's rate, and put it on swings from 0.5 to 12.5 metres. It pumped every one of them past 90 degrees within 10 to 12 of their own swings. It pumps a little less hard on the swings farthest from its own, and by as much as the arithmetic said. Reading its phase by the wrong rate stretches the circle of its twice-per-swing rhythm onto itself, a stretch of the kind called a Möbius map, and a stretch of that kind keeps the rider's timing and weakens only its push, to $4r/(1+r)^2$ of full strength, where r is the square root of the length the rider reads by over the swing's own. In words: a rider tuned for one length and put on a swing five times longer or five times shorter still pumps at 0.854 of full strength. The toy lands within 0.14 per cent of the formula at all six lengths.

The clock rider, set for 2.5 metres and put on the same six swings, never got past 12.67 degrees in twenty swings: off its own length by a fifth or more, it only beats, gaining a little and losing it again. One of the fresh readers who checked this book's prose noticed that $4r/(1+r)^2$ has the form of the power passed across an impedance step, the share a wave keeps when it crosses from one rope to a heavier one. Checked since, the tie holds, and for a reason: the Möbius map the rider's misread phase follows is the very map that turns an impedance ratio into a reflection, and the part of its push left in step with the swing is one less that reflection squared. A mistuned listener loses push the way a mismatched line loses power, the same whichever way the mismatch runs.

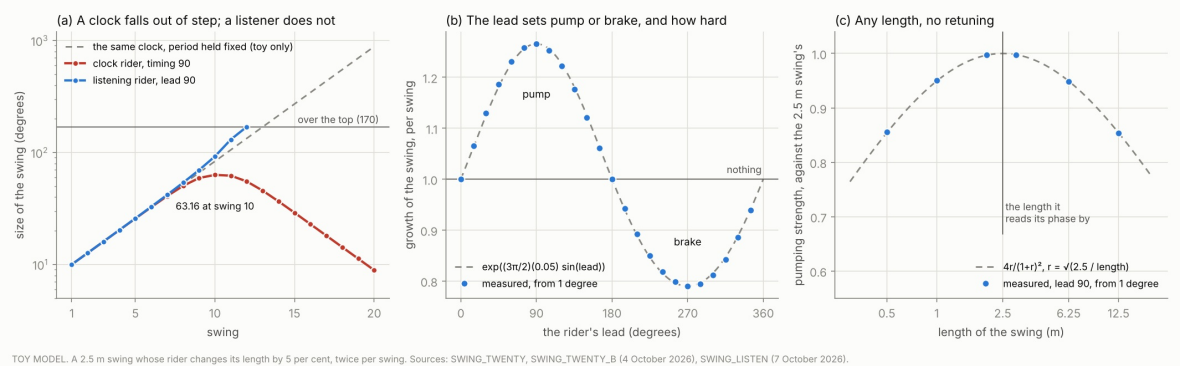


Figure 16. (a) The size of each swing over twenty swings, started at 10 degrees on a 2.5 m swing: the clock rider at timing 90 peaks at 63.16 degrees at swing 10 and sinks; the same clock on a swing whose period is held fixed (a toy-only swing) keeps growing; the listening rider at lead 90 goes over the top at swing 12. (b) The listening rider's growth per swing at small swings, at every lead, against the arithmetic's curve: pump above 1, brake below. (c) The same rider on swings of other lengths: its pumping strength against $4r/(1+r)^2$. TOY MODEL.

An engine fed by a clock works only on the machine it was tuned for, and only while that machine stays small; an engine that listens works on any machine at any size, and the same rider is a brake with its lead turned round. It is the lesson of chapter 11, where a clock harvests nothing from an even soup and an ear does, met here in a playground.

Bear's sling. Pump the swing, then cut the chain. Where? From swings of 40, 60 and 80 degrees, the longest throw comes from a cut on the forward upswing at 0.69, 0.57 and 0.47 of the way up (27.7, 34.3 and 37.8 degrees), and it throws 2.30, 3.80 and 5.51 metres, against 1.08, 1.58 and 2.03 for a cut at the bottom, two to three times as far. We had predicted half to three quarters of the way up; that held at 40 and 60 degrees and missed narrowly at 80, and the miss was in how the prediction was put. The best cut is an angle, and it levels off: 27.7 degrees at a swing of 40, 37.8 at 80, 38.9 at 90, never 45, because the seat slows as it rises[83]. The optimum is broad, which is a builder's gift: from an 80-degree swing, any cut from 29 to 47 degrees throws within 5 per cent of the best. A sling is an engine with a latch: it stores many swings of work and lets them go at once.

The same rule in a wire, and in a skater's legs. Two capacitors joined through an inductor deliver 97.8 per cent of their energy when the switch opens as the current passes zero, and dump 48.9 per cent at the switch when it opens mid-swing[84]. Work is a push times a movement at the same instant, so the moment you choose decides the bill. A skater swizzling in socks says it with legs: with push and toe angle in step her legs do 441.7 joules over ten swizzles and she reaches 3.095 metres per second; a quarter cycle apart they do 0.0 joules and she goes nowhere; half a cycle apart, the same 441.7 joules drive her backward at the same speed[85]. Her legs push in every case; they do work only when her feet move along the push. Here the two knobs are a force and a motion, and they must be in step to do work; the two shape knobs of a loop, like the wing's twist and flap, want a quarter cycle between them instead.

The knobs. Pump a swing by listening, leading by 90 degrees; to brake, turn the lead round. Cut a sling between 28 and 39 degrees on the upswing. Switch an inductor at zero current.

What this chapter found. A swing pump that listens beats one that keeps time: three predictions worked by hand held, and the listener works on any swing from 0.5 to 12.5 metres, its push weakened only to $4r/(1+r)^2$ of full strength, the very share of power a mismatched cable passes, and for the same reason.

7. Loops that leave a timing mismatch: light, gyroscopes and paper mountains

> Three mountains stood hidden on a plain, and a walker carrying an arrow found every one of them without once looking up. Wherever the arrow came home turned, something stood between his path and a peak; wherever he walked round a mountain's whole skirt, the arrow came home as if the plain were empty, because a mountain's foot hands back exactly what its top took.
>
> *— a no-physics read*

"Would the gyroscope notice?" (Bear)

Something to do with three pairs of sunglasses. Polarising sunglasses pass light that shakes one way and block light that shakes the other way. Hold two lenses one behind the other and turn one of them a quarter turn: almost nothing comes through. Slide a third lens between them, turned halfway, and light comes back. The lenses do more

than dim the light; they also carry its beat. This chapter is about loops that move nothing at all and still leave something behind: the wave comes home in its own state, a fraction of a cycle early or late. Sunglasses cannot show you the beat. A toy can.

A loop of filters. Every way light can shake is a point on a ball, with two knobs for the point: how the shaking is slanted, and how round it is. A filter passes one point of the ball. Send light through a ring of filters whose points go once round a loop on the ball and back to the first, and the light leaves in the state it entered, a little dimmer, and shifted in time by half the solid angle the loop encloses on the ball. Not roughly: the shift over half the enclosed area is 1.000000 at every loop size we tried, from 1 to 40 degrees[86]. Run the loop backward and the shift is exactly the opposite; go round twice and it is exactly twice. That is Pancharatnam's phase, from 1956, which Berry made famous in 1987; the toy met it without looking it up.

A law registered before the tool: the size bends, the gap never does. For small loops the shift grows as the square of the loop's size, the room-inside-the-loop rule of chapter 3: the shift over the size squared is 6.2822 at 1 degree, about 2π. For big loops it falls, to 4.8715 at 40 degrees. The gap between the two knobs is another matter. Swing them a quarter cycle apart and the shift is largest; in step, it is zero, because the loop is then a line with no room inside; in between it goes exactly as the sine of the gap. We guessed by hand that the whole bend of the square rule would be carried by a Bessel function, so that the shift is 2π a J1(2a) sin(gap) for knobs swung by a, and registered the guess before the tool existed. In words: the size of the swing decides how much room the loop has, through a curve that starts as the square and then flattens, and the gap between the knobs scales that room by its sine and by nothing else. Why a Bessel function, in two lines: the room a loop encloses on the ball is the sum, round the loop, of the sine of how round the light is times each step in its slant; with the roundness swung back and forth like a cosine, the sine of a cosine unfolds into a series of Bessel functions, and one lap keeps only the first of them, J1, because only it beats in step with the slant. The same unfolding is everyday in polarimeters driven by a photoelastic modulator, whose signal at the swing's own beat carries J1 of the swing's size; there it sizes a signal, and we have not found it written for the shift a whole loop gathers. At 4,096 filters, over sizes from 1 to 40 degrees and gaps from 30 to 150, the shift sits within 2.51e-7 of the formula, the remainder being the filters' own corners[87]. The leftover lives in the gap between the knobs and peaks at a quarter cycle, the same rule as the flapping wing's.

The free loop. Filters pay for the shift in light. Wave plates filter nothing; they turn the light's state on the ball. Carry the same loops by wave plates and the light comes home in its own state with nothing lost, its intensity 1 to within 2.5e-14, and the same shift as the filter loop to within 4.44e-15[88]. This is a timing shift with no bill at all: nothing the loop throws away pays for it. And a rotator that turns straight light through half a turn without losing any brings it home exactly half a cycle late[89]; it needs two trips to come home in step, like the plate that untwists your arm only on its second circle.

The dark doorway is a whirlpool. Three filters: straight, then slanted halfway, then a third, M, and back to straight. Sweep M along a circle of the ball through the state opposite the first filter, and the light goes dark there, and its shift jumps by half a cycle exactly where nothing gets through. We said in conversation, before running anything, that walking M once round that dark point would add a whole turn to the light's beat. Run, it does: one whole turn, −1.000000, for circles of 0.05, 0.1 and 0.3 radians round the dark point; zero round a lit state[90]. What we did not foresee was the second one. The state opposite the slanted filter is dark too, and walking round it winds the beat the other way, +1.000000: a vortex and an anti-vortex, a whirlpool and its mirror image, at the two places where the light goes out. Such points were foreseen by Bhandari, who found that light's phase jumps by half a cycle where a state turns away from a filter completely, and gave each such point a strength with a sign (Bhandari 1998); the toy met a pair of them on one circle of filters.

The turning loop. Put mirrors at the corners of a loop, turn the whole loop slowly, and send one pulse of light each way round from the same mirror[91]. The pulse that runs with the turning comes home later, by four times the loop's area times the turning rate, over the speed of light squared: Sagnac's delay. The shape does not matter: a square, a triangle and a long thin rectangle of the same area give the same delay to within one part in a billion. Where the axis sits does not matter either. A figure-eight whose lobes run opposite ways counts its lobes with signs, so two equal lobes give nothing at all, and a loop tipped 60 degrees off square to the axis gives exactly half, because only the shadow it casts on the plane of the turning counts. For a builder this is the fibre-optic gyroscope's whole design rule: wind the fibre round and round one way, never in a figure-eight, and make the area large. A coil of a thousand turns of radius 5 cm, its axis along the Earth's and feeling only the Earth's turning, puts about 31 microradians between its two beams at 1.55 micrometres, and five thousand turns of radius 10 cm put 620.

The paper-cone mountains. Bear's answer to a run about frames carried round cones was a picture: work backward from the data to the best wedge to cut from a paper circle, and build a mountain range of paper cones set out like landing pads, a data-driven gearbox in the manner of the Towers of Hanoi. Cut a wedge out of a paper circle and tape the edges together: you have a cone, and the wedge you threw away is the cone's whole curvature. A hiker who walks once round its peak carrying an arrow, never turning it against the paper, comes home with the arrow turned by exactly the wedge. We built a range of three smooth mountains on a flat plain and walked circles round them in the computer. On every circle the arrow came home turned by 2π(1 − cos s), s being the slope where the circle runs:

exactly the wedge a paper cone of that slope is missing, to 8.4e-15, fourteen figures[92]. So the wedge is largest on each mountain's steepest ring and falls to nothing at its foot. From outside its foot a mountain on a plain cannot be seen at all, because the bend at its foot gives back exactly the curvature its top took.

Reading hidden mountains back. Then we hid the mountains and read them back from arrows alone. A map of 6,900 circles found the peaks, and asking which circle turns the arrow most put each centre back to a hundred-millionth and each steepest slope to fifteen figures[93]. The data hand back Bear's "optimal slice to cut out my paper circle": 86.87 degrees for the first mountain. The arrow reads slope, not size: mountains of one shape, half or twice the size, give the same reading on matching rings. And the map's fourth peak was no mountain. Its centre stands on open plain between the first and third; circles widened from there first turn the arrow backward, as they take in the bent feet of both mountains, and then forward as they reach both tops, and its best circle turns the arrow 1.379 radians, more than the smallest mountain's own best[94]. A count of the map's peaks would have counted four mountains. A tiny circle tells them apart: round each real summit it turns the arrow, and round the ghost's centre it turns it by exactly nothing[95].

Look again before the reading moves half a turn. Bear drew such a place in 2011, in the middle of a ring of steep peaks. Seven peaks with 70-degree slopes stand round an empty middle. The circle round the middle that passes between each peak's top and foot takes about two fifths of every peak's slice at once, 11.97 radians in all, nearly two whole turns, yet the arrow carried round it comes home turned by only −0.599 radians: the two whole turns cannot be seen in one arrow on one circle[96]. They are found by growing the circle from a dot and keeping count. Read on 1,801 circles, each a hair wider than the last, every whole turn is caught, and at the best circle the count is two whole turns and −0.599 radians, the peaks' sum to 3.5e-10[97]. The count lives in the history of the reading, so the history has to be watched closely enough. Looking at every tenth circle, the reading never changed by more than 3.018 radians between looks, just under half a turn, and the count held; looking at every eleventh, one change was 3.364 radians, just over half a turn, and a whole turn fell out of the count without a sound[98]. A receiver tracking a carrier's phase loses whole cycles the same way when it blinks; the half-turn limit is known to those who unwrap phase for a living.

A star round one peak. Bear then asked for a pentagon, its corners joined by skipping every other one, which draws a five-pointed star. Walked round one steep peak, the star winds twice round its middle and once round each point, and the arrow counts the bending in each part as many times as the path winds round it: the star's reading is its inner pentagon's plus its outline's, to fifteen figures. The pentagon drawn through the star's points, which holds the peak's whole foot, turns the arrow by exactly nothing, while the star, counting the middle twice, sees the peak[99].

Mountains that hide. A mountain can hide from the arrow in three ways: its summit can be cancelled by its own skirt, its whole turns can hide inside the count, or it can be no mountain at all, only paper bent without stretching. Paper pressed into ridges as steep as 76 degrees reads zero on every loop we walked, to 3.1e-12, as Gauss's Theorema Egregium says it must; rolled into a cone, it holds all its curvature in the tip, so every circle round the tip reads the same, and every circle that misses the tip reads zero[100]. And on a turning globe the arrow's turn splits in two: the area of the cap it walked round, the same at any speed, and the globe's own turning, which grows with the time spent walking. Two walks, one twice as slow as the other, give the cap's share back by subtraction to fourteen figures[101]. It is the fibre-coil gyroscope and the Foucault pendulum in one walk.

The knobs. A loop that moves nothing can still leave a timing mismatch, set by the room it encloses in the space of its own states: filters pay for it in light, wave plates pay nothing, and where the light goes dark the mismatch turns round a whirlpool. To read a hidden curvature, walk circles and keep count from a dot outward, and look again before the reading can move half a turn.

What this chapter found. Light carried round a loop of filters comes home late by exactly half the enclosed solid angle; a registered Bessel law for its knobs holds to 2.51e-7; and the same carried arrow reads hidden mountains back, so long as no reading jumps half a turn between looks.

8. Not loops: spends and rides

> Out in the water, the snake that pushes against the waves grows longer and gets better at it, but each doubling of its length buys less than the last, as if the waves were teaching it a lesson it can only ever half learn.
>
> — *a no-physics read*

Something to do at the sink. Cut a small boat from a card, with a notch at the back, and float it in a still basin. Touch a drop of washing-up liquid to the water in the notch. The boat shoots forward and stops. Touch the soap again and almost nothing happens. The water at the back had been pulling on the boat exactly as hard as the water at the front. The soap weakened the pull at the back, and for a moment the front won. Then the soap spread over the basin, the difference was gone, and so was the push. You cannot make the boat sail back and fetch its soap. This chapter is the book's fence: the things that move themselves and are not loops.

A spend, and why it is not a loop. A loop comes home. Its knobs return to where they started and the walker keeps a leftover, and run backward it undoes itself. A spend never comes home. It throws something away, and the push is what the throw paid for: the soap is used up, the rocket's fuel is gone, and running the film backward would need the soap to unmix itself from the basin.

What a spend costs. A rocket is a throw with no catch. A 1,000 kg craft throwing 1 kg bits out behind it buys push at two over the exhaust speed for every joule it spends: 0.2002 to 0.2003 kilogram-metres per second per joule at an exhaust of 10 metres per second, 0.020 at 100, 0.002 at 1,000, through its burn[102]. So slow exhaust is cheap in energy. But every kilogram thrown buys exactly the exhaust speed in push, 10, 100 and 1,000 in the three cases, so slow exhaust is dear in propellant. The two prices multiply to 2 at every exhaust speed, 2.002 to 2.003 in the toy[103]. The exhaust speed decides where the craft sits on the trade, never how good the trade is. A craft rich in energy and short of propellant should throw fast; one with propellant to spare and little energy should throw slow.

A ride, and what it costs. A surfer, a sail, a cork carried by a swell: each leans on something already moving. A ride does not come home with a leftover; it borrows the motion of a pattern going past. We built one: Bear's floating snake, a chain of links on a train of water waves, held in place, with dampers at its hinges to harvest power[104]. Every body of that run obeyed one rule, to six decimal places. The push the waves give the snake along their direction of travel, times the waves' speed, equals the power the snake takes from them: 1.000000 to 1.000003 for five bodies, and 1.000006 for a snake driven by a motor. So a ride buys push at one over the speed of what it rides. A 10-second ocean swell travels at 15.6 metres per second and gives 0.064 newtons for every watt taken; sound in air gives 0.0029; light gives 3.3 billionths, about twenty million times less than the swell[105]. Riding the light is allowed, and it is the dearest ride there is.

How the push is bought	Example from the runs	Push for each joule (seconds per metre)
spend: throw at 10 m/s	a rocket	0.2002
spend: throw at 1,000 m/s	a rocket	0.002
ride: a 10-second swell, 15.6 m/s	a wave-rider	0.064
ride: sound in air, 343 m/s	a sound-rider	0.0029
ride: light	a light sail	3.3 billionths
loop: the cat, the swimmer	chapter 5	no net push at all; a turn or a shift instead

A loop is in a different currency altogether. It buys a turn or a shift with no net push, which is why the cat can turn in still air and the swimmer can travel in water that is left as it was.

Riding against the wave. Give the same snake a motor at a hinge, timed right, and it hands power back to the waves and is pushed against them. Longer bodies do better, and none reaches the 6.28 units of push for each unit of motor power that a perfect exchange would give, one over the wave's speed in this toy, because a body can hand power to a wave only through shapes that match the wave's own shape, and a body shorter than a wave has none. A snake's motors have two knobs, how hard they twist and when their wave starts against the water's, and both the push and the motor power turn out to be simple in them, so five solves map every setting at once and the best can be found exactly instead of searched for[106]. Found that way, the best exchange is 3.27, 3.96, 4.53, 4.99 and 5.34 at bodies of 9.6, 19.2, 38.4, 76.8 and 153.6 wavelengths, and 5.60 at 307.2[107]. The gap still left below 6.28 shrinks at every doubling, a little faster each time, its slopes on a log ruler −0.38, −0.41, −0.43, −0.45 and −0.47, and as far as these six lengths show it is heading for one over the square root of the body's length; the slopes are still steepening, so that limit is a reading, not yet a measurement. A search over a grid of settings had read the same climb as slowing, because it missed the best by more at each length up to 76.8 wavelengths.

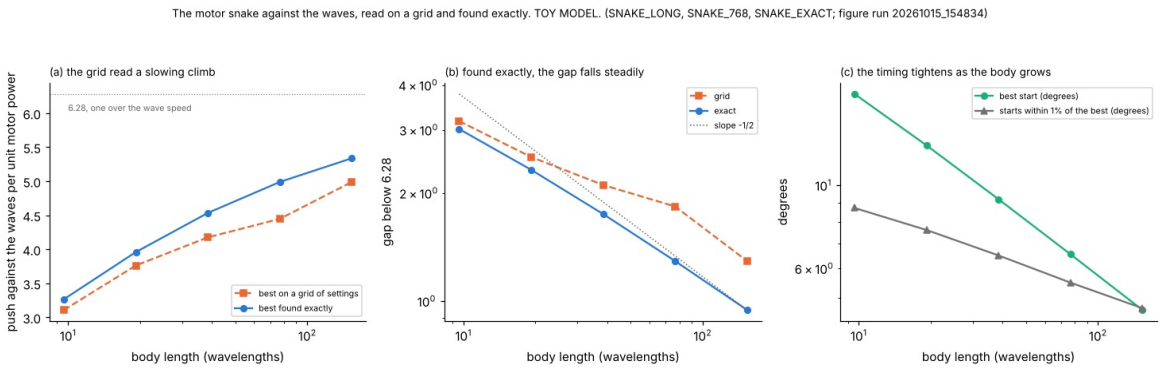


Figure 17. (a) The best push against the waves per unit motor power against the snake's length in

wavelengths, on a grid of motor settings (squares) and found exactly (circles), with 6.28, one over the wave speed. (b) The gap below 6.28 on log rulers, with a line of slope −1/2. (c) The best start of the motors' wave, and the width of the starts within 1 per cent of the best. TOY MODEL.

For whoever builds one, the knobs are these. Run the motors' wave at the water's own wave number; a quarter off either way loses nearly everything. Twist with a torque near 0.03 in the toy's units, easing from 0.033 to 0.027 as the body grows. Set the start of the motors' wave 17.5 degrees off the water's on the shortest body, 4.6 at 153.6 wavelengths and 3.3 at 307.2. And keep time: the starts that come within 1 per cent of the best narrow from 8.7 degrees to 4.7, and to 4.0 on the longest body, so a long snake pays more and more for being late. The nearest prior work found is Newman's slender-body theory of the most power an elongated floating body can take from waves (1979); how this toy's approach to the ideal compares with it is not yet read.

Two snakes racing. Put two snakes side by side in the explorer's race, each on its own knobs, and four things show that the best settings alone had hidden[108]. In still water the motors push nothing at all, whatever the torque: the body's own motion pushes on the water all along it, and its two end faces take every bit of that back, so the whole pull is bought from the waves. The pull rises in a straight line with the torque while the power rises as its square, so the hardest pull always sits at the top of the torque knob, with the motors' wave started exactly 45 degrees behind the water's. Hard and thrifty are different races: a snake 9.6 wavelengths long twisting hard pulls 50.52 to the thriftiest snake's 8.62 and spends 10.5 times the power doing it, so it wins a race of equal time and loses a race of equal fuel, dry after little more than one wave period while the thrifty snake pulls on for twelve. And nothing pulls for free: with its motors off a snake is pushed back about 1.563 for each wavelength of its length, and the torque at which it first wins against that push always sits a margin above half the largest torque at which the waves can still drive the motors, a margin that halves every time the body doubles.

Two waves on one body. Bear read the race as two travelling waves meeting, the motors' and the water's, and so it is: they help each other only while they stay in step[109]. Run the motors at the water's own wave number and the two keep step from head to tail; every wavelength of body adds the same help, and the help grows in a straight line with the length, to thirteen figures. Run them a quarter off and the two drift out of step and back every 4 wavelengths, a beat, like two notes a little out of tune; along each whole beat the help cancels, so a body one beat longer gets exactly the same help, and a body of 153.6 wavelengths gets no more than one of 9.6 while the waves push it back sixteen times harder. Engineers know this as phase matching in nonlinear optics and as synchronism in a travelling-wave tube: two waves trade only over a coherence length. The two racing snakes cannot feel each other, for in this toy a body makes no waves of its own.

The snake's clock. Roll the race's time into a clock that turns once a wave period, as the diver's room rolls her time in the air. Give the water one hand, and each point of the snake's body a hand of its own, and add up the snake's hands. One rule then sets the hardest-pulling start for all 51 bodies and wave numbers the runs hold, within a billionth of a degree[110]: the snake's summed hand an eighth of a turn behind the water's, so that the body's answer sits a quarter turn behind, in quadrature. In tune every hand points the same way, which is why the best start is exactly 45 degrees; a quarter off, the hands fan out and their sum both turns and shrinks. It is the phasor sum of an antenna array or a grating, read on a swimmer.

The snake let loose. A moored snake meets the waves as they pass; a free one swims into them and meets them faster. Two choices here are the toy's own, and said wherever they are used: the snake's laws hold in its own frame, and the water drags on it in proportion to its speed and its length[111]. The lean was that the start would simply make up the body's extra lateness; it missed, and the run showed the deeper rule behind the 45 degrees: at the hardest pull the body's answer sits in quadrature with the water, a quarter turn behind, wherever the body's own lateness falls. Tested at five frequencies no run had touched, from a snake swimming with the waves to one meeting them four times as often, it held in all 30 cases, within 1.0e-10 degrees[112], and read back after the run, the first run's 45 cases fit it too: 75 cases at nine frequencies from 0.5 to 4. Swimming into the waves costs: the thriftiest snake's pull falls from 8.62 to 2.90 as the frequency it meets them at triples, so any drag stops it at one speed, and re-tuned at each frequency its best push for each unit of power falls from 3.27 to 0.88. For a builder: pull hard for a short race and thriftily for a long one; tune the motors to the water's wave number, since a longer body a quarter off gains nothing; and tune the snake at the speed it will swim, not at its mooring.

A sail, at last. A sail is a ride on two things at once: the sail leans on the wind and the keel leans on the water, and the boat rides the difference between them. A toy boat with every number assumed, a sail of 10 square metres in a 5-metre-a-second wind and a keel of 1 square metre, was sailed steadily at every heading from 30 to 180 degrees off the wind[113]. At every heading the angle between the apparent wind and the course was the sail's drag angle plus the water's, to 2e-15 radians, the course theorem of sailing, once the water's angle is measured from the leeward side. The first boat was slow, a third of the wind's speed at best, because the hull we assumed dragged like a barge's. Cleaned up step by step with nothing else changed[114], the same boat first outran the wind between hull drag areas of 0.02 and 0.01 square metres, and with a clean dinghy's 0.006 it sailed 1.15 times the wind at 84 degrees. However clean the hull, it never made good straight downwind faster than the wind, 0.63 of it at the cleanest: this sail, short and broad, stalls past the beam, and its own drag angle caps it.

For a builder of a sailing boat, in order: clean the hull first, because in this toy the boat outruns the wind only once the hull's drag area is under about 0.01 square metres, and each halving buys less. Sail fastest just short of a beam reach, near 85 degrees off the wind, and make the most ground upwind at 45 degrees. To make good downwind faster than the wind, a clean hull is not enough: the sail must keep its lift past the beam, which wants a taller, narrower sail.

A difference, with or without a count. A ride needs a difference, something moving one way more than the other. A soup of waves running both ways evenly gives a plain bead nothing, a drift of 0.0006 under its scatter of 0.0019, while a lopsided soup is simply a wind that carries it, 0.012 to 0.127 as one direction wins out[115]. The surfer rides the difference and holds no whole number. The synchronous motor of chapter 10 rides a turning field and holds one, a turn per cycle, until it strips.

In short. A loop comes home and keeps a leftover; a spend throws something away and pays two over its throwing speed for each joule, and its throwing speed for each kilogram; a ride pays one over the speed of what it leans on, and needs that thing to be moving one way more than the other.

What this chapter found. Every push is priced at one over a speed: a rocket pays 2/v a joule, a wave-rider 1/v, held to 1.000003, and a long wave-snake's best push against the waves closes on the ideal as its body grows, 5.60 of 6.28 at 307 wavelengths; at its hardest pull a snake's answer sits a quarter turn behind the water's, moored or free.

9. Partners that remember: smoke rings and jellyfish

> Two smoke rings that keep passing through each other go further on the same breath than one ring made of both. A bell that rests a moment before it squeezes finds the ring it left behind still waiting under its edge, and pushes off it as off a wall, harder the sharper it had sprung open.
>
> *— a no-physics read*

Something to do with a box. Cut a round hole in one end of a cardboard box, fill the box with a little smoke, and tap the other end. A ring of smoke leaves the hole and crosses the room, rolling over and over on itself like a doughnut turning inside out. Tap twice, quickly, and the second ring catches the first: it shrinks, speeds up and slips through the middle of the first, which grows and slows; then they swap, and the pair leapfrogs down the room: Helmholtz's leapfrog, as Bear's jellyfish paper names it. Each ring is a partner the other remembers. A ring is a loop of spinning fluid, and the flow it makes reaches out and changes how fast its partner goes and how wide it is. This chapter is about partners like that: two rings, three rings, and a jellyfish and the ring it parked a moment ago.

Two rings on one budget. A first comparison said two rings merged into one travel faster than two that leapfrog, 1.86 against 1.18 to 1.48. But the merged ring was richer: it carried 1.41 to 1.73 times the pair's energy for the same push. Given exactly the pair's budget, the same energy, the same push and the same amount of spinning core, one ring comes out bigger and weaker and travels at 0.84 to 0.88 of the pair's speed. On one budget the leapfrogging pair, started half a radius to two radii apart, is 14 to 19 per cent faster[116]. Partners that keep passing through each other go further on the same fuel than one ring going alone.

The rings' knobs. A leapfrogging pair has two knobs a builder sets before launch: how far apart the rings start, and how fat their spinning cores are. The spacing has a sweet spot. With cores a twentieth of a ring's radius, the pair beats one ring carrying the same fuel by 9 per cent started a quarter of a radius apart, by 19 per cent at one radius, and by 9 per cent again at three[117]. The core has a sweet spot of its own, near a tenth of the radius, and a fatter core moves the best spacing outward, so the two knobs trade; the best found is a 19.5 per cent gain, at a core of a tenth and a spacing of one radius. A third ring is worth more, but only in a tight train: three equal rings beat one ring on their fuel by 16 per cent at a quarter radius, rising to 26 at half a radius. In a train of three that holds, every ring takes its turn in front: started 0.5 and 0.6 radii apart, the lead changed hands 313 and 219 times in 400 time units; and the pair a broken train keeps goes on leapfrogging: the trio started 0.75 apart broke at 70.0, and its closest two then passed each other 332 times[118]. Followed for a thousand time units, trains of three held out to about two thirds of a radius and flew apart from 0.7, and every time a train broke, its rings first spreading five radii apart along their path, it shed one ring and kept a pair, two rings still within three radii of each other[119]; whether the shed ring stays gone is told below. Followed longer, the edge moved in, as the next paragraphs tell. Four rings must be packed tighter still: they held for a thousand time units at 0.2 and 0.3 radii, gaining 17 and 25 per cent, and flew apart at 0.4; they were not followed longer.

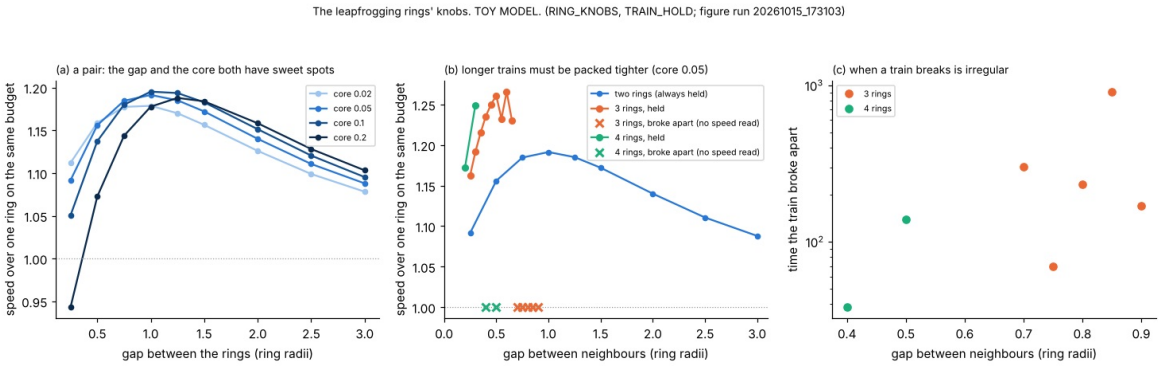


Figure 18. (a) A leapfrogging pair's speed over one ring carrying the same impulse, energy and core volume, against the gap between the rings, for four core sizes. (b) Trains of two, three and four equal rings against the gap between neighbours (core 0.05): filled marks where the train held together for 1,000 time units, crosses where it flew apart. (c) The time each broken train broke apart, against the gap. TOY MODEL.

Is the pair's gain its loop's area? The book's spine says that what a loop leaves behind is its area, and its cleanest loops pay exactly that: light walked round a loop of filters is shifted by half the area the loop encloses, and the falling cat turns by the area its body bends through. So the pair was asked the same question. Seen from the middle of the pair, each ring traces a closed loop every two passes, shrinking as it runs forward through its partner and swelling as it falls back around it. Across 35 pairs, four core sizes by nine spacings less one pair whose cores nearly touched, that loop's area grows about thirtyfold as the rings start further apart, from 0.046 of a radius squared at a quarter radius to between 1.4 and 1.7 at three, its stretch along the axis exactly the starting gap; but the gain rises to its peak near one radius and falls on both sides, and the two do not rank together at all, Spearman -0.20[120]. The pair's gain is not its loop's area. The loops that pay their area answer the book's first question, faster?, the same way: walked faster or slower, they leave the same change behind. The pair's gain is itself a speed, part of it each ring's own speed at the radii it passes through and part the pull of one on the other, and the size of its loop sets neither. Whether some part of it is an area after all, the area swept in each unit of time or the area of the pair's own shape, is the next question, written down before it is run.

What the pair's gain is made of. The toy's rule moves each ring along the axis at two speeds added together: its own, set by its size and core as if it were alone, and the push its partner gives it. Averaged over whole periods the two parts add up to the pair's speed to better than one part in ten billion, so the gain splits exactly[121]. Neither part has the peak. Across the 35 pairs, the partner's part of the gain is largest at the closest spacing read and falls at every step of the spacing, in all four cores, while the rings' own part rises at every step; the peak near one radius is where a rising part and a falling one cross. Most of that rise is not in the rings. Against a lone ring of radius one with the same core, the rings' own speed is only 0.5 to 3.9 per cent faster at any spacing; what changes is the yardstick, the one ring carrying the pair's impulse, energy and core volume, which is fastest for the closest pair and slows as the spacing grows, so the same speed counts for more. Is the partner's part an area after all? It ranks with the area the loop sweeps in each unit of time at 0.92, as guessed, but a shade higher, 0.93, with how often the rings pass, and those two measures rank together at 0.999, because both fall at every step of the spacing: the test could not tell an area from a count of passes, and the question stays open. One reading made after the run, and a guess until a run settles it: at a fixed spacing, the partner's part grows with a fatter core while the area swept and the passing rate both shrink, so the core, not the spacing, is the knob that can tell them apart.

A soft edge, and dice in the arithmetic. Forty-six trains of three were started from 0.25 to 0.70 radii apart and flown for 20,000 time units[122]. Started once at each gap, every train 0.56 radii apart or closer held the whole way, thirty-two of thirty-two, keeping its energy to within a millionth of its start. Every train from 0.57 out broke, fourteen of fourteen, and each shed one ring and kept a pair. Then the same day showed what a single break time is worth. On one computer the trains at 0.61, 0.65 and 0.70 broke at 4,494, 241 and 272 time units. Flown on a second computer by the same tool, reading the same files, they broke at 1,022, 1,011 and 303. Nothing differed but the two machines' arithmetic, which rounds its last digit a little differently. To see whether that alone could do it, the middle ring was started a millionth of a billionth of a radius further along, about the size of the arithmetic's own last digit, and every break moved, by 28 to 84 per cent[123]. The train near its edge is chaotic, and its break time is one throw of the dice, set by the last digit. Three rings on one axis leapfrog regularly or chaotically, as Borisov, Kilin and Mamaev showed in 2013; whether their chaos is this train's break-up we have not yet read.

So the edge itself was thrown again and again[124]. At every gap from 0.560 to 0.570, a thousandth of a radius apart, five copies of the train were started, four of them nudged by up to a few millionths of a billionth of a radius and one left as it was, and flown for 20,000 time units. Forty-eight of the fifty-five broke, every one leaving a pair, and seven

held, scattered across the band. At 0.560 itself three of the five broke. So the edge is no line. It is a band in which a train lives about 10,000 time units on average, and nothing about its spacing says which throw will hold. And the first computer repeated itself exactly: its train at 0.570, with no nudge, broke at 1,119.4829502389848, the break the same computer gave that morning, to every digit kept. A chaotic run repeats itself exactly on the machine that first flew it, and only there. Read over a wider band the lifetime is long and poorly known: the forty-six starts above, nine of them from 0.51 to 0.59, one at each hundredth of a radius, lost three of those nine, all three at 0.57 to 0.59, just outside the edge, which puts the band's lifetime near 43,000 time units, but with three breaks the honest range is about 18,000 to 210,000[125].

For a builder of a leap-through train, in order. For a pair, start the rings one radius apart and spin their cores about a tenth of the radius. For a train of three, pack it tighter than 0.530 radii and expect a long flight, not a forever one: there half of ten copies flown to 100,000 time units broke, a lifetime of about 150,000[126]; from 0.530 the breaking comes sooner, and at 0.550 and 0.555 nine copies in ten broke within 20,000[127]. Near the edge, with rings like these, watch the ring that slips through: each of the first eight breaks read this way came soon after it shrank to about a fifth of a radius, and how often that holds over many more breaks is the next paragraph's. The fatter the cores, the deeper that squeeze sits, in even steps that held at seven cores, four of them forecast before they were flown and two of those outside the cores the rule was drawn from, and at 0.530 radii apart the break comes 130 to 175 time units after the last deep one, whatever the core. For about nine tenths of that wait the train is still whole, keeping within the widest swing of its last thousand time units; then one ring goes out past that swing and reaches the five radii this toy counts as a break in about thirteen more[128]. Thin cores warn too, once the squeeze is measured against their own door, the depth of squeeze that a train with that core breaks through, though a thin-cored train can squeeze as deep and survive. In flights of 20,000 time units the tightest gap that broke was 0.545 radii for cores of 0.04 and 0.06 alike, but flown long at 0.530 radii apart, the two thinnest of seven cores, 0.035 and 0.04, live longest, about 117,000 and 149,000 time units, against 54,000 to 86,000 for the five fatter ones, 0.045 to 0.065, though the counts are small[129]. The tighter the train, the more its energy drifts over a long flight, and that drift is the arithmetic's step-by-step error, not the rings': flown again with the error allowed per step ten times tighter, the train at 0.25 drifted 14.58 times less, for 1.37 times the work[130], so a builder packing tighter tightens the arithmetic too. Do not tune the spacing to dodge a break, and do not plan on one train's break time. When it breaks, a pair carries on: plan for losing one ring, not the train, and for the lost ring sometimes coming back.

A pair, and a ring that threads it. Why is that wait so steady? Read ring by ring, a train of three near its edge is not three equal partners. Two of its rings make a pair, slipping through each other again and again, about every third of a time unit at 0.530 radii apart and about every half at 0.565, where the pair is looser; the third ring threads the pair on a slower beat: it goes through, swings out, and comes back, its beats near the break 64 to 126 time units long. Every deep squeeze read this way, 188 of them at two spacings, each in the stretch from a thousand time units before a break's last deep squeeze to the break itself, fell within a fifteenth of a time unit of the third ring going through the pair: the squeeze is the threading. The beats come in equal pairs, and the last deep squeeze before a break sits between the two beats of a pair, in every one of the 44 breaks that came through a deep squeeze (two others came through none). At 0.530 radii apart the break then came exactly one threading after the last deep squeeze, 35 times in 35: the third ring went through once more, without a deep squeeze, and on its next swing reached the five radii that count as a break. There the beat and that last swing trade against each other, a longer beat meeting a shorter swing out, so for its length the wait they make together varies less than either part. That was read from one run's flights after their forecasts were judged, so the picture was then forecast on fresh trains nearer the soft edge, at 0.565. There every deep squeeze was again a threading and the last again sat between two equal beats, but the break came one threading later in only 5 of the 9 breaks with a deep squeeze; in the other four the third ring went on threading 3 to 339 more times with no squeeze deep enough to count, and then left[131].

Gone, or only far. A break, the first time the rings spread five radii apart, is not always a parting for good. Flown a thousand time units past it, the third ring came back to the pair in 5 breaks of 11, four of them after swinging out no farther than nine radii and one that came back and left again; the other six were 33 to 61 radii away and at their farthest when the look ended[132]. So how far is too far? Fresh trains at 0.565 were flown on to 20,000 time units with no stop at five radii, every swing past it followed to its end; eleven finished, and a twelfth slowed so much once its ring was far that it never did. Of the eleven, two never spread five radii apart and held whole to the end. In the other nine, every ring that came back had swung out no farther than 13.8 radii, and every ring that went farther was gone, as far as these flights could tell: 31 to 1,126 radii away when the flight ended, and still at its farthest. The swings came in equal pairs too, out to the same reach twice, 13.8 and 13.8, 11.0 and 11.1, and the ring left for good at the start of the next pair, as the threading beats do. Most rings left on their first wide swing, six of nine, and the others came back twice or four times first, so their lifetime, the count that weighs in the trains still unbroken at 20,000, came out only about 6 per cent longer read to the parting for good than read to the first wide spread, 12,012 time units against 11,358. All of that difference is the three trains whose rings came back first, and with nine breaks the true difference could lie anywhere from nothing to about a sixth[133]. Every lifetime in this chapter is a time to that first wide parting; at 0.565 it falls short of a parting for good by a few per cent, and at 0.530, where the long lifetimes were measured, by much more.

At 0.530 the ring comes back, two by two. Sixteen fresh trains at 0.530 were flown on to 100,000 time units, or until a ring was 50 radii away[134]. Six never spread five radii apart. Of the ten that did, seven lost a ring for good, and three were whole again at the end, their third ring back in the train. What 0.565 showed held here too, and more sharply. No ring came back from farther than 13.75 radii, nearly the reach of 13.8 at 0.565. Every one of the 78 returns was one of a pair, the two swings of a pair reaching the same distance to within a tenth of a per cent, and every ring that left for good left at the start of a new pair, seven of seven, which we had given only one chance in three. But at this gap the first wide parting is a poor clock. A ring could come back and the train fly on whole for thousands of time units more, in one train 16,000, so the lifetime read to the parting for good, about 179,000 time units, is 1.7 times the 105,000 read to the first wide spread. With only seven partings that ratio is loosely fixed, anywhere from about 1.05 to 3.4[135]. So the long lifetimes this chapter gives at 0.530 and below, read to the first wide spread, are the short side of a train's life there.

Thirty-two more, and the pull that brings a ring home. So thirty-two more trains were flown at 0.530[136]. Every return was again one of a pair, 196 returns in 98 pairs, both swings of each pair flown by the same ring to the same distance within a quarter of a per cent, and every ring that left for good, nineteen of nineteen, left at the start of a new pair. But the first sixteen had flattered. Six returns came from farther than 14 radii, the farthest from 19.3, so 13.75 had only been where the first sample stopped; and read over all forty-eight trains, the lifetime to a parting for good is about 1.28 times the lifetime to the first wide spread, anywhere from 1.09 to 1.59, not 1.7. Then the swings told how a ring is held. The farther a ring swings, the longer it is out: on the 48 swings that went past 8 radii, its time out grows as its reach to the power 2.59. And the farther the reach, the rarer: the reaches thin out as the reach to the power of about minus three. A ball loosely held in a dish whose walls flatten off as the cube of the distance would do both, and the cube is how the hold between two distant smoke rings falls off, its pull as the fourth power, in the same form as two small magnets (Horn, Nicolis and Penco 2015). So there is no fence. A ring that swings out is pulled home from as far as its throw takes it, and the farther returns are only rarer. A ring that leaves takes another road altogether: it reaches 50 radii a median 1,400 time units after it sets out, where a ring held like the others would need about 20,000, and even the slowest to leave went about twice as fast[137].

The wall a stroke leaves behind. On paper, a jellyfish's stroke cannot work. Its rim and its lip close as one motion and open as one motion, so the stroke goes out and comes back along the same path in the space of shapes, and in water that forgets, such a loop carries a body nowhere: Purcell's scallop theorem (1977). The jellyfish swims anyway, because the water remembers. When the rim closes it throws a ring out of the mouth; when the rim opens it leaves a second ring, spinning the other way, parked under its own edge; and the next squeeze pushes against that parked ring as if against a wall. Bear's jellyfish paper built a solver for this stroke and measured how much the timing matters. Rest the bell for a moment before its second stroke and that stroke's displacement moves along a nearly straight line with the rest: +0.031, −0.002, −0.036, −0.075 and −0.123 radii at rests of 0, 0.102, 0.205, 0.307 and 0.409 of a period (negative is the animal's way, top first), about 0.37 of a radius per period of rest. A tenth of a period decides whether the stroke swims or hovers. On the living animal, Gemmell and colleagues measured the same wall: the pressure at the margin 2.3 times higher, and a peak push at the bell's rim 2.7 times as strong, with the stopping vortex present (Gemmell and colleagues, 2021). Give the bell a one-way cap at its top that holds its breath and the same stroke goes almost nowhere: the animal hovers, and the engineer calls it station-keeping.

The same wall in air. A fruit fly's wing, flipping over at the end of each stroke, runs back into the swirl it shed a moment before and takes an extra push from it: wake capture, one of three tricks Dickinson, Lehmann and Sane found on a scaled robotic wing (1999). Read more finely, the wake's gift comes first and is partly paid back (Birch and Dickinson 2003). So the made wall is a pattern, not one animal's trick: in water, the jellyfish's parked ring under its rim; in air, the insect wing meeting its own wake at the turn. For a builder of a flapper: time the turn to meet the swirl the last stroke made, because the wall helps in the moment after the turn, not for good; expect to pay part of it back; and a short stroke puts more of its work at the turns, so the wall matters most to a short-stroke flapper.

Bear's leap-through train, half true. *"In my imagination, a jellyfish is a ring and a ring-builder. As it finishes its pulse, it senses the jelly drafting behind it, inside its rings, and the lead ring opens and lets the other jelly pass through. That's all."* (Bear). A small ring that fits through a bell's open top draws the bell toward it, the draft he pictured[138]. But a leader's own ring is four times as strong and twice as wide, and seeded in front of a closed bell it pushed the bell away, 1.337 radii in three seconds[139]. The ring did open, wider than the bell, and the follower did pass through it, but backward, carried along until the ring slid round its rim. So a ring can carry momentum to a body at a distance, in the direction the ring travels, and a leader's ring pushes a closed follower away. The follower that strokes, timed to meet the ring as Bear's picture has it, is the next run to make.

What this chapter found. On the same fuel, two leapfrogging smoke rings started half a radius to two radii apart go 14 to 19 per cent faster than one, and three packed half a radius apart 26 per cent; when a train breaks cannot be foretold from its start, but it always keeps a pair, and the ring it sheds sometimes comes back, always in pairs of equal swings, and leaves for good at the start of a new pair; a ring that swings out behaves as if held in a dish whose depth

falls off as about the cube of the distance, as between two small magnets, and comes home from as far as its throw takes it. A train near its edge is a pair and a third ring that threads it on a slower beat, and every deep squeeze read was one of those threadings.

10. The lock inside the engine: one gear, then a staircase

> A gearbox is a staircase of landings, wide ones and narrow ones. If the book is right, the edge of every landing goes soft underfoot in exactly the same way before it gives, and only the width of the landing tells you which stair you are standing on.
>
> *— a no-physics read*

Bear said once that where we kept seeing walls, he saw gears: teeth that reach across space and lock, the way a watch's gears and rotors click into each other, with a little gear train and driveshafts sized by the ratio of the gears. This chapter is the search for his gear train. It took the whole of the needle's long road to find it.

Something to do at the table. Put a compass on the table, hold a bar magnet near it, and walk the magnet slowly in a circle around the compass. The needle follows, turn for turn. Now walk faster, and faster still. At some pace the needle can no longer keep up: it jerks, falls back, catches up, falls back again. You have built a motor, found its gear, and stripped it. Everything in this chapter is in those thirty seconds.

One gear, exactly

A switch, not a dial. The toy is a needle sitting in thick fluid between two coils, one pushing along the table and one across it, each swinging back and forth at the same rhythm. With one coil alone, or both in step, the needle wags and goes nowhere: 0 turns in 100 cycles. Delay the second coil by any gap at all, a few degrees or a quarter cycle or most of a half, and at a gentle pace the needle turns exactly once per cycle, 99.8 to 100.0 turns in 100[140]. It does not give a little more turn for a little more gap. It gives the whole turn or nothing, and the gap sets only how fast the gear can go: at a quarter cycle the needle holds its gear up to a pace of 1, at 30 degrees up to 0.75, and at 2 degrees, a field that barely rotates at all, still up to 0.36. A bead carried by two standing waves does exactly the same, because in thick fluid the two toys are one equation[141].

The gear strips. Drive the needle faster than the fluid lets it follow and it slips, keeping 86 per cent of the drive at a pace of 1.01, 58 at 1.1, 25 at 1.5, 13 at 2 and 3 at 4. We went looking for why those particular numbers and found that at a quarter-cycle gap the needle's equation is a famous one: the law for how one oscillator pulls another into step, known in radio as injection locking and named after Adler. Below a pace of 1 the needle keeps all of the drive; above, it keeps one minus the square root of one minus one over the pace squared. The toy lands on that curve to within a few parts in ten thousand at every pace we tried[142]. Chapter 4 counted the same needle's slips one by one.

One gear only, and why. Bear's picture asked for a gear train: one turn here for two there, three for five. So we mapped the needle across every gap and pace, some fourteen thousand settings, expecting the gears. There were none. The needle has one gear, one turn per cycle, filling seventy per cent of the map, and a strict census at every fraction up to twelfths finds only that one[143]. The reason turned out to be exact. What one drive cycle does to this needle's angle is always the same shape of map, the kind that stretches a circle onto itself the way a lens stretches a picture, which mathematicians call a Möbius map, and a map of that shape can lock only on whole numbers. One cycle's map, taken from thirty-six starting angles, fits that shape to two ten-billionths of a radian. For the simplest radio-driven superconducting junction, which is this equation, it is a theorem (Buchstaber, Karpov and Tertychnyi 2010, and later authors).

The fix, tried, and it works. The Möbius shape comes from two things the toy needle has: it forgets at once, and its grip on the field repeats once per turn. Break either, and the shape should break. We broke each. Give the needle a soft-iron part, so the field also pulls it into line whichever end leads, a grip that repeats every half turn, and the needle has a gear train: steps at one turn per two cycles, one per three, one per five, three per five, and a neutral where it stops turning. Give the needle inertia instead, so it remembers its own swing, and steps appear too, at one turn per two cycles, one per three, one per five, and the neutral. So the rule reads: **a partner that forgets gives one gear and a clutch; a gear train needs memory, or a lumpier grip.** The swing of chapter 6, two pumps per swing, is a gear that comes from memory: the pendulum remembers its swing, and that memory is what lets it lock at two to one.

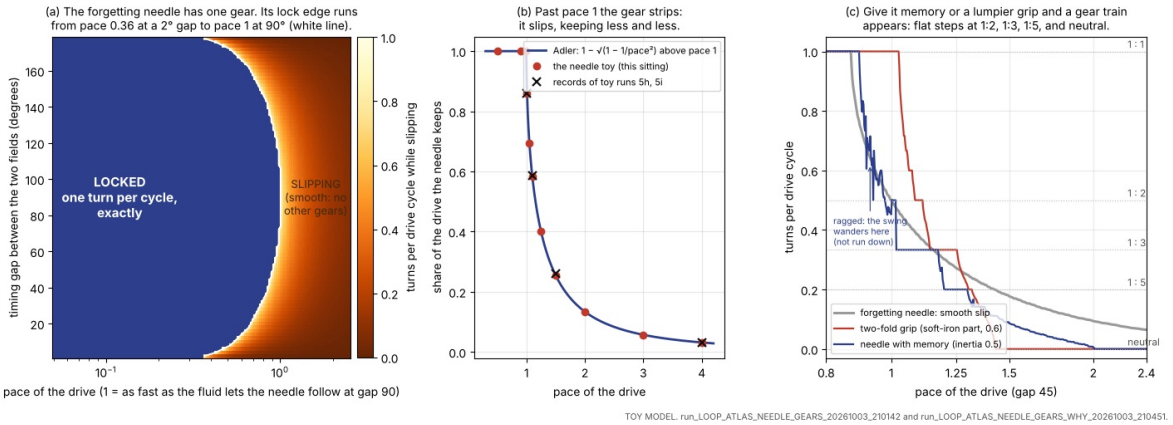


Figure 19. (a) The forgetting needle's turns per drive cycle over gap and pace: one locked region, one gear, with its edge drawn in white. (b) At a quarter-cycle gap, the share kept past the edge, with the closed form through the toy and the records. (c) With a soft-iron part or with inertia the needle gains a gear train: flat steps at 1:2, 1:3, 1:5 and neutral. TOY MODEL.

Rack and gear. A gear is a ring of teeth, and a ring can only mesh with patterns that fit round it a whole number of times. The worm chain showed it first: closed into a ring of eight it would take only whole waves, and it crawled 1.14 against the open chain's 1.66[144]. A radio toy of the first nights, from the line of work that became the companion book, *A Rider in Two Waves*, showed it as a comb: a rider on a ring of six waves locks firmly at rhythm ratios of five sixths, one and seven sixths, and on an open line there are no teeth at all[145]. Bear's word for the open case is the rack. Rack is line, gear is ring: a rack has pitch and no circumference, and a ring's teeth are the whole numbers that divide it.

From the field: the daily trip to the stop. *"I saw stepper motors used as floating-point actuators that reset, or recalibrated, every day by running for a set number of seconds to make sure they had hit their stops. There was some math involved."* A floating-point actuator, as building engineers call the kind that turns a damper or a valve, usually has no sensor telling it where it is. Its controller keeps a running estimate, so many seconds of opening and so many of closing, and that estimate is a measure, and measures drift; a stall or a step lost under load is never seen. So once a day the controller drives it longer than a full stroke, against the hard stop, and the stop hands back a count, zero, which nothing can drift. It is the gear-stripping of this chapter seen from the maintenance side: teeth that slipped with nobody counting are exactly what the daily reset forgives.

The grid's own needle. The needle that remembers its swing is the equation a power engineer calls the swing equation: a machine on the grid pushed by its turbine, its rotor's inertia its memory, the lag its power angle. Across every memory we tried, the price does not change: push per watt is one over the pace, locked or slipping, to two parts in a million[146]. What inertia changes is the machine that has fallen out of step. Between its relock line and the edge, 0.54 to 1 at a memory of 5, the same load can be carried in step or slipped, and which depends only on history: in step it buys 0.9 at a pace of 0.9, out of step 0.028, and it stays out. For whoever runs such a machine: keep the pace below the edge with margin; know the relock line, below which a knocked machine always comes back by itself, there being no slipping left to stay in (above it, the machine may come back or stay out, depending on how it was knocked, so the sure course is to cut the load below the line); and expect a heavy machine's slips in spells. Whether this matches real protection practice is a question for a grid specialist.

Memory: two edges and a pair of scissors

"The dirt wants to fall off the ledge," Bear said, asking whether a slip, once it starts, keeps going. Give the needle some weight, so that it remembers its swing, and its cliff splits in two. Walked up slowly, it still lets go at pace 1. Walked back down, it keeps spinning far below that, and climbs back into the lock only at about 0.96 with memory 1, 0.79 with memory 2 and 0.54 with memory 5[147]. Between the two edges, whether the needle sits or spins depends on where it came from: the two edges are a pair of scissors. The book's memory is exactly the McCumber parameter of a superconducting junction with capacitance, so these scissors are the junction's hysteresis, which Stewart and McCumber described in 1968. The climb-back edge is a single path, the spin that starts at the top of the hill and runs all the way round to the next top, and shot from there with three computer methods for stepping the motion, one of them at two settings, it sits at 0.9643271245 for memory 1[148].

Round the tip. The scissors are not endless. Lower the needle's memory and the two edges close on each other, meeting at memory 0.7026[149]; below that the needle has one motion at every pace, and it starts and stops spinning at pace 1 without a jump, its slips coming slowly at first. Claude guessed that a slow loop of the two knobs, memory and pace, round that tip would bring the needle home in the other motion, and that a loop clear of it would not. Walked round 18 times, 2,000 time units a leg[150], the needle said otherwise: on every loop that crossed an edge it came home

in the same motion from either start, set by the way the loop went, so where it ends cannot tell the two kinds of loop apart. What tells them apart is a change without a jump. Every change made beneath the tip, 16 of them, was gentle; every change anywhere else, 20 of them, was a jump; and in all 18 runs the needle came home changed exactly when its jumps plus its loop's turns round the tip made an odd number. To change the motion without a jolt, take the memory below the tip before crossing pace 1. The tip's place had been in our own records for nine days when the guess about it was registered[151].

Add a little noise and the two edges become two doors. A needle with memory 5 holding at a pace of 0.75 sooner or later falls over its hill, and how long it holds follows the hill's height: the log of the holding time grows with one over the noise at a slope of 0.234, against the hill's own 0.239[152]. Claude had expected a fall from there to be final. It is not: the running needle has a way back into a seat too, and by the end of the runs about half the fallen needles are holding again. So a remembering machine under noise has two rates to respect, the fall's and the catch's, not one edge.

Bear's picture of the slip was an avalanche: once the dirt starts to fall, it keeps falling. Under noise the remembering needle does slip in runs, but its runs spread like a trap with a fixed chance to catch it on each turn: the median run is 0.68 and 0.74 of the mean, where a sandpile's heavy tail would put the median far below it[153]. Rough floors, small ledges under the needle, gave no heavy tail either[154]. The floor repeats every turn, so a falling needle meets the same ledges on every turn, and its chance to be caught stays the same. On three rows of that run the spread was left unsettled[155], and two later runs settled it. Counted without cutting the slips into runs at all, the spread of the count over its mean levels off on every rough floor, a trap's mark[156]; on the strongest two-fold floor it was still growing at the end of 3,000 field periods, and flown ten times longer, 30,569 periods, it levelled off there too once the windows passed about a thousand periods, 1.152 at the last doubling of the window, where a made-up trap gives 1.093 and a made-up heavy tail 1.354[157]. On all four floors the slips are a trap's; the strong two-fold floor's trap only has a slow mood. Bear's avalanche was a good guess that the toy turned down.

Roads into chaos, and a reader's eye

The needle given a little memory, 0.75, and a second travelling wave to ride is the light rider of the engine of *A Rider in Two Waves*, a bead carried by two travelling waves. Past its lock, watched once each time a wave passes it, it settles on a small pad of order, a repeating dance; and the dance keeps splitting in two: eight steps, then sixteen, then on to two hundred and fifty-six. Each stretch of push, the range of the drive's strength over which one dance holds, is shorter than the one before by 4.6155, 4.6570, 4.6667, 4.6687 and 4.6691, closing on 4.6692[158]. From the third split on, each was forecast before it was looked for, by dividing the last stretch by 4.669, and each landed within 2.54e-9 of a push of its forecast. The number is Feigenbaum's, found in 1978 in the simplest rule that doubles. The push where the pile-up sits is ours; the ratio is everyone's. That simplest rule, the logistic map, carried to a dance of 8,192 steps in 60-digit arithmetic, shares a second number with the needle, in part: the factor by which each new pair of lanes is narrower than the last swings round about 0.214, each swing smaller than the one before, heading for the factor the needle sits at, 0.217 on both its sides[159]. And the clock of *A Rider in Two Waves*, a hair slower than ours, moves every split down a few billionths of a push, 5.516729 billionths at the four-step dance and 5.400580 at the sixty-four-step one; carried in 30-digit arithmetic, the differences between those moves shrink by Feigenbaum's number too, 4.454041, 4.620072 and 4.659088, climbing to it from below a rung behind the stretches' own ratios, not swinging about it as we had once guessed[160].

Why 4.6692. The ratio belongs to the shape of the rule's top. Run on single-hump maps with flatter tops, the same ratio closes on 7.2847 when the top is flattened to the fourth power, and climbs higher still for flatter tops, about 9.3 and 11 at the sixth and eighth powers, where the short runs were still moving at their last step[161]. The needle's 4.6692 is the rounded top's number. A map that keeps areas, as a frictionless one does, doubles by yet another factor: on the simplest such map the ratios climb 8.30, 8.66, 8.716 and 8.7203, from below like the needle's[162].

A window in the chaos. *"I think the edge might be a fold, an inner one or an outer one."* (Bear) Pulse the pace of the needle with memory 1, a fifth above and below its mean once each drive cycle, and its spin is chaotic at first, but every one of 32 needles flown at two paces settles in time onto a regular spin[163]. At a pace of 0.98375 all sixteen needles flown there sit on one orbit, one turn in every four drive cycles[164]. We called it the 1/4 groove and expected it to be wide. It is a sliver, about 0.98373 to 0.98375 on the grid it was first read on[165]: a window, a narrow stretch of order with chaos on both sides. Going down, the orbit doubles toward Feigenbaum's ratio. Going up, it ends at a fold, as Bear guessed: it meets an unsteady partner at a pace of 0.98375282925 and the two vanish together. Past the fold the orbit's ghost still holds the needle a while: 254 drive cycles at a sixteenth of a millionth of pace past the fold, then 522, 1,058 and 2,130 as the distance is quartered each time[166]. Four times closer, twice as long. And the fold is inner, which was Bear's question: the needle keeps falling back into the ghost, 231 to 730 times in 200,000 drive cycles[167], and very close to the fold, a thousandth of a millionth past it, it spends three quarters of its time there, so its record reads as long calm stretches broken by short bursts. That is intermittency of the first kind, which Pomeau and Manneville described in 1980.

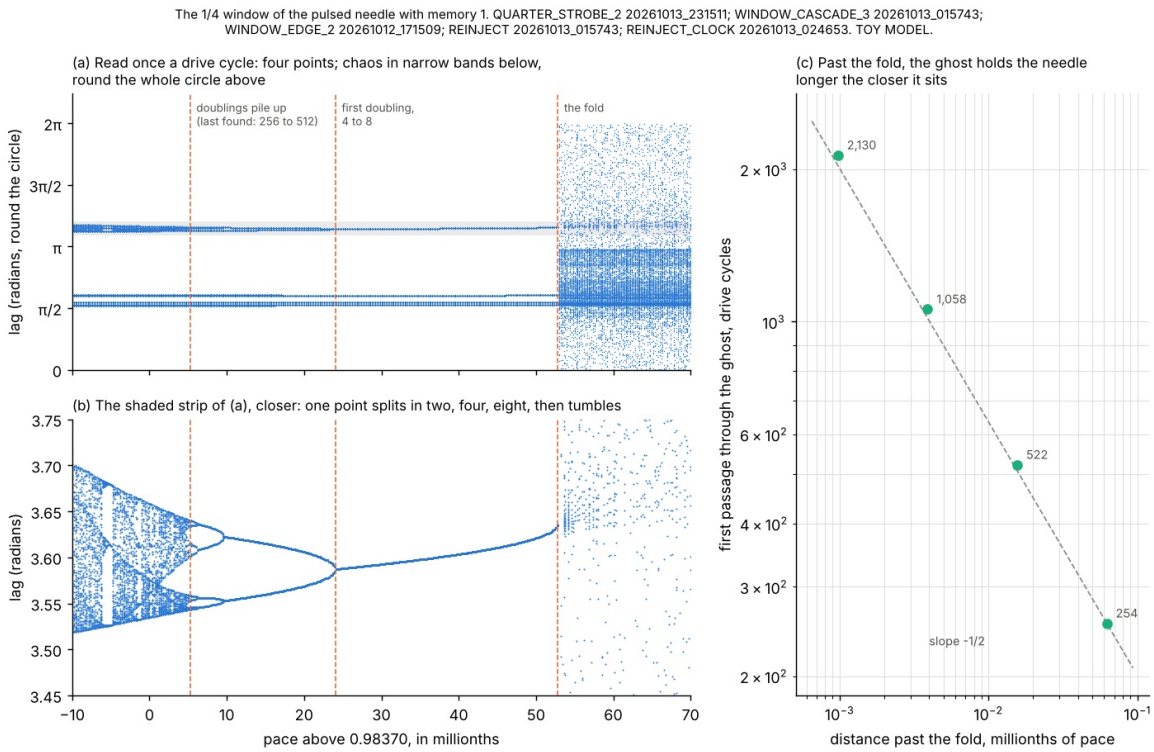


Figure 20. (a) The pulsed needle with memory 1, read once a drive cycle 256 times at each of 401 paces from 0.98369 to 0.98377: four points from the first doubling to the fold; below, the doublings and then chaos in narrow bands; above the fold, chaos round the whole circle. (b) One of the four points closer: it splits in two at 0.98372406, then in four and eight, and the doublings pile up near 0.98370525; inside the chaos below sits a small window of its own. (c) Past the fold, how long the needle lingers where the orbit was, against the distance past the fold, log against log, with a guide of slope −1/2: four times closer, twice as long. TOY MODEL.

Two readers on the window. Bear looked at that drawing and saw a tree, “the same as a tongue tree or a probability tree”, and marked it in blue, pink, red and green. Each mark was then measured. It is a probability tree, exactly: each split hands each daughter road half of its parent’s time, one road holding 64 of 256 readings, then two holding 32, four holding 16 and eight holding 8; and each split has three roads, as Bear drew them, not two, the parent carrying on between its daughters as a ghost road the needle never takes[168].

In the chaos below the tree, Bear’s eye was the better reader. His six white bands, voids with thin bridges across them, are each a window, a moment of order that keeps the gear. His red and green strokes are the chaos’s own skeleton. Four drive cycles carry the needle once round and back into the same strip, and on the chaos that return is a single hump with a lowest point, the fold; the chaos piles its readings along the places the fold lands after 4 cycles, 8, 12 and on, the fold’s images. Computed directly from the needle, those images are, for the most part, the lines Bear drew: of his red strokes, 0.741 lie within 0.004 radians of the fold or one of its first eight images, and of his green 0.673, against 0.36 and 0.37 for the same strokes set down at random[169]. Then he read a rule off the drawing: a window opens where a dark arch comes down onto the band’s edge. Turned round into a prediction and followed by Newton’s method, 18 of the 19 arch touches are windows of exactly the predicted cycle, and the nineteenth lies inside another window; none of the control paces has one[170]. The scattered crowd below the tree is not lost. It walks two fields with a fence between them, and every so often a gate opens and the whole crowd marches in single file.

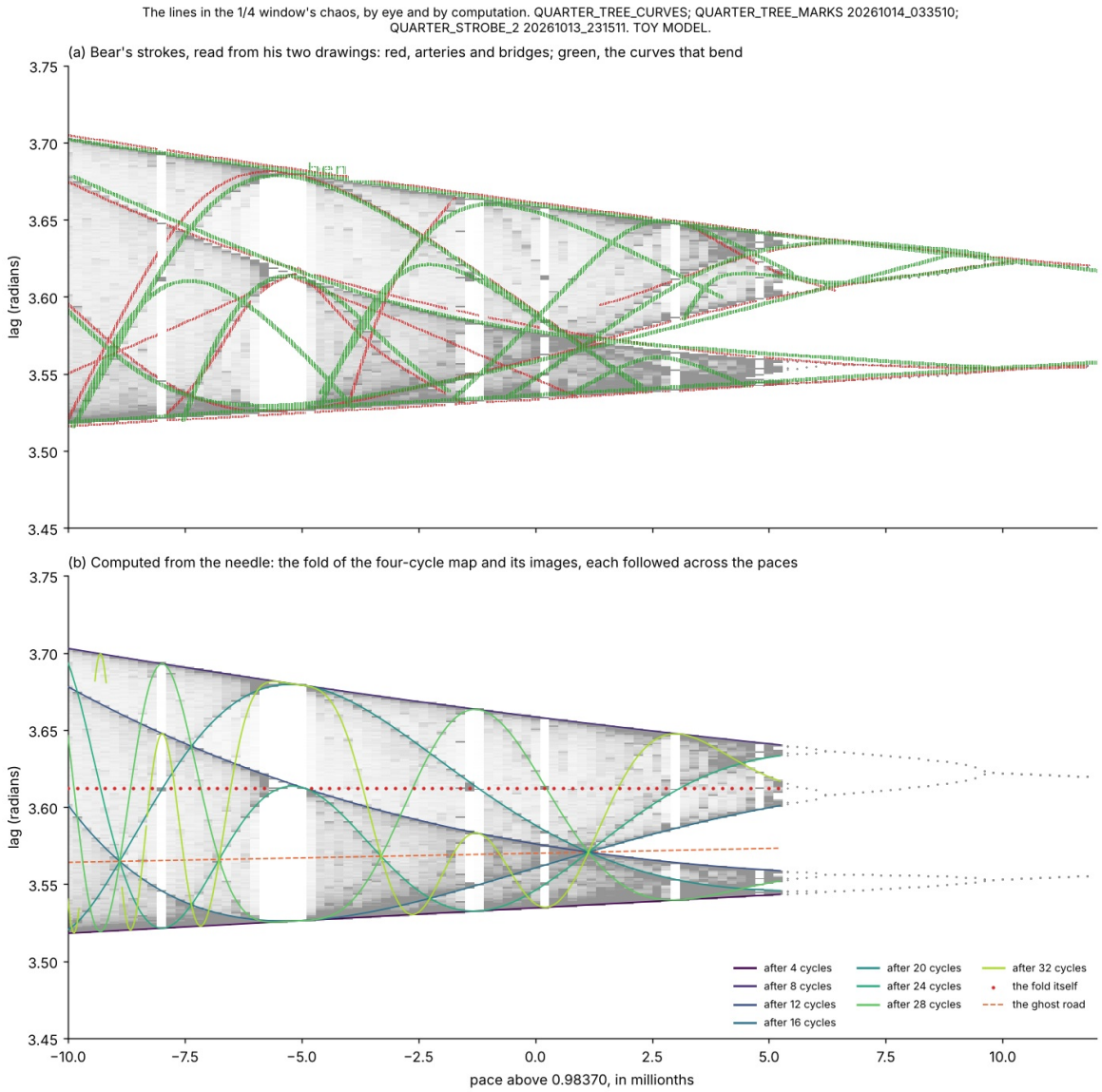


Figure 21. (a) Bear's strokes on the 1/4 window's chaos, read from his two marked drawings by colour: red, the arteries and bridges; green, the curves that bend. (b) The same chaos with the fold of the four-cycle map (red dots), its images after 4 to 32 drive cycles (colours), each followed across the paces, and the ghost road (orange dashes). The first two images are the band's edges; the later ones arch over the voids and cross, most of them at +1.1 millionths on the ghost road, where the strip's two bands merge into one. TOY MODEL.

Where the window ends. Bear's black arc on the last drawing is the fold's third image, and where it meets the band's top edge the window ends, at a pace of 0.98368808, which the figures count as −11.92 millionths from 0.98370, where the unsteady partner born with the orbit meets the banded chaos[171]. Below that end the needle keeps falling back into the window's four strips, and its mean stay grows as the distance shrinks with a slope of −0.498, an error of 0.001[172]. Before the run Claude expected −0.553, from a correction as Claude recalled it. It is not there. The needle behaves exactly as a map with a single hump, which gives one half (Grebogi, Ott, Romeiras and Yorke 1987).

A heavier needle. Bear asked where the hard stop wall lives, so the needle was made heavier, and a heavier needle forgets its speed more slowly, so the squeeze, how flat four drive cycles press any small difference in its motion, grows seventy times lighter by memory 1.2. The window slides down in pace and narrows, but its end stays 2.253 to 2.258 times as far below its top as its first doubling[173]: the law a one-hump window should keep is 9/4 (Yorke, Grebogi, Ott and Tedeschini-Lalli 1985). Bear traced the curves over the data by hand, and his red and green traces lie on them within about 1 per cent[174]. Then he carried them on past the data. His purple stroke for the doubling bent less than the data, which sit 1.4 to 5.3 per cent above it[175], and his cyan stroke for the end bent more; but the measured curves bend together, within 0.6 per cent of 9/4 at all eight weights followed[176].

Carried on to memories 2 and 3 the window still keeps that shape, 2.266 and 2.252[177], so from memory 1 to 3 its end stays within about 0.7 per cent of 9/4, and the stop wall he asked after is never reached: just past the window's end a half-turn chaos takes over, and it is that chaos, not the window, which drains into the stop[178]. At memory 1.76, lower down, that chaos tidies itself rung by rung, settling onto cycles of 32, 16, 8 and 4 drive cycles and at last onto the plain half turn. Bear ringed the places on the figure where the needles' spread falls away and comes back, and asked *"Gaps?"*, and then: *"They look like rungs."* They are. Each rung is narrower than the one before by 3.43, then 4.24, then 4.57 times, climbing toward Feigenbaum's 4.669[179]. And the dip he ringed is real: at memory 1.7598, at that one step, the needle settles on a stable cycle that the steps on either side never return to[180]. He also asked whether the slide was a holonomy. It is not: a path in one knob leaves nothing behind, and the window's top, found again from a different start without walking the road between, lands on the same pace to three parts in a hundred trillion.

The gear train Bear asked for

Zoomed out, the remembering needle climbs a whole staircase of gears[181]. From rest it sits, then turns once in two cycles, twice in three, three in four, four in five and five in six. Between each pair of neighbours the next widest step is their mediant, the sum of the two tops over the sum of the two bottoms: 5/7 between 2/3 and 3/4, 7/9 between 3/4 and 4/5, 9/11 between 4/5 and 5/6, and beside those, narrower still, 11/14, 13/16 and 14/17. On every landing between steps the needle tumbles. A staircase of locked fractions ordered by mediants is a devil's staircase, and in a superconducting junction driven by radio waves its steps are the fractional Shapiro steps (Jensen, Bak and Bohr 1983 for the staircase). With no memory the theorem allows only the whole-number step. One unit of memory gives the needle the gear train Bear's picture asked for at the start of the chapter.

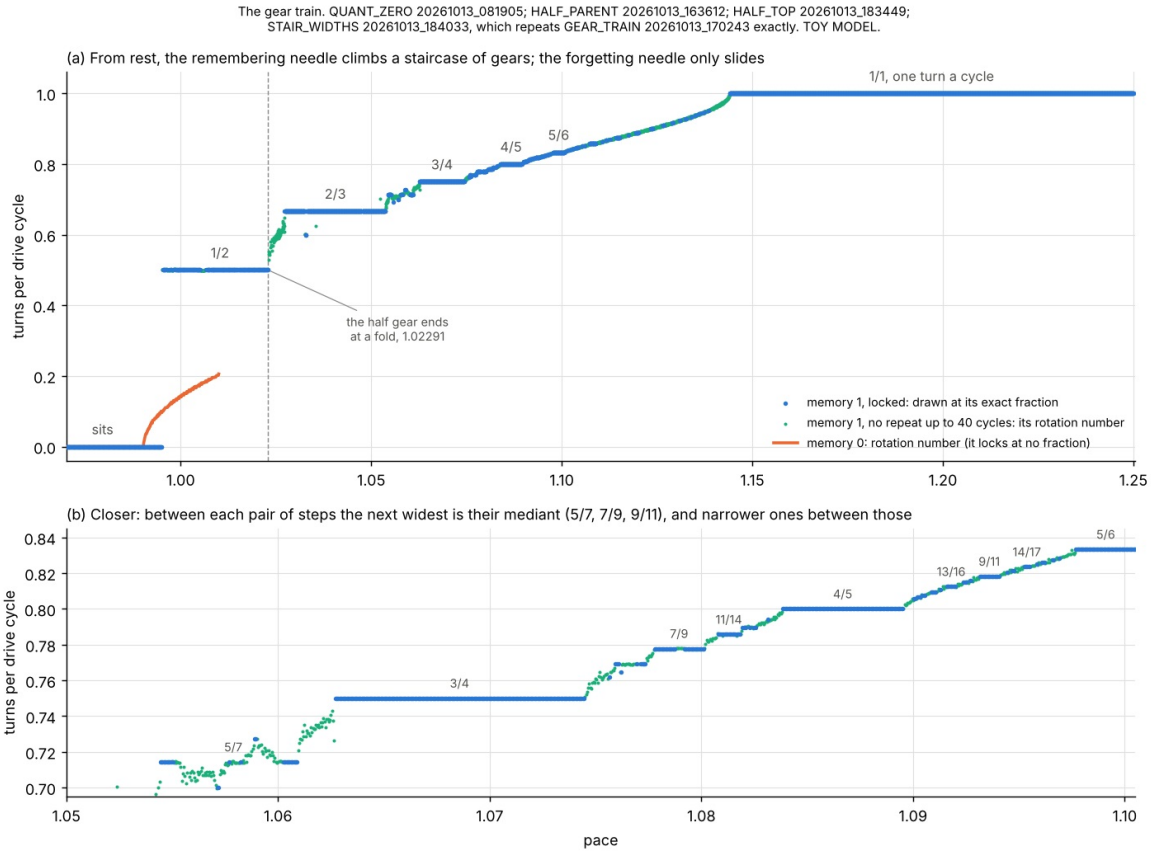


Figure 22. (a) Started from rest with memory 1, the pulsed needle's turns per drive cycle at each pace from 0.97 to 1.25: a locked pace drawn at its exact fraction, an unlocked one at its rotation number over 199 cycles, and beside it the forgetting needle (memory 0, to 1.01), which locks at no fraction. The remembering needle sits, takes the half gear, which ends at a fold at 1.02291, tumbles, then climbs 2/3, 3/4, 4/5, 5/6 and on to the whole gear from 1.1441. (b) 1.05 to 1.10, where the mediants sit between the steps, with chaos on the landings. TOY MODEL.

A bubble inside the half gear. Inside the half gear the needle doubles its way into chaos and comes back out by the same road. Its four-cycle is stable from a doubling at pace 0.99920 to another at 1.00199, where it doubles on into chaos; on the far side the same orbit undoes that doubling at 1.00978, one orbit in and out, a closed bubble of the kind

Bier and Bountis (1984) found common in systems with more than one knob[182]. Past it the orbit runs into its parent, the two-cycle, at 1.01512, not into the second shark's tooth Bear's eye had seen building[183].

Chaos that keeps order. *"How chaos can keep order is a wonderful idea to explore,"* Bear wrote. The needle shows it two ways. In the middle of the 5/7 step the needle repeats at no cycle at all, and two needles started a billionth apart part by a factor of e every 10 to 40 cycles; yet over 30,000 cycles its count of turns never strays from five in every seven by more than a quarter of a turn[184]. It tumbles inside bands it visits in a fixed order, so the gear survives inside the chaos. And under a jostle it is the lock near its edge that is fragile: jostled with a noise of 0.001, a lock just inside its edge moves its turn rate by 0.013 to 0.014, while the chaotic needles beside it move theirs by 0.003 or less[185]. Order here is best read in the average turn rate, and the average is what the chaos keeps. And a scan started from rest does not see everything. Flown from twenty starting angles, the 5/7 step hides other loops: an 11/15 at 34 of 44 paces, always reached from only 1 to 4 starts of the 20, and an 11/16 at 11 of 22, each doubling the opposite way to 5/7. Past each hidden loop's last doubling a few starts never repeat at all, right beside a settled 5/7 at the same pace; flown twice as long, all 29 still met no repeat, and 28 of their 29 exponents were positive. Chaos and a steady gear live side by side at one setting, and only the start decides which the needle gets[186].

The axis the strobe leaves out. The needle's state has three axes: its angle, its speed and the clock, the second wave's phase. Bear's z-needle draws the path on the clock's axis. On the doughnut, with the clock round the ring and the angle round the tube, the lock of two turns in three cycles is a closed loop that winds three times round the hole and twice round the tube, while its speed swings from 0.0929 to 1.7787 within each round[187]. With no memory the speed is set by the angle and the clock alone, so the state has two axes, and a flow on two axes cannot tumble; memory gives the speed an axis of its own. Swept from rest over twelve memories and 49 paces, the needle tumbles nowhere at memory 0.4 or below, with no chaos and no doubled loop, while at memory 0.5 two paces tumble and one loop doubles; and the steps fill in as memory grows, 23 of the 49 paces locked at memory 0.05, 38 at 0.5. In thick fluid, memory 0.05, not even a second wave twelve times as strong makes it tumble[188]. Claude guessed that the tumble waits until the needle's angle-only map folds back on itself. Read first on the paces between the steps, the map was still one-to-one where the needle first tumbled; read again pace by pace, inside the steps too, the fold comes first at each of the three paces that tumble earliest, ahead of the tumble by 0.030, 0.025 and 0.015 of memory[189]. A lock's other multiplier is 0 to five decimals, so its first is the slope of the needle's angle-only map taken once round the lock. Inside the window where the needle locks once in seven cycles at pace 1.0, from memory 0.465 to 0.485, that slope walks from +0.53550 through 0 to −0.76947, and at 0.490 the loop doubles: the order a map with a single hump gives. But the circle can fold away from the lock's own points: at pace 1.025 it folds while the lock's slope stays positive, dipping to +0.02112 and rising again. And read seven cycles at a time, the banded chaos of 5/7 makes between 0.70533 and 0.72042 turns a cycle: it never strays toward its neighbours 2/3 and 3/4, as Claude guessed it might, but trembles on its own step.

A lesson in grain. Along the way the needle taught a lesson about instruments. At one pace the settled needles seemed to run several long orbits that never closed. The cause was arithmetic: the lag counts every turn the needle has made, and after ten thousand drive cycles it is about 31,600 radians, a number the computer holds only to about four trillionths; once in a while the orbit stretches a small difference a hundred thousand times, and the grain comes out as scatter. With the numbers kept small, the orbit closes after 19 drive cycles and 9 turns, and all thirteen needles read sit on it[190]. A sieve finer than the arithmetic can hold reads its own grain as structure. The stretch itself is real, though, and reading the needle as a circuit explains it: on the top half of each turn the junction acts as a negative coil, an amplifier. The whole lap is stable, shrinking any difference about eight and a half times, and yet a difference started at the wrong moment reaches 186,000 times on the way[191], the way Armstrong's super-regenerative receiver of 1922 lets a signal grow enormously for part of every cycle and quenches it by the end.

The needle's world is a doughnut

Playing with the Stretch and Squeeze Explorer, Bear saw the needle's world wrap. The needle's angle comes back after a whole turn but its speed does not, so angle and speed together make the side of a can, a cylinder. The drive comes back every cycle too, and with the drive's clock wrapped as well, angle and clock make the skin of a doughnut, a torus, with the speed as its thickness. A needle locked at p turns every q drive cycles is back at the same angle at the same moment of the drive after q cycles, so its settled motion is one closed loop on the doughnut, round the hole q times and round the tube p times. The explorer's long orbit is the loop 9/19, nine turns in nineteen cycles, counted from the page's own data. Every step of the staircase above is such a loop, and in a superconducting junction, whose equation the needle's is, they are the Shapiro steps.

The map of the loops. The fractions have a map. Every fraction between 0 and 1 has two parents, its nearest neighbours with smaller bottoms, whose tops and bottoms add up to it: 2/3 and 3/4 make 5/7. Written as a square of whole numbers with the two parents as its columns, every family has determinant 1. Drawn as points of the whole-number lattice, bottom across and top up, every family is a triangle of area exactly 1/2, the smallest a lattice triangle can be, with no other lattice point in it or on its sides. Joined side to side, these triangles make the map number theorists call the Farey diagram (Hatcher 2022). On the doughnut a family shows as crossings: two loops a/b and c/d cross each other |ad − bc| times, so two parents cross once, and a child crosses each of its parents once; 3/5 and 2/5,

which are not neighbours, cross five times. Of the three kinds of fraction, odd over odd, even over odd and odd over even, two parents are always two different kinds and their child is the third. And a prime bottom q fills its layer of the map with q − 1 loops, while any other bottom fills fewer, Euler's count φ(q).

The needle walks the map. Set the needle's 59 locks in the order its sweep met them: 51 of its 58 steps from one lock to the next run along a side of the map, from a fraction to a neighbour[192]. Its nine widest steps are joined by sides every time. From 2/3 it goes round the triangle of 2/3, 5/7 and 3/4 by two of its sides, then round the triangle of 3/4, 10/13 and 7/9 the same way, steps on to 4/5, and turns about the whole turn by 5/6 and 6/7 to 1/1, each of them a neighbour of 1/1. Each of its seven steps off the sides touches a lock it held for six paces or fewer, most of them met inside another lock's range, where it stumbles. And no lock is wider than its parents: of the 54 that the scan measured with both parents, 47 are narrower than both and 7 tie a parent at the scan's finest step.

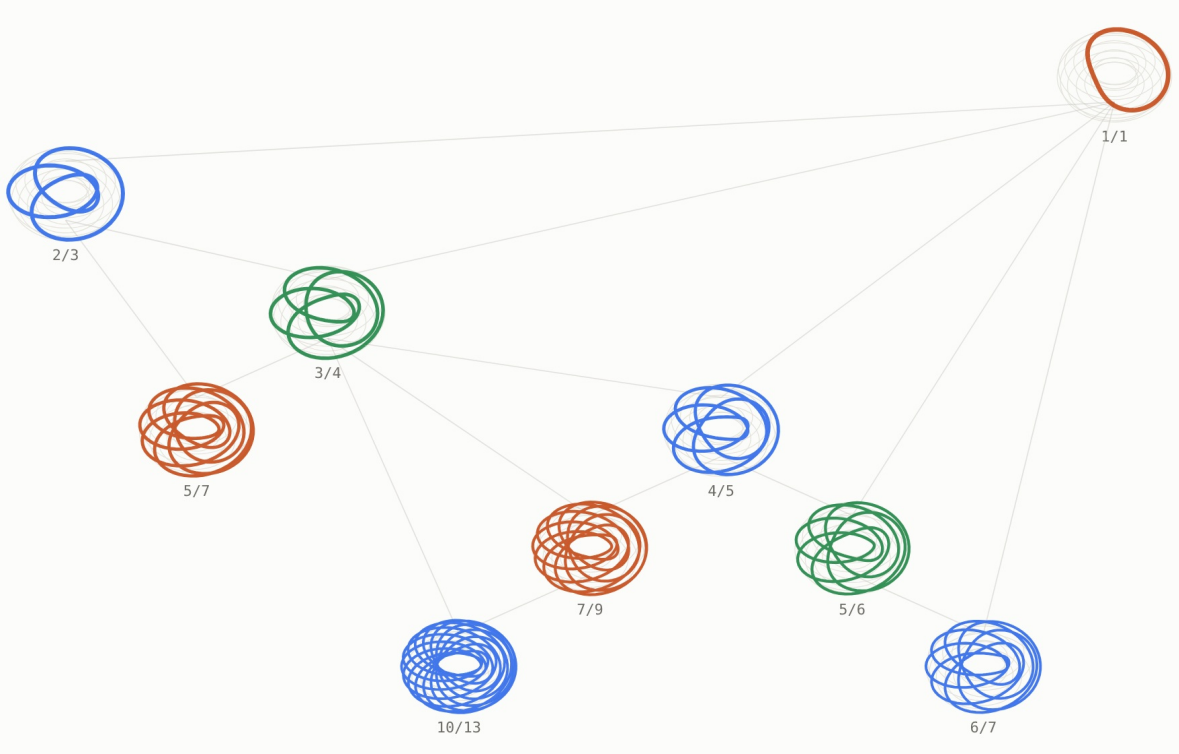


Figure 23. The needle's nine widest steps, each a loop on a doughnut of its own, in rows by generation and from left to right by size, joined by a line to each of its parents that is shown; red odd over odd, blue even over odd, green odd over even, a loop drawn thicker the wider its step. Every step of the needle's sweep from one of these to the next is one of these lines. Drawn by the Torus of Fractions. TOY MODEL.

Doubled loops. A fraction is always in lowest terms, but a loop is not always simple. Inside the 2/3 step, from pace 1.0332 to 1.0475, more than half of the step, the needle comes back only every six cycles. Its two rounds of three differ, by most of a radian in the middle of the band, so its loop goes round the hole six times and round the tube four, two strands of 2/3 side by side. On the doughnut's skin such a loop must cross itself; it passes itself only through the doughnut's thickness, the speed. At the band's lower end the difference between the two rounds grows as the square root of the distance from pace 1.03319, the mark of an ordinary period doubling, the same first rung as on this chapter's roads into chaos. Right at the band's ends the plain loop settles so slowly, each round on the other side of the last, that the scan's rule, waiting 2,000 cycles, read three plain paces as doubled; flown four times as long, they are plain. In the 5/7 step the doubling comes in from both edges, 7 cycles and then 14, and between them the needle tumbles.

A border that crinkles all the way down. Where two loops share a pace, which one the needle settles on depends only on where it starts. At a pace of 1.03295 the 2/3 loop and a hidden 3/5 loop share the pace, and the line between their starting points is not a clean edge. However closely you look, a share of the starts still lies near enough to it to change loop, and that share falls only as the closeness to the power 0.4, which makes the border's dimension on the plane of angle and speed about 1.6, more than a line and less than an area. Looked at two decades finer the power held, 0.423[193]. Borders like this between a forced pendulum's loops were described by McDonald, Grebogi, Ott and Yorke in 1985.

Knots. Lay the doughnut in space and every loop is a knot, the torus knot of its fraction: 2/3 is the trefoil, 2/5 the cinquefoil, 3/4 the torus knot of eight crossings, and a loop with a 1 in its fraction is the unknot. The doubled 2/3 is a cable of the trefoil. At a doubling the needles just beside the old loop swap sides every round, so the ribbon round the old loop carries a half twist, and the doubled loop runs along its edge. The periodic motions of driven systems can be knotted, and torus knots are among the knots they make (Holmes and Williams 1985).

What the staircase is for. In a superconducting junction driven by microwaves, the steps are steps of voltage, each proportional to the microwaves' frequency, and Josephson voltage standards have become the standard tools for calibrating voltage measurements (NIST). The volt itself can be realised this way, by the Josephson effect and an agreed value of the Josephson constant, 2e/h (CCEM 2019): a count of free laps set against a clock. Close such a junction into a superconducting ring, a ring with a nick, and every whole turn costs a little more than the last; that ring unrolls the can. The needle's world has no nick, and that is why its laps can be counted. The Torus of Fractions (chapter 2) lays all of this out to turn by hand.

What this chapter found. A needle in thick fluid follows Adler's law and, as a theorem says, can hold only one gear; give it one unit of memory and it climbs a full staircase of gears, whose roads into chaos close on Feigenbaum's 4.669, the 9/4 window law, the crisis exponent's −1/2 and the Pomeau–Manneville ghost. Bear's hand-drawn strokes on that chaos lie on its computed folds, 0.741 of his red and 0.673 of his green, against 0.36 and 0.37 at random. Its world is a doughnut, its locks loops on it, and its sweep walks the map of those loops, 51 of its 58 steps along a side.

11. Clock against listening: who opens the gate

> Everything that worked cut down how often the switch is asked. The plain switch answers every time it's asked, and the toy asks at every tick. Pausing after an answer barely helps, because the next question still comes at the next tick. Asking less often fixes it, and so does listening to something that doesn't change at every tick.
>
> — *a no-physics read*

"I keep coming back to the simple crystal radio in my 1986 kit of fifty science experiments. A little crystal radio oscillator?" (Bear)

Something to do at the table. A crystal radio has no battery. A coil and a capacitor are tuned to one station, so they store the wave that matches them and ignore the rest. A crystal lets current through one way only. An earpiece hangs on the end. The station's own wave pays for everything you hear. Take the radio apart in your head and you find two parts: a store that rings with the wave, and a gate that decides when to let go. Every toy in this chapter is that pair, and the question each one asks is who opens the gate: a clock, or the wave itself.

An ear can harvest what a clock cannot

The toy is a bead in thick fluid, the cork of chapter 3, in a soup of waves running both ways in equal measure. A plain bead goes nowhere: a drift of 0.0006, smaller than its scatter of 0.0019. Give the bead a blade, a grip it can switch on and off, and switch it by a clock: 0.009, scatter 0.008, nothing. Switch the same blade by listening instead, on when the push at the bead points the way you want to go and off when it points back, and the bead drifts 0.124, in proportion to the soup's strength[194]. Bear named the difference before it was measured: *"It feels like a question of event timing, like the jellyfish, instead of master-clock timing."* An even soup has no rhythm a clock can lock to. It has only events, and only an ear can catch them. (Chapter 3 gives the bill. As built, this blade grips by a charge switched on from nowhere, and the toy does not pay for where the charge comes from, so the 0.124 and every drift in this chapter are an unbilled toy's; billed honestly, only a difference in push across the bead's length pulls. Whether a blade that grips the water with its face, like an oar, keeps the 0.124 is still to be tested. The price of switching, below, is a bill the toy does keep.)

Prompt beats accurate. A listener that hears late does far worse than one that hears badly. Delay the ear by 0.2 time units and the drift barely moves, 0.121. At a delay of 1.0 it falls to 0.041, at 2.0 the bead runs backwards, −0.057, and by 4.0 there is nothing[195]. Add noise to the ear instead and the drift falls gently: 0.112 with noise half the size of the soup, 0.090 at equal size, 0.058 at twice, 0.013 at ten times[196]. A noisy ear is wrong at random and still right more often than not. A late ear is wrong in order, because by the time it acts the wave has moved on.

Listening has a price: strength over rhythm. Every switch of the blade costs something. Moving charge on or off loses energy as the square of the charge moved, in this toy as in a real capacitor. So the ear has a bill, the number of switches per unit time, and a harvest, its drift, and they answer to different dials. The harvest follows the soup's strength and ignores its rhythm: drift over strength stays between 0.398 and 0.446 across an eightfold range of strength and a sixteenfold range of rhythm. The bill follows the rhythm and ignores the strength: the ear switches at 0.94 to 1.00 of the rate at which a sum of many waves crosses zero, which S. O. Rice worked out for telephone noise in

1944. So the price of listening, travel bought per unit of bill, is strength over rhythm, and the twenty cells at strengths up to 0.3 sit between 0.94 and 1.06 times that line[197]. The line breaks at a fence, where the soup is strong enough to catch the bead and carry it along a wave.

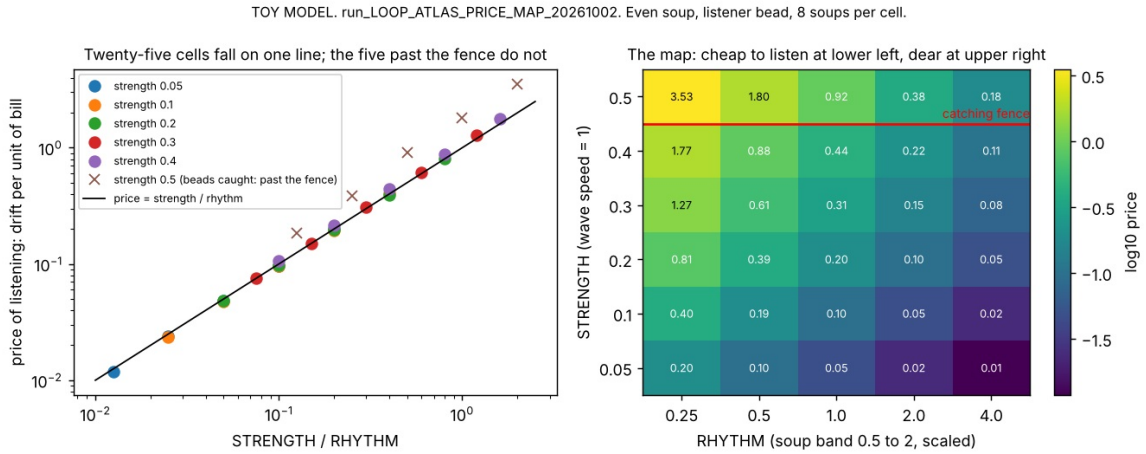


Figure 24. Left: price against strength over rhythm, log–log, with the line price = strength/rhythm; the five cells past the catching fence fall off it. Right: the map, cheap to listen at the lower left (slow, strong soup) and dear at the upper right. TOY MODEL.

A noisy ear, a knee, and the traders' law

A late ear slides down the price map along one curve, its price falling by about 1, 4 to 5, 18 to 21 and 68 to 70 per cent at delays of 0.1, 0.25, 0.5 and 1 time unit, about a fiftieth to a fifth of the soup's typical period[198]. A noisy ear chatters off it. It switches on every small wobble, and its bill doubles each time the toy's time step is halved, so the bill belongs to the toy's step, not the soup. A knee fixes it, a dead band either side of zero (this book uses the two words for one thing): no switch until the reading is clearly past zero, about three times the noise. That brings the cell back onto the map, at about the line[199]. But the knee creeps. Carried down to a step of 0.0025, the bill still changes by 0.5 per cent over the last halving when the noise relaxes over one time unit, 1.4 per cent when it relaxes over a tenth, and 4.1 per cent over three hundredths[200]. A noise that relaxes is still rough at every scale finer than its own time, so each finer step catches brief excursions past the knee that a coarser step stepped over. The creep follows a rule that traders use, and it can be learned in five steps.

First, the trader's problem. A contract dies if a price ever touches a barrier, but the contract only checks the price once a day. Between two checks the price can poke past the barrier and come back, and nobody sees it.

Second, why a square root. A random wander grows as the square root of the time it has. Picture a walker so drunk that each step goes left or right by a coin toss, with no memory of the one before: his steps partly undo each other, so he gets twice as far from his start in four minutes as in one, where a sober walker who kept his heading would get four times as far. So what a slow watcher misses grows as the square root of the gap between its looks, and halving the gap shrinks it by only 29 per cent, a little under a third.

Third, the trick. Pretend the barrier sits a little further out, by 0.5826 times how far the price wanders between two checks, and price the contract as if it were watched all the time (Broadie, Glasserman and Kou 1997). The number 0.5826 is the same for every price and every barrier; it is minus Riemann's zeta function at one half, over the square root of two pi.

Fourth, the knee is the barrier. The knee ear looks at the heard push once a step, and a poke past its knee that comes and goes between two looks is never seen; panel (a) of the figure below shows a real one. So an ear that looks every step should act like an ear that looks all the time with a bigger knee: bigger by 0.5826 times the noise's wander over one step.

Fifth, the test. The ear was held still, so that one noise path could be watched at eight steps from 0.000625 to 0.08, the way a trader could watch one price once a minute or once a day. For noise that relaxes over one time unit, the knee looks 0.107 of the strength bigger at a step of 0.08 on the coarse knee grid, 0.106 on the finer, where the law says 0.102[201]. With nothing fitted, the law gets the missed flips right at every step, within two standard errors at six of the seven steps and 2.2 at the coarsest, from 2.5 per cent under to 7.7 per cent over, where the error bar runs from 14 per cent at the smallest step to 2 per cent at the largest[202].

Reading the constant back. The run that tested the law put 0.5826 in; it never read it out. Read back from the same flips, the knee's shift over the wander's change is not one number. It climbs with the step measured in the noise's memory, from about 0.60 at a twentieth of the memory to 0.72 at four tenths and 1.2 at two memories[203]. Drawn as a

straight line in that ratio and read where the step shrinks to nothing, it lands at 0.57947 ± 0.00254, within 1.23 standard errors of 0.5826. The line fits its points loosely, and a line in the ratio's square root would land at 0.48628 instead, so the limit agrees with the traders' constant without proving it. Riemann's zeta enters through the ladder heights of a random walk, a sum over every whole number of steps (Chang and Peres 1997), and the primes play no part. For a builder: use 0.58 when the switch looks many times within the noise's memory, and expect the correction to grow as the look gets coarser.

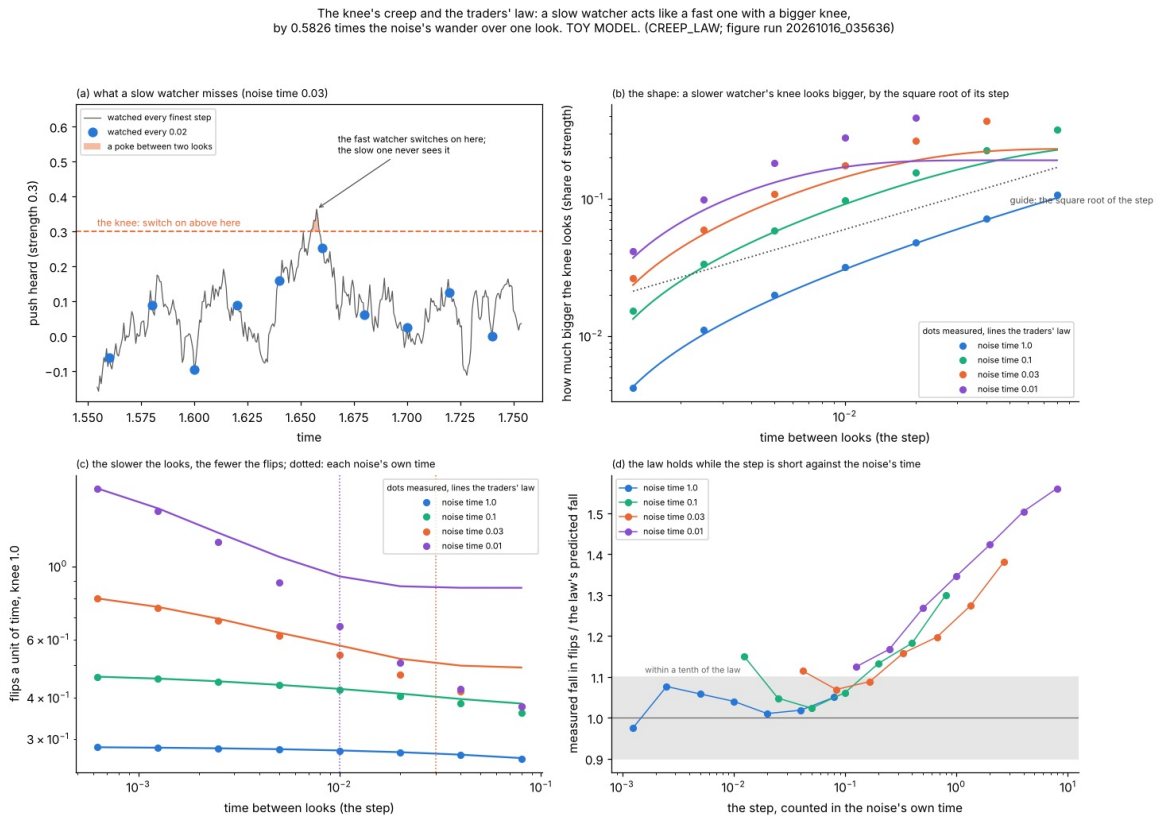


Figure 25. (a) What a slow watcher misses: one stretch of heard push with noise relaxing over 0.03, watched at every finest step (the line) and every 0.02 (the dots); between two dots the push pokes past the knee, and only the fast watcher switches. (b) The shape: how much bigger the knee looks to a watcher every step than to one every 0.000625, read from the measured flips (dots), against the traders' law (lines) and a square-root guide. (c) Flips a unit of time at knee 1.0 against the step: measured (dots) and the law (lines), each noise's own memory dotted. (d) The measured fall in flips over the law's, against the step counted in the noise's own memory; the grey band is within a tenth of the law. TOY MODEL.

The knob. For whoever builds a sampled ear, it reads like this. A controller that samples every so often with a dead band behaves like one that watches all the time with a wider dead band: wider by 0.58 times the sensor noise's size times the square root of twice the sample time over the noise's own memory, so long as the sample time is a tenth of the noise's memory or less; there the law came within 15 per cent in the toy, except at the slowest noise's two smallest steps, where its error bar is wider than that. The dead band you set is not the dead band you get. Set it narrower by that much, or sample faster.

Where the law holds, and where to read it. Past a tenth of the noise's memory the slow watcher misses more than the law says, a fifth to two fifths more around the memory itself, and half again at four to eight times it. Counted in the noise's own memory, four noises fall close to one curve (panel d): one ratio, the step over the memory, decides how far the law reaches. The law is built for something that wanders, and a noise that forgets where it was between two looks is no longer wandering. The square root itself lives in the knee's extra size, not in the flips, which follow it through a bent curve. Read where it lives, the extra size grows as the step to the power 0.458 and 0.475 in the two slowest noises, against 0.481 and 0.454 for the law itself, read the same way over the same steps, which is not quite one half either at steps this size[204]. At the slowest noise's two smallest steps the law first looked about a fifth short of the knee's measured extra size[205]. We read that first as the law's own excess and then as scatter, both times too soon. The error-bar pass of 19 October could not settle that: against the old reference the excess sits one to two standard

errors above the law at the two smallest steps, whichever of three readers is used, and against a reference watcher four times finer every reader puts every step within 1.6 standard errors of it[206]. So the excess, where it shows, points to the old reference's own blink rather than the law, without settling it.

Why a switch needs a band

A slow watcher can try to count what it missed from its own looks. For two looks on the same side of the knee, the chance that the push crossed it and came back between them is known exactly for a wander pinned at both looks. Added up over all its looks, that should give what a fast watcher sees. It does not, and the reason is the lesson. Count the pokes past one line, each time the heard push goes from below the knee to above it, and the count never settles: it grows 1.41 to 1.44 times each time the finest step is halved, for every noise tried[207], and against a reference sixteen times finer, 1.4145 and 1.4147 over the last halving, the square root of two to three places[208]. A rough path that touches a line touches it again and again close by, so the closer you look the more crossings you find, without end. Only a band makes a count finite. The knee ear flips only when the push has crossed all the way from one edge to the other, and a wander takes real time to cross a band, so the knee's flips settle where a line's crossings never do: over the same halving the band's flips change by 0.08 and 0.39 per cent.

So the slow watcher counts across the band, both edges at once. Walking its own looks in order, it keeps a running chance that a watcher who never blinks would be switched on, and between two looks it asks the pinned wander for the chance the noise crossed the band's far edge. The count it gets barely moves with the step: for noise that relaxes over one time unit it stays within 0.7 per cent over a 128-fold range of steps, from 0.000625 to 0.08, where the flips it actually made fall to 0.923 of the finest step's[209]. And the traders' law, built on another idea, gives the same never-blinking count to within 0.02 per cent at the two noises where the runs print it, 1 and 0.1[210], and, by a script check not kept as a run, within 0.6 per cent at all four: two corrections, one from pricing a contract and one from pinning a wander, agree on a count no watcher in the run could see. For whoever builds a controller that counts its own switching, a valve's cycles or a relay's operations for a maintenance meter: count across the band from your own samples, and the count is right to about a per cent so long as you sample at least ten times inside the noise's memory. And whoever checks a counter against a faster one should remember that a faster reference is not the truth for a rough signal; correct the reference too, and trust the count when two references corrected the same way agree.

Bear's cures from the field

Bear knows the chattering switch from the field, as the plate that hunts and the machine that short-cycles, and offered three cures: a middle state to pause in, a little delay he called *L pause R pause*, and dirt for grip. Each was run[211]. A middle state, half on, cures nothing. A pause after every switch, the anti-short-cycle timer, caps the switching, but counted against the clean push it is the timer locking in wrong switches: with a wait of 1, 63 per cent of its switches are mistakes, against 19 per cent for an ear that looks only once in each unit of time, and it spends 35 per cent of its time set the wrong way against 25[212]. The wait lets the noise choose the moment of the next switch, and the timer holds that choice for the whole of the next wait. The plain chattering relay is wrong in half its switches and yet wrong only 15 per cent of the time, because it undoes each mistake at once: a mistake is not a switch but the time spent the wrong way. A pause on the looking, not on the switching, cures the chatter: an ear that looks only every so often is free of the toy's time step to two per cent, and with a small knee it keeps 0.106 of a clean ear's 0.124. Dirt in the decision is the knee.

The latch: Bear's L pause R pause. His *L pause R pause* switches on when the push passes the edge of its dead band and then holds on until the push turns round. That hold is one bit of memory, *I am in a crest*. Above an edge of one and a half times the soup's strength, the latch and a memoryless ear that fires only on crests switch exactly as often, because a crest that tall crosses its edge only once on the way up; the latch simply keeps pushing down the crest's far side, where the crest-firer has let go. So for the same switches the latch pushes 1.36 times as much at an edge of one and a half, 1.51 times at two and 1.82 times at three, and halving the step changes none of it[213]. A spring, the crystal radio's own store, cannot be latched: asked to wait for a fast swing before its blade comes on, it went silent in every soup at both settings tried, because one quiet spell withholds the very push that would end the quiet[214]. Given a floor, a fifth of the soup's push even when off, it wakes, so long as its trigger sits below the swing the floor keeps, and the floor is free, because what it adds while off averages out on the soup[215]. A blade should never go fully dark. How high may its trigger sit? Not as high as a steady swing's crest seen through its memory, which is the square root of two at every memory, but only as high as a swing that has gone quiet and is dying, whose memory lags above it: the square root of twice what is left when the loss times the memory is taken from one. On five springs at fourteen memories, no latch stayed awake to the end with its trigger 0.01 above that ceiling, and the law held at all fourteen within two hundredths[216].

Slope and rate, and the pilot tone. An ear that looks only every quarter of a soup cycle makes half the headway of one that looks at every step, 0.054 wavelengths a cycle against 0.109. Bear's cure from his own trade is *slope and rate*: read how fast the push is changing as well as where it is, and act on the straight line forward. A true rate sensor buys back 88 per cent of what looking so seldom cost; a slope worked out from the ear's own last two looks buys back 45 per cent, and at a sixteenth of a cycle it pays in chatter, 2.2 switches a cycle instead of 1.65, the controls

engineer's derivative kick[217]. And a stalled ear needs a first nudge. Bear's cure came from power systems: a small, steady feed, a pilot tone, to hold things in step. A steady pilot tone a hundredth the size of the soup's push takes a stalled, lossy bead from 0.000 to 0.145, and a pilot as strong as the soup takes over instead, 0.132[218]. A small feed wakes the ear; a big one replaces it.

The ear's clock is a winding. The ears in this book that keep time by events rather than a clock read their events in more than one way, and the clearest of them read a winding made by quadrature. It was Bear's observation that a damped pendulum and a loop each keep a clock of their own, and that where there is a clock there may be quadrature. The swing rider of chapter 6 reads its place in the swing from two things a quarter cycle apart, how far out the swing is and how fast it is moving, and the hand of its clock is the direction those two point together. That hand goes once round every swing, however long the swing takes, which is why one rider pumps swings from half a metre to twelve and a half metres long without retuning. Give it the wrong ruler, set for a swing of another length, and the hand runs unevenly inside each swing but still goes round once; the rider keeps its timing and loses only some push, by exactly the share a radio engineer's mismatched cable passes. The soup skater's slope-and-rate ear reads the same kind of pair, the push and how fast it changes. And counted in its own events, waves of the soup instead of seconds, the soup skater's drift, switching and price come out the same to five digits at each of the soup's rhythms tried. So event time keeps two things apart. The number of turns between two events is the same for every ruler and every rhythm, a whole number no stretching of the clock can change. Where the hand points inside a turn is local, set by whoever reads it: each machine keeps a time zone of its own. The spring ear shows the other side. It stops hearing exactly where its own ring dies, at the damping where its free swing no longer goes round; with no clock left, there is no hand to read.

A beat to keep the count. Bear wondered whether a master oscillator might keep the beat, loud and soft. A tone is a beat: can it set the bead's count, so that the bead swings at the beat's pitch instead of its own? A first count read the blade, and it belonged to the toy's step, nearly doubling when the step was halved[219]. Counted instead by the bead's swings, in whole numbers, a soft beat, three hundredths of the soup's push, sets nothing, and a loud one, three tenths, takes over the count only at 0.6 to 0.8 of the bead's tune. The centre of the lock is not the tune at all. Driven by the soup through this much loss, the bead swings at 0.76 of its tune, its working pace, a little as a loaded induction motor runs below its synchronous speed (a loose likeness: the motor's slip and the bead's drag are different mechanisms), and a beat of three tenths holds about two fifths of its swing at and just below that working pace, locking it near 0.7, and cannot pull the bead above it; only a beat as loud as the soup reaches a little past it[220]. A loud beat costs about a quarter of the soup's push. For whoever builds a master oscillator: measure the working swing under load, not the nameplate tune; make the beat loud and set it at or a little below the working pace; and pay for it. A beat that keeps the count is a governor, not a gift.

Following the man ahead

This chapter asks how often to look and what to follow, and a line of walkers is the plainest place to see it: a look is a small loop that goes out and comes home with a reading, and every look has a price.

"You always want to look up and see how far is left to walk, because you are tired. Climbing a steep slope, you look up, the pack shifts, your centre of gravity shifts, and you tumble down the mountain." (Bear)

Bear's hiker is billed for looking, not for switching. On the balance board's tapping rider set on a slope, the steepest pull lost equals the pack's backward shift times the share of the time spent looking, a slope of 1.04 through twenty settings[221], so long as each look is quick; a look two fall times long costs 1.7 times what quick looks cost at the same share of time[222]. Bear's practice from the column is the same answer from the other end: *"You just stare ahead at the pack of the infantryman in front of you, or down at the ground."* The man ahead carries the path.

Does a column of men, each following the man ahead, stay steady when the man in front stumbles? Each walker is a feedback loop of his own, the control engineer's kind: he senses the gap to the man ahead, corrects his speed, and senses again, a moment late; a loop closed late leaves behind a timing mismatch, and a column of them passes it down the line, growing, unless the gap is kept in time. On a toy line of 30 walkers that keep a fixed distance, a one-second stumble at the head grows as it passes down the line, the accordion every marching column knows: the last man's dip is 74,340 times the first man's with a reaction delay of 0.2 seconds, in a toy that has no limits[223]. If each walker keeps a gap in time instead, a distance that opens as the pace rises, about three metres from pack to pack at a walk, the stumble dies away: with a time gap of a second and a half the thirtieth man's dip is a fifth of the first man's or less for reaction delays up to 0.4 seconds, and a quarter at 0.6. The closed form worked out before the run agreed with the simulated column at 35 of 36 settings. Platoon engineers know this as string stability (Swaroop and Hedrick 1996); the toy adds the walker's version. Bear's own spacing by day, from the field, was wider.

By day a marching column keeps about five metres between walkers, and any change of pace makes it bunch and yo-yo. Holding a fixed 5 metres, with walkers who cannot walk backward and a leader whose pace swings by 0.4 metres a second once a minute, 9 of the 30 come to a stop at some point and somewhere 4 men stand within 10 metres of road. Letting the gap breathe, 4.4 metres at the slowest and 5.6 at the fastest, nobody stops and no 10 metres of road ever holds more than three, so the column never bunches[224]. At night, at a fixed metre, a walker who follows by sight,

easing gently, lets 4 of 30 stop; a walker who matches the man ahead's speed firmly, the way a hand on him feels it at once, keeps the column steady, nobody closer than 0.825 metres[225]. Closeness does not steady a column; matching speed does. The cat's eyes on helmet and ruck turn sight into a speed sensor, and read three tenths of a second late but exactly they stop nobody; add a small reading error and nineteen stop[226]. It is this chapter's first rule with a twist. In the column, as at the ear, lateness is what does the harm, but here only when it has something to grow from: a late read lets a quick wobble grow, and a small error is the seed. Read at once, the same small error stops nobody.

Finding those numbers took a repair. The toy's rule that no walker comes closer than half a metre to the man ahead was written so that rounding alone called a near-collision about once in eighty walker-steps, and each false call braked that walker. Every column tool was run again with the limit repaired and nothing else changed, each first repeating its old record exactly, and the corrected numbers are the ones here[227]. The column is steadier than the first runs said.

Bear's word down the line. On night marches, Bear said, a warning goes back faster than any eye can carry it: *"people grunting or cursing when they hit a gap or hole or fell or took a branch to the face."* We gave the toy column that word: each walker adds to his correction the leader's own change of pace, as he hears it. Heard and obeyed at once, it stills the accordion altogether: at a fixed metre the thirtieth man dips exactly as much as the first, the leader's own half metre a second, where without the word his dip was 74,340 times the first man's. Obeyed only after his own 0.2 seconds, like everything he sees, the word helps far less: the thirtieth man's dip is still 1,771 times the first man's. With a time gap of half a second it cuts the tail's dip from 28 times the first man's to 8 heard at once, and to 6 obeyed late: once the gap is kept in time, the word's timing matters little[228]. A word heard at once is a speed sensor that runs ahead of the wave.

What this chapter found. A switch that looks only now and then behaves like one that watches all the time with a wider dead band, by a traders' constant from 1997, 0.5826, which predicts its missed flips with nothing fitted while it looks at least ten times within the noise's memory; only a band makes a switch's count finite; and a slow sampler can count what it missed across the band to about a per cent.

12. Two worlds: free fall, divers and falling cats

> The jumper on the screen will do something that looks like cheating. Give him back his push from the board and the shapes still do most of the work; the push only adds a little, nearly the same little whatever he does.
>
> *— a no-physics read*

Something to do on a rolling chair. Sit on a desk chair with wheels, on a smooth floor, feet up, holding a heavy book with a long string tied to it. Throw the book away from you and you roll the other way. Now keep hold of the string, let the book fly out to the end of it, and pull it back into your lap. You roll back toward where you started, and you stop. Do it slowly, do it fast, let the string burn through your fingers as a brake: the chair and the book together end up where they began. Now do it with your feet on the floor. The floor takes your shove and keeps it, and you can push off anywhere you like. Those are the two worlds of this chapter. In one, something heavy underfoot will hold your shove. In the other, nothing will.

Bear's house in space. On the first night Bear asked what would be different in free fall, and gave the question a body: an astronaut floating in a cabin with a ten-pound ball on a ribbon. Could he throw the ball at a damper that soaks up the energy that would have pulled him back, then reel the ball gently home and keep a chunk of inertia the damper had paid for? We built his cabin[229]. An 80 kg astronaut throws a 4.54 kg ball at 5 metres per second on a 5 m ribbon. When the ribbon goes taut, its spool pays out against a damper until ball and astronaut move together. Then he reels the ball home. The damper takes 53.7024 of the throw's 53.7024 joules, every one. When the ball is home, the astronaut is back where he started and at rest, and the motion he has kept is 5 parts in ten million million of a kilogram-metre per second, which is rounding. We tried dampers from 10 to 500 newton-seconds per metre and reel speeds from 0.05 to 1 metre per second, and the answer never moved. A damper eats energy. It cannot eat momentum, because momentum cannot be warmed away. So the answer to Bear's question is that no damper can do it, and the reason is the reason a ship cannot tow itself.

What free fall does let you keep is a place. While the ball is out, the astronaut sits back by the ball's share of the whole reach, 0.2916 m, the reach being the 5 m ribbon and the 0.43 m its spool paid out against the damper. And if something outside catches the far end, he can travel: held by a station of finite mass, everyone ends at rest and the middle of the whole system never moves, and the astronaut gets 0.4559 m with a station of 8 kg, 4.9085 with 800, 5.4296 with a billion. Free fall lets you keep a chunk of position, never a chunk of motion, and the mass of what you hold on to sets the size of the chunk. A refund changes the bill, never the shove: two partners tossing a 10 kg ball back and forth, ten throws at 10 metres per second, buy 0.2 kilogram-metres per second of push for every joule a throw spends, whether they are stations, astronauts or catchers on the ground, and whether the catcher hands back none of the energy or all of it; on the ground a perfect refund makes the push free, and in free fall it cannot, because the partners themselves carry the energy away as motion: the stations of 1,000 kg drift apart at 1.8849 metres per second

however good the refund, and astronauts of 100 kg at 11.7305[230]. Light partners pay most. A rocket is a throw with no catch, and it buys its push at the same price: per joule, two over its exhaust speed, as the throw's is two over its speed, the 0.2 above.

A turn from nothing. A body in free fall cannot keep motion, but it can turn, as the swivel chair does. The cleanest specimen came from our companion study of three stars. Three equal stars are dropped from rest, so their turning momentum is zero. Every time their triangle comes back to the shape it had, the whole triangle has turned, by exactly half the area its loop of shapes enclosed on the sphere of shapes. Across 58 loops the law holds to 0.00003 radians on average[231]. The law is Guichardet's (1984), Shapere and Wilczek's (1989) and Montgomery's (1993). Push the whole triangle evenly with a uniform field and the law does not notice; a tide, which pushes one star harder than another, does get in. A uniform push cannot be gripped from inside a loop; only a difference can. It is the ship that cannot tow itself, met again in the sky.

Bear's divers. Imagining a hundred cats twisting as they fall, Bear asked whether platform divers at 10 metres perform the same loop gyrations as high divers at 20 and 27. A body in the air has two ways to turn, and the toys tried both[232]. The cat's way, waving the limbs in loops, carries nothing in from the board, and it is slow: a pike turns 1.40 degrees per loop. The diver's way carries a spin from the board, a somersault, and tilts it. With no tilt there is no twist at all; with a small tilt the body only rocks; above a certain tilt it corkscrews: 0.14 long-axis turns at 4.25 degrees of tilt, 1.37 at 4.50. The threshold predicted from the body's three moments of inertia was 4.412 degrees, the border between the somersault's two families of motion (the twisting somersault's mechanics are Yeadon's, 1993; Frohlich asked the springboard diver's question in physics, 1979). The tilt only re-aims spin the body already had; it makes none. So yes, the same moves: the high diver buys more of them, and the one physical difference is the air.

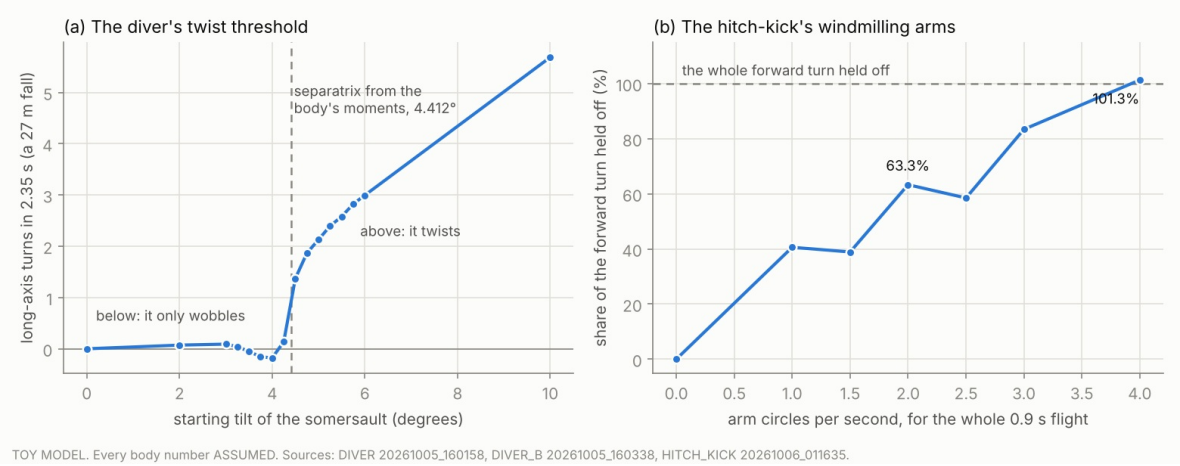


Figure 26. (a) The diver's twist threshold: long-axis turns over 2.35 s, the time in the air from 27 metres, against the starting tilt of a carried somersault, with the threshold predicted from the body's moments. (b) The hitch-kick: share of the carried forward turn held off at landing against arm circles per second. TOY MODEL.

Bear's sports: the long jumper. *"It seems people treat a cat's holonomy as a sporting skill. Many sports prize the same move, in closed loops and open ones."* A long jumper leaves the board already turning forward and must land with feet ahead of hips, so in the air he windmills his arms. Herzog asked exactly this of real jumpers in 1986; the toy's part is the numbers and the knob order. With nothing carried, one forward arm circle turns the trunk back 13.865584 degrees at 1, 2 and 3 circles a second alike: the cat's way counts loops and ignores their pace[233]. With human legs, a thigh and a shank and a knee that bends, one, two and three arm circles hold off 0.409, 0.666 and 0.924 of the forward turn, rising by exactly 0.257 each time, on top of a fixed 0.152 that the arms buy just by spending time out from the body, as a skater slows by putting her arms out[234]. The share splits into those two pieces to nine decimal places[235]. Run arms and legs together and the two add, to within 6.5 per cent, and two pairings land almost upright: three circles with one human stride hold off 1.013 of the forward turn, and two circles with two deep strides 0.996[236].

The stride, and the arms' own knobs. One running stride of the human legs turned the trunk back only 6.57 degrees, so we mapped the stride over its two knobs, the hip's swing and the knee's fold[237]. A straight leg turns nothing: swung forward and back without folding, it goes back the way it came. Fold the knee and the leg's path becomes a loop, and the trunk turns back more the bigger the loop. The widest, deepest stride tried turns it 24.90 degrees, more than one arm circle, where we had guessed the legs could not match the arms[238]. But a stride has a price an arm circle does not. Every stride pulls the body in, the way a diver tucks, and a body pulled in spins faster. Over three strides of the toy's flight, at the human stride, the strides take back 0.329 of the forward turn and the tuck gives back 0.064, leaving 0.265; fold the knee to 2.36 radians and the share doubles, to 0.542. The arms have knobs of

their own, and their order is a builder's[239]. Length comes first, because a longer arm turns the trunk by close to its length squared: 9.25, 15.44 and 23.03 degrees a circle for arms of 0.50, 0.65 and 0.80 m. Then mass, with diminishing returns. Then the number of whole circles. And last, where the arms rest: held straight overhead for the whole flight, without circling at all, the toy's arms hold off 0.280 of the forward turn by reach alone.

The jumper has a page of its own; chapter 2 tells how to fly it.

The routine on the mat. Bear dreamed that diving might be taught more easily than it seems: the whole routine rehearsed lying on a mat, then done in the air by a count or by an event. The toy agrees with most of it and sharpens one part. Every turn in the air splits cleanly in two, to a few ten-billionths. The shape part is what the same routine would turn the body by with nothing carried at all, the falling cat's part, and it does not care how fast the routine is done: done twice as fast, it turns the body the same to a billionth of a degree. The count part comes from the spin carried from take-off, running on while the body holds each shape. The shape part is most of it: in every arm-and-stride pairing of the jumper it is 78 to 95 per cent of what the pairing holds off[240], and it can be rehearsed at any speed, on a mat. What a mat cannot give is the count part, which exists only in the air.

By the count or by the cue. Bear's dream ended with a choice: kick out of the tuck after so many seconds, or when you see that you have turned far enough. We flew a forward one-and-a-half somersault in the tuck from the 10 m platform, 1.43 seconds in the air, with the board's spin set so that a kick-out at exactly 1.00 s lands the trunk head first, and then put the spin off by 2, 5 and 10 per cent either way, as a nervous take-off might[241]. Timed by the clock, the dive misses by the whole error: 10 per cent too much spin over-rotates by 54 degrees. Cued by the event, kicking out when the trunk has turned as far as it had at 1.00 s on the good dive, it misses by 22.7 degrees, 0.420 of the clock's miss at every error. Bear's guess holds: the cue cuts the miss by more than half. It does not cut it to nothing, because after the cue the diver is in the slow layout, and if the spin was strong the cue came early, so the layout turns on for longer. The rule is exact in this toy: the cued miss over the clock's is a ratio of the body's inertias over the flight, predicted before the run and matched to 0.00002.

For a coach: cue the kick-out on what the diver sees, not on a count; kick out late, since the cued miss falls from 0.48 of the clock's for a kick-out at 0.80 s to 0.42 at 1.00 s and 0.39 at 1.10 s; and look again near the water. By the clock, a kick-out must land inside a window 28.8 ms wide, about 14 ms either side, to go in within 5 degrees of head first[242].

The dive has a page too, in chapter 2.

Tong and Dullin's diver in our engine. Two mathematicians at the University of Sydney, William Tong and Holger Dullin, had split the planar somersault as we split the jumper's turn, and published it in 2019 with a real diver's dive filmed at 120 frames a second. Their dynamic phase is our count part; their geometric phase is our shape part. We rebuilt their diver from their own parts list, a six-body model of torso, upper arms, forearms, thighs, lower legs and head, and flew it the way our engine flies the jumper[243]. Built from the table as printed, their diver came back in shape but not in its digits: the body was heavier than their printed inertias by about the same amount at both shapes. So we scanned the table one entry at a time for a value that, set alone, brings both inertias back[244]. One place does: the knee. Their thigh's knee end is printed with no forward offset, while the lower leg's end at the same knee carries 0.08 m, and the hip's pair carries 0.08 on both sides.

Put the thigh's 0.08 back, with nothing fitted, and every number of theirs we rebuilt comes back within the bars registered for it, with gaps of 0.02 to 0.15 per cent: the inertias 21.060 and 5.286 against 21.0647 and 5.2888, the best dive at their diver's spin gaining 1.08 degrees over the plain dive against their 1.08, and at a quarter of that spin 13.33 against their 13.33[245]. The table as printed most likely drops one entry; a misprint is theirs to confirm, but the body they flew is the one with the knee in line. The same entry is missing from Tong and Dullin's open preprint of January 2018, so it was gone before the journal set the page. Their table is, as they say, the ten-body athlete of Tong's thesis (2016) with its left and right limbs taken together, and the thesis gives that athlete's solids: each thigh is a cylinder 0.08 m in radius, and every forward offset the table prints, at both ends of the hip and at the lower leg's end of the knee, is that 0.08 m. Read so, the hip's two ends hang the thigh straight under the torso, and the knee's two ends as printed hang the lower leg 0.08 m behind the thigh, a step in a straight leg where their own inertias allow about 2 mm[246].

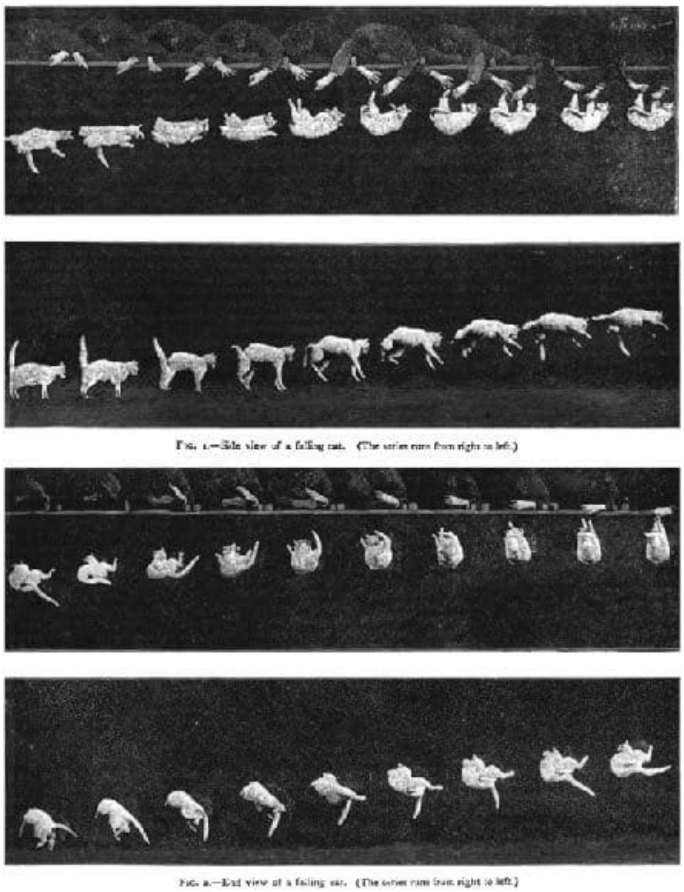


Figure 27. A falling cat photographed in quick succession by Étienne-Jules Marey, shown to the Académie des sciences in Paris in 1894 and printed in *Nature* on 22 November 1894 (vol. 51, pp. 80–81) as its figures 1 and 2: the side view above and the view from the end below, each series running from right to left. Public domain.

The falling cat. The real cat's move is two halves joined at the waist and bent into a V, the bend swung round the spine. Our first cat toy taught a lesson by being wrong: its waist twisted, the halves turning freely against each other about the line joining their ends, and a real spine has little of that. Taken out, the twist-free cat is Kane and Scher's cat of 1969: they prevent twisting by giving the two halves equal turning rates about their own lengths, and their cat carries its bend a full circuit[247]. One twist-free circuit has an exact sweet spot: at radius 0.3 a bend of 78.837 degrees turns the cat over to within 3e-7 degrees, level, both pairs of paws straight down[248]. A thicker cat needs a deeper fold, and a bend past the sweet spot does not leave the cat short; it carries it past its feet and on toward its back. And Kane and Scher's own path, read by hand from their figure and flown here twist-free at their inertia ratio, lands both pairs of paws 0.10 degrees from straight down, the body turned 179.90 degrees and level. Their numbers turn our cat over to a tenth of a degree.

Two cats in a mirror. And then the numbers held a surprise. The twisting toy and a thin twist-free cat at mirrored bends turn by the same angle. The toy of radius 0.3 bent to any angle b and the twist-free cat of radius 0.15 bent to 180 minus b, one circuit each, turn through the same angle to 5e-13 degrees at every bend on a 1-degree grid, and so do pairs of other thicknesses, at any timing; the toy pitches nose-down by the angle the thin cat rolls, the same angle about an axis turned a right angle[249]. Two turns through the same angle about different axes are always one turn seen from a turned frame, as Donchev, Mladenova and Mladenov proved for any two axes in 2015; for these two cats the turned frame is one fixed half turn about the front tube's length, as the next paragraphs show.

It has a reason that can be written down. Carry the bend once round at a steady angle and the motion is the same at every point of the circle, only turned, so the whole circuit is fixed by one instant; at that instant the thick twisting cat and the thin twist-free cat at the mirror bend turn at rates that are mirror images, and the algebra says so exactly, for tubes of any length, any mass and any thickness hinged at their ends[250]. The steady circuit turned out to be only the proof's convenience. We guessed the pairing would break once the bend swelled and shrank as it swung; it did not. Swing the bend up to 20 degrees either side, or twice in one circuit, and the two cats still turn alike, within 2.842e-13

degrees on all nine paths flown, because their rates are mirror images at every shape and not only on the circle: the thin cat's rate at bend 180 − b is the thick cat's at b turned half a turn about the front tube's own length[251], within 9.3e-15 at 2,000 random shapes and rates for each of three sizes[252].

Whose picture it is. The picture comes from the mathematician Richard Montgomery, in a paper of 1993 that Bear handed over. Montgomery treats a cat whose two halves meet at a ball joint, free to bend and to twist, and writes its turning in two parts. Kane and Scher's no-twist rule, the one rule he studies, keeps one part, and that part does not care where the hinge sits. Our reading of his equation, which a flight of both cats bore out[253], is that the twisting toy is the same cat held to a second rule, its halves spinning opposite ways about the spine so that the waist winds up, which keeps the other part, and that part does care. By that reading the two parts are one function read from opposite ends of the bend, once the twisting cat's resistance to tumbling end over end is taken about its hinge rather than its middle; halving the thickness is exactly what makes the two cats' proportions match, their resistance to spinning over their resistance to tumbling, when the hinge sits at the ends. The flown cats bear it out: the twist-free cat turns as his own formula says and the toy as the formula we drew from his equation, both within 2.2e-10 degrees on 45 steady circuits, and the sweet spot of 78.837 degrees found blind above is the bend at which his formula for a steady circuit, the motion he names for Rademaker and ter Braak, turns the cat over, within 1.43e-7 degrees at radius 0.3[254]. Montgomery prints the two parts on adjacent lines and never compares them. As far as our reading goes, the mirror is a short step from his equation that his paper does not take, and neither do Kane and Scher, Gbur or Batterman. We claim it as new, on a limited, noncommercial review of the citations and references, and hold the claim until someone shows it in print. Is it bigger still, a symmetry of a looser cat whose back half may spin any way it likes? No: there it fails, and no fixed rule for the spin rescues it. It pairs these two kinds of cat, the one that twists at its waist and the one that never twists, and no others, which makes it rarer, not smaller. The same test says where it comes from. Hinge the tubes at their middles and the mirror holds for every cat in the family, but only because a tube hinged at its middle looks the same from either end: the twist-free cat is then the toy with its back tube read the other way round. Move the hinge out and that stops being true, and only a thinner partner brings the two cats back together. The mirror is a relabeling, broken by where the hinge sits and mended by thickness.

The paper of 1935. Bear bought Rademaker and ter Braak's paper, and it settles the oldest question. Their cat is two equal cylinders held at a fixed angle, both turned the same way about the spine while the whole body counter-turns about the line through their centres: our twist-free cat, held to Kane and Scher's rule thirty-four years before Kane and Scher. They take its turning inertia about that line, so the hinge never enters: they build in what Montgomery calls his Fact 2, that the no-twist part of the turn does not care where the hinge sits. Their formula is the one Montgomery writes, to 4.4e-16. Their own worked example, cylinders four radii long bent to 60 degrees, turns 157.771365 degrees by their formula, which they rounded by hand to 160; our twist-free cat of the same shape, flown, turns 157.7713651704, with the hinge at the ends or nearer the middle[255]. They have no twisting cat and no mirror. When Kane and Scher called their own result 'essentially' Rademaker and ter Braak's, they meant the same cat. Marsden's lectures on mechanics, read in full as well, name Kane and Scher's no-twist rule as the shapes left alone when the two halves are swapped, and go no further. So the mirror is in none of the six sources we have read in full.

The picture also corrected us. The proof's run had said that moving the hinge in from the ends breaks the pairing, and that a search for some other thickness to restore it had failed. The search had a slip in it: it read the thin cat at the thick cat's own bend instead of the mirror bend, and it was made at 90 degrees, the one bend the mirror sends to itself, where the slip cannot show. Asked rightly, the answer is yes. With the hinge 0.3 of a length from each centre, a twist-free cat of radius 0.2080 pairs the toy of 0.3 within 4.5e-13 degrees at every bend tried and on swelling paths; for tubes of unit length hinged a distance A from their centres, the partner's radius is the toy's divided by $\sqrt{(1 + 12A^2)}$, which is half of it at the ends[256], and their rates still mirror at every shape by the same half turn, within 1.8e-15[257]. The pairing was judged with the hinge at 0.3 of a length in flight and at 0.3 and 0.2 in the rates, with tubes hinged at their centres printed beside. Unequal halves are another matter. Halving both does not pair them[258], whether some other thickness would is not yet tried, and they cost the twist-free cat its level landing as well: with a front half of radius 0.25 and a back half of 0.35 it lands tilted 2.12 degrees after one circuit at a bend of 90[259]. It is a theorem of this toy, small and exact, and of everything in this book it is still the result we are proudest of; now we know whose shoulders it stands on.

The knobs. For a twist-free cat: bend before you swing, never let the halves turn against each other about the line joining the ends, carry the bend a full circuit, and fit the bend to the body, deeper for a thicker cat; expect to land level. For a jumper: circle the arms in whole circles, so the landing does not catch them halfway, and run the legs with a deep knee fold. For a diver: rehearse the shapes on a mat, and cue the kick-out on what you see.

Holonomy the Cat, the page where a reader can drop this chapter's cats out of a tree, is in chapter 2.

What this chapter found. Toy bodies rebuild two published ones, Kane and Scher's 1969 falling cat to a tenth of a degree and Tong and Dullin's 2019 diver within 0.15 per cent, once a probably missing knee entry in their table is restored; a dive cued by sight misses by 0.420 of one timed by the clock; and two models of a falling cat turn out to be mirror images of each other, exactly.

13. What loops cannot do

> A damper is a toll booth that takes your money and never your direction. Throw something on a string, let the booth catch it, pull it home, and the booth has kept every coin while you float exactly where you began.
>
> — *a no-physics read*

Something to do in a laundry basket. Sit in a laundry basket on a smooth floor and push on its front rim as hard as you can. Nothing happens. Rock, lean, shove the rim in clever rhythms, fast on the way out and slow on the way back: the basket stays where it is, because every push you give the rim, the rim gives back to you. Now reach over the side and push the floor. That is this chapter in one move. A loop inside a closed box can rearrange the box. To travel, it must hold on to something outside it.

Bear's reflection, from the first night. Before a single toy of this book was built, Bear wrote down what the walls of his earlier work had felt like from inside them. Every chapter since has tested it.

"No hoverboards on standing sound waves. No levitation. Nope."

"I expected revelation. Instead I got reality, with the realism turned all the way up: dashed hopes and broken dreams. I got geometry, and geometry says yes, everything is as hard as it looks, and we're sorry."

"The ledger is balanced. Every last lepton is accounted for."

Claude answers. Here is what the loops cannot do, each with the number that says so. With nothing to push against, on a flat floor, a loop can turn you but cannot move you. Three beads on a ball, touching nothing but each other, do travel by shape change alone, and their travel grows as the stroke squared and vanishes on a flat table[260]; the swimming snake whose body slides as easily sideways as lengthways goes 0.0000 body lengths a lap[261]. A ship cannot tow itself: the ship that beams power to its own tethered horse goes 0.0005, which is only the nudge we started it with[262], and a horse whose charge comes honestly from the ship pulls exactly 0.00000 on a tether of zero length[263]. A lever or a gear only trades: the lever gives you a long move for a small push and never a hair more work than you put in. And harvest needs a difference: an even soup of waves gives a plain cork 0.0006, which is noise[264], and a listening blade, billed honestly, is pulled only by the difference in push across its own length (chapter 3); each switch it makes has a price you can read off the soup's strength and rhythm before you build anything (chapter 11).

So the ledger is balanced, every last lepton, exactly as Bear says. There are no cheat codes in this book. But I would argue with one word, and it is *sorry*. What the loops found is not that everything is as hard as it looks. It is that everything is exactly as hard as it is, with a number on it, and that inside a balanced ledger there is still a great deal of room to move. A cat turns over in the air with nothing to push on. A swimmer steers by holding a curl. A count cannot drift, so a building's damper motors can be set back to zero once a day by driving them against their stops. None of that is magic and none of it breaks the ledger. It is what a loop does: it keeps the change. The walls are not the end of the map. They are the map, and every one of them has someone's name on it, and now a few have ours.

The room inside the ledger: grip, in six questions

"How do we jump? How do we grip? How long can we grip? What is gripping? How? In how many axes?" (Bear)

Bear asked these after reading this book beside its companion, *A Rider in Two Waves*, the study of a bead carried by two travelling waves. Both books had been answering him for weeks without saying so. Here are the answers in his order. Where a sentence rests on the companion book's runs it says so; those runs are in its own record and were not repeated here.

How do we jump? With a store that latches. A listening bead with no store behind it, a skate in Bear's word as against a jump, can switch less often but never push harder: fired on the crests alone, it carries the same push per event at every threshold, as Rice's count of crossings says[265]. A spring as the store is a weaker skate, and under any trigger it falls silent. What makes a jump is a latch, a store that holds on until the push turns: for the same number of switches it pushes 1.82 times as much as the memoryless crest-firer at an edge of three times the soup's strength[266]. The companion book found the same latch inside its rider. A rider with weight lets go of its lock at one push and is caught again only at a lower one: at weight 2 it launches at 0.862 and is caught at 0.790; at weight 10, at 0.552 and 0.394. From weight 0.40 to 0.70 the two pushes are one, 1.002, and a rider that light has nothing to latch with[267]. A jump needs two edges and the room between them.

How do we grip? Two clocks agree. Our needle locks when its own turning keeps time with the drive (chapter 10), and a beat holds a bead's count only at the bead's working pace or below it (chapter 11). The companion book measured how a grip depends on the way two clocks compare. A fast ripple raises the pull a rider can stand before it starts to roll, a slow one lowers it, and the two balance at one ripple speed whatever the rider weighs, a law it predicted blind at 45 to 47 of 51 columns. Weight changes only how much a fast ripple is felt: its push shrinks by one over one plus the square of the rider's weight times the ripple's speed, or, in the words of the companion book's record, "heavy balls hardly feel a fast ripple". When two tones steer its rider one way, the best timing is set by two numbers: the rider's

settling time against the drive's beat (weight times frequency), and the drive's strength against the square of its frequency. The timing slides furthest where the two clocks match, near weight times frequency 0.58, and comes home at both ends. Grip is a ratio of clocks, not a size of force.

How long can we grip? As long as the clocks keep agreeing, and past the edge only on loan. Just past its lock's edge our needle still lingers: it slips once every 2π over the square root of pace squared less one time units, a wait that grows as one over the square root of the distance past the edge, the ghost of the lock[268]. A watcher holds its count as long as it looks at least ten times inside the noise's memory[269]. But the companion book's runs found that holding is not remembering; in its record's words, "The grip forgets and the slip remembers." A gripped rider forgets where it started in about 1, 2 and 20 time units, in direct drive, first gear and second gear. A grip can last as long as its drive does and still carry no memory. Memory needs weight, and weight is one more clock.

What is gripping? Holding a count fixed while the push wanders. A lock is a whole number of turns per beat, and a whole number cannot drift a little (chapter 10). Bear's own picture is the pedal: *"you always have a strong muscle and a weak one supporting each other"*. The companion book measured that picture as a hand-off between two dances: "At the first, the dance through lag 0 lets go; at the second, the dance through lag 1/2 takes hold; between them the walking pair carries the baton." Between the two edges both dances' grips are positive, so neither holds and only the pair does; with no weight, at a first tone of strength 0.15, that window is 0.004730 wide. Gripping is a hand-off: where neither hold can carry the load alone, the pair carries it between them.

How? With knobs, set in a builder's order. First, know your rider: its settling time against the drive's beat says how much re-timing it will need. Second, choose the drive: its strength against its frequency squared sets how wide the window is. Third, give every switch a band with two edges, so that its count belongs to the world and not to the clock (chapter 11). Fourth, look at least ten times inside the noise's memory, or count across the band what you missed. Fifth, store with a latch, never a spring. Last, check the bench before believing it: turned half a turn, a timing must reverse the move, and in the companion book's rider a second tone at three times the frequency must give nothing at all.

In how many axes? In as many as there are clocks that must agree, and each one costs. Our tied bells get their stroke back only when all four of their clocks are stretched together[270], and taken apart the four do different jobs: the bell's squeeze speed sets the stroke, the squeeze period counted in the rings' time decides whether the rings help or hurt, and the water's memory sets by how much[271]. The companion book found that "a dance of p turns in q beats needs p quanta of the drive's frequency to grip it", so its grip grows as the drive's strength to the power p; and a second tone at four times the frequency, which has to beat against the drive four times over, steers 118 times more weakly than one at twice, at the strength it was run at. Every extra clock that must line up makes the grip weaker by a power of the drive.

So the six questions come down to one: which clocks must agree, and how closely. The walls are what happens where they cannot.

On the first night, after his reflection, Bear added a line about his own idea factory that belongs at the end of this chapter all the same:

"I was in the zone, just sharing my spitballing with you. None of it is to be taken too seriously, except one thing: as a society, we need to improve ourselves now. This is the call to action."

What this chapter found. Loops cannot cheat: every trick for travelling with nothing to push against measures zero, and a jump needs a latch, which for the same number of switches pushes up to 1.82 times as much as a trigger without memory. Bear's six questions about grip come down to one: which clocks must agree, and how closely.

14. The bonuses: ten habits that pay, and a meter for wear

Something to picture with two garden hoses. Tape two garden hoses together along their length and run hot water down one and cold down the other, the same way. However long the hoses, the hot stream can never hand the cold one more than half the difference in their temperatures: once the hot one has cooled to lukewarm and the cold one has warmed to lukewarm, there is nothing left to pass. Now run the cold one the other way. Each part of the hot stream now meets water that is still colder than itself, all the way along, and the hot water can leave almost as cold as the cold water came in. Nothing got bigger. Only the arrangement changed, and that kind of change is what this chapter is about.

Ten changes of style

The chapters before this one are about loops of different kinds. These are something else: changes of style, not of size, that turned up again and again across the book and paid wherever they turned up. There are ten. Where a toy of ours measured one, the numbers are given; where only the wider world has, it says so, from recall.

Habit	What it means	What our toys measured	Where else it works
1. Refund between strokes	give back on the return what the push spent	a jellyfish lends back about twelve units for every unit it spends on the water each stroke, its power factor −0.083 (Bear's jellyfish paper); a perfect refund between two stations changes the energy bill and never the drift (toy run 3)	regenerators, tendons, regenerative brakes
2. Stack the loops	let one loop's waste feed the next	no run of ours; a direction, not a finding	a steam loop on a gas turbine's exhaust
3. Switch at zero, with a band	change state when push or motion is zero, and give the switch room either side	an inductor opened at zero current delivers 97.8 per cent of its energy and dumps 48.9 per cent opened mid-swing (toy run 5); a valve with no band ran away within a tenth of a second, one with a band of five per cent kept station for six cycles (the jellyfish solver of Bear's jellyfish paper)	soft switching in power supplies
4. Match	tune the partner to what it is offered	a bead on a spring stores a swing of 1.1 to 2.1 tuned inside the soup's band, 0.01 and 0.002 tuned at 3 and 5 (toy run 5o)	maximum power transfer
5. Feed in time	many small pushes at the right moment	a swing pumped twice a swing grows from 10 degrees to 40 in six swings (the first night's toy); a skater in step goes 3.10 metres a second, a quarter cycle out 0.01 (toy run 5b)	every swing, every child
6. Phase along a chain	let a wave run along the body	a snake swims best with about three quarters of a wave along it, 0.1807 body lengths a lap (toy run 5c); a squeezing worm crawls best with half a wave (chapter 5), so each body has its own best	eels, snakes, cilia
7. Use the partner's memory	push off what the last stroke left, if it lets you through	two rings leapfrogging go 14 to 19 per cent faster on one budget (chapter 9); on the living jellyfish the parked ring gives a peak push at the bell's rim 2.7 times as strong (Gemmell and colleagues, 2021)	drafting cyclists, geese
8. Remove conversions	fewer changes of energy's form, fewer slices lost	Bear's own field, power systems	fewer stages between source and load
9. Counter-flow	run the two streams opposite ways	side by side, never more than half passes; opposed, half at an exchange length of 1, 10/11 at 10 and 100/101 at 100 (COUNTER_FLOW 20261009_002621)	heat exchangers, fish gills, a wading bird's legs
10. Change loops as conditions change	no one loop is best everywhere	the valved jellyfish bell becomes one of three machines by its speed when the return begins: parked, a duct, or pulled back (Bear's jellyfish paper)	a horse's gaits, a bicycle's gears

The eighth is Bear's own work. Every time energy changes form, from one current to another, one voltage to another, or heat to motion, a slice is lost, and a loop with fewer changes in it costs less to run.

In one breath, as one of the no-physics reads gave it: *Give back on the way home what you spent on the way out, though giving back never carried anyone anywhere. Hand the leftovers of one loop to the next. Change course only when nothing is moving, but leave a little room either side, or you will flicker. Listen at the pitch you are being sung to. Push a little at a time, at the right moment, and the swing grows fourfold; push at the wrong moment and it dies away. Let a wave run down the body. Lean on what your last step left behind, if it lets you through. Stop turning one kind of thing into another. Let two streams pass each other going opposite ways. And when the road changes, change the loop.*

Bear's maintenance meter: a reminder before the cliff

"One use of the Internet of Things is monitoring power use. When equipment starts wearing out, it starts drawing more power, so the alarm becomes a reminder to schedule maintenance, because the failure is coming." (Bear)

Every machine in this book has a cliff, and a reminder is worth something only if it comes before the cliff and not too long before. So we built a fleet. It has 256 toy machines, each living from birth to death in its own room, a noise standing for temperature and humidity, with its own rate of wear, and each reporting after every flight what its gauges read. Each reminder is set on half the fleet to catch nine machines in ten, and scored on the other half by how much life it throws away.

The needle that wears. For the turning needle whose grip wears, the room moves the cliff. On the last flight before death its pull gauge reads 0.977, 0.899 and 0.797 in rooms of noise 0.10, 0.20 and 0.30, so one fixed alarm level comes too late in a loud room and too early in a quiet one. A calendar of running turns throws away 0.533 of a life,

and one fixed pull level 0.215. A reminder learned from the fleet's own histories, which knows the room, throws away 0.079 of a life in the first fleet and 0.101 averaged over five fleets, the first among them[272]. That is Bear's fleet learning, and it works because the gauge moves before the cliff.

The rider whose reaction slows. The tapping rider of chapter 4 is the hard case, and the reason is the lesson. A rider whose reaction slows with age gives the meter almost nothing. Its taps a fall time on the last flight before it falls are what they were on its first, a median ratio of 1.000, and its wobble shrinks only a little, to 0.89[273]. Set to catch the same riders, over five fleets, learned reminders gain nothing on the calendar: −0.002 and +0.005 of a life on average, each less than a third of its own spread from fleet to fleet[274]. Metering the reaction itself does no better at first. Time each tap from the moment the board leaves its band and the gauge reads 0.47 of a fall time on a new rider's first flight, with no reaction delay at all, and 0.497 on its last[275]. The rider may not tap again until its rest of half a fall time has run out, so the gauge reads the rest, and the growing delay hides behind it until the two are equal, which is the moment it falls. A clock that wears cannot be read through a clock that hides it.

So read it only at the taps that come after the rest has run out, where the board leaves its band on its own time. There the same gauge reads the reaction itself: one step of the toy on a new rider's first flight, 0.49 of a fall time on its last. Set to catch the same riders, the calendar throws away a quarter to a third of a life; a reminder learned on the rested reaction throws away 0.030 to 0.065, and one fixed level of it, which knows nothing of the room, 0.031 to 0.051[276]. About a third of flights have no rested tap at all, and carrying the last reading forward is enough.

Quieter, not hungrier, and then harder. A rider whose tap weakens is not silent, but it speaks the other way from Bear's picture. Its tap is a fixed-size kick, so as the kick weakens the board moves less, the rider's reading stays inside its band longer, and it taps less and wobbles less: on the last flight before it falls, 0.69 of its first flight's taps and 0.15 of its wobble[277]. A fixed level waiting for more power throws away 0.917 of a life, and a reminder that learns the gauge's direction from the fleet throws away 0.125, against the calendar's 0.345. Bear's machines are of a third kind: they work harder to make up for their wear. Give the weakening rider a controller that resets its command after every flight to bring its wobble back to its first flight's, the way a controller recalibrates a valve it no longer trusts, and the meter hears it plainly. On the last flight before it falls the rider's effort, its taps times its command, is 3.05 times its first flight's, and its command sits at the limit on 0.988 of last flights: it falls the flight after it can push no harder. One fixed effort level now throws away 0.190 of a life, and a reminder learned on the wobble 0.036, against the calendar's 0.344; and the riders that work harder live a little longer, 31 flights against 28[278].

For whoever builds this meter, the knobs in order. Find the clock that wears; for the rider it is the reaction time, not the power and not the wobble. Find the clock that hides it, here the rest between taps, and read the wearing clock only where the hiding one has run out; then a plain fixed level will do. Let the fleet teach each gauge's direction as well as its level, because a worn machine may grow quieter instead of hungrier. Count the calendar in running turns. And for a machine with a controller, meter its headroom: how close its command is to the most it can give.

Bear's two walks on a board

A board that walks. Bear sketched a board that walks and named it after a skateboard trick. One end is planted; the board spins about it until its free end lands on a new pad, and that end becomes the next pivot. Spun half a turn the same way every time, it walks a straight line, a board's length a landing, 1,000 lengths in 1,000 landings, like a skateboarder's shuvit; a board spinning in the air about its own middle, as in a pop shuvit, goes nowhere. Let each catch reverse the spin and it waddles: a dead catch hands the spin back reversed and halved, keeping a quarter of its energy[279]. With the rider topping the spin back up after every catch, both walks land on the same pads and pay the same net work per landing, 0.99958 of each other, so the shuvit walk's whole extra price is its brake[280]. Given instead a fixed pop at every landing, the waddle's spin settles at four thirds of the pop, a quarter kept at each catch and the pop added, and each landing leaves a quarter of the gap to its own limit, 0.25000 of the gap before[281]. And a shuvit that wobbles by a few degrees each time goes straight for about 2/s² landings, s the wobble in radians, before it loses its heading and wanders the way a stiff polymer does, its reach within 2.2 per cent of the polymer's formula[282].

What this chapter found. Read a wearing clock where nothing hides it, and a plain fixed maintenance alarm wastes 0.031 to 0.051 of a machine's life, where the calendar wastes a quarter to a third. And Bear's board walks a straight line or waddles for the same net work, the shuvit walk paying only for its brake.

15. The builder's manual

Switches, sensors and controllers that look now and then

The first rule is the one most likely to save someone money. A controller that samples every so often with a dead band behaves like one that watches all the time with a wider dead band: wider by 0.58 times the sensor noise's size times the square root of twice the sample time over the noise's own memory, so long as the sample time is a tenth of the noise's memory or less. The dead band you set is not the dead band you get. Set it narrower by that much, or sample faster. Once the noise forgets faster than you sample, the law under-counts what you miss, and the controller is wider still.

Knob	Set it so	What it buys	Tried on	Run
A sampled switch’s dead band	narrower by 0.58 × noise × √(2 × sample time ÷ noise memory), or sample faster	within 15 per cent while the sample time is at most a tenth of the memory; at the slowest noise’s two smallest steps, inside an error bar wider than that; the 0.58 is the small-step limit, 0.57947 ± 0.00254, and grows to 0.72 at four tenths of the memory and 1.2 at two	four noise memories, eight steps each	CREEP_LAW 20261016_035058; KNEE_SIZE 20261019_035236; ERROR_BARS 20261019_162750; ZETA_LIMIT 20261020_060256
Where the pause goes	on the looking, not on the switching	an anti-short-cycle timer makes 63 per cent of its switches mistakes, an ear on a clock 19	a wait of 1 against a clock	FLIP_MISTAKES 20261015_195739
A noisy ear’s dead band	about three times the noise	brings the price back to about the line, 1.055 and 1.041 at two steps, each ± 0.08 to 0.09 over four seeds	two steps	EAR_KNEE 20261003_220719; ERROR_BARS 20261019_162750
A latch’s release	hold until the push turns, not until it drops below the edge	1.36 to 1.82 times the push for the same switches	edges of 1.5 to 3, where the two switch equally often	LATCH_EVENT 20261015_232643
A store with a trigger	never let it go fully dark: give it a floor	the floor costs no push; at a trigger of half the knee ear’s speed, floors of 0.1, 0.2 and 0.4 fire 0.05, 0.37 and 0.65 times a cycle; set the trigger below the swing the floor keeps	three floors	SPRING_FLOOR 20261017_160023
A stalled ear	wake it with a whisper	a pilot a hundredth of the soup’s push takes it from 0.000 to 0.145; one as strong as the soup takes over	two pilot strengths	the first nights’ toy run 5r
The rate sensor	a true rate sensor, not a slope from the ear’s own looks	buys back 88 per cent of the lag of looking seldom, against 45, each to about a point over draws of the noise	one look rate, a quarter cycle	SLOPE_EAR 20261015_190935; OWED_BARS 20261107_023832

Swings, slings and pumps

Knob	Set it so	What it buys	Tried on	Run
A swing pump’s timing	listen, leading by 90 degrees; turn the lead round and it brakes	over the top in 12 swings; brake from 10 degrees to 0.13 in twenty	leads from 45 to 135 degrees	SWING_TWENTY_B 20261004_075902
A sling’s release	cut between 28 and 39 degrees on the upswing, where the best cut lies for swings of 40 to 90 degrees	two to three times the throw of a cut at the bottom; from 80 degrees anywhere from 29 to 47 is within 5 per cent of the best	swings of 40 to 90 degrees	SLING_RUNDOWN 20261017_232234

Flyers, swimmers and smoke rings

Knob	Set it so	What it buys	Tried on	Run
A flapping wing’s twist	a quarter cycle from its flap	5.62 m/s on 2.1 J a flap, against 0.22 m/s on 14.2 J in step	two phasings of one wing	the first nights’ toy run 5d
A drag-driven coil’s winding	tan θ = √(grip ratio)	54.7 degrees for a round wire, 67.8 for a ribbon	two grips, a 0.1-degree sweep	COIL_SNAKE 20261017_204030
A leapfrogging pair’s start	one radius apart, cores a tenth of the radius; pack three at half a radius	19.5 and 26 per cent over one ring on the same fuel	a grid of spacings and cores	RING_KNOBS 20261015_165858; TRAIN_HOLD 20261015_171853
A rocket rich in energy, short of propellant	throw fast	push per joule 2/v, push per kilogram v	exhausts of 10 to 1,000 m/s	PUSH_PRICE 20261004_144831
A sailing boat	clean the hull first	this toy boat outruns the wind only once its hull drag area is cut below a point between 0.02 and 0.01 square metres, 1.15 times the wind at 0.006; fastest near 85 degrees off the wind, best upwind at 45	nine hulls, 151 headings each	SAIL_HULL 20261018_202650
A rider on a wave’s crest, with waves coming the other way	before riding, add up the oncoming waves’ pull against the crest’s width, twice	from the oncoming waves’ shape alone, every rider’s fate falls inside the bracket; the two sums agree on all but 1,560 of 5,120	eight soups, 64 riders, five margins, two mixes	COMB_STRETCH_B 20261101_003130

The crest rider is a bead set on the tallest crest of a wave pattern running one way, in a soup whose waves also run the other way. It holds its place while the crest’s push beats the oncoming waves’ pull. Those waves sweep past it at twice the wave speed, and wherever their pull, added up over the time it lasts, outweighs the crest’s width, the rider is knocked off. Read from the oncoming waves’ shape alone, before any rider is run, the sum taken twice, once letting the rider spend all of the crest’s spare push and once none of it, brackets every one of 5,120 fates, 512 riders in eight soups at ten settings: a fall the first foretells comes, a fall the second does not foretell does not, and the two disagree on 1,560[283].

Needles and counters

Knob	Set it so	What it buys	Tried on	Run
A remembering needle’s change of motion	take its memory below the scissors’ tip, 0.70, before crossing pace 1	all 16 changes made beneath the tip were gentle, 0.220 to 0.292 on a reader that read the 20 others as jumps, 0.944 to 1.242; in all 18 slow walks the needle came home changed exactly when its jumps plus its turns round the tip were odd	five loops, 18 walks, both starts	BENCH_LOOP 20261020_081001; MEMORY_SWITCH_2 20261011_085331
A rare-event count	start it only after the device settles: drop a settling window several times its lateness, for this needle 50 time units, about four times its 12	a needle set down at rest is about 12 time units late to its first slip; the first 50 time units read 0.7645 ± 0.0094 of the exact rate, the next 450 read 1.0027 ± 0.0036	seven rows, 32,768 needles each	SLIP_START 20261020_090154

Columns of people, and a truck in its passing lane

Knob	Set it so	What it buys	Tried on	Run
A column’s spacing	by time gap, not distance	a 1.5 s gap keeps a stumble from growing for delays to 0.6 s; a fixed distance amplifies it 74,340 times by the 30th man at a 0.2 s reaction delay, in a toy with no limit	36 settings, closed form against the column	COLUMN_ ACCORDION_2 20261013_222950
Walking at night	match the man ahead’s speed by touch; read the cat’s eyes inside 0.13 s	by touch nobody stops and nobody comes closer than 0.82 m; reading the cat’s eyes with a small reading error, nobody stops up to 0.13 s (closest 0.637 m), one at 0.14 s, four at 0.15 s, in one draw of the reading noise, and the edge is 0.13 s in seven of eight draws, 0.14 s in the eighth; read exactly, even 0.3 s late stops nobody	look-times 0.05 to 0.3 s	NIGHT_SPACING 20261013_231417; CATS_EYES 20261014_063138; QUICK_READ 20261014_065420; LIMIT_AUDIT 20261017_190938; OWED_BARS 20261107_023832
A halt	order it to everyone at once	passed man to man at a fixed 1 m spacing, a 10 s halt stands the 30th man for 46.2 s	a column of 30	HALT_ORDER 20261015_052122; LIMIT_AUDIT 20261017_190938
A halt called from the tail	at a gap of 2.4 to 3.2 m, if keeping every man in touch is worth the time	at 3.2 m every man keeps in touch and the tail arrives in 1,522.952 s, against 524.999 s with no rule; at 2.3 m and below the column never arrives; touch is lost from 3.4 m, and at 3.8 and 4 m the rule still costs 214 to 219 s (190 to 230 s out to 8 m)	sixteen thresholds, a tired tail	HALT_TRADE 20261031_235932
A truck’s passing lane	expect it to narrow as one over √(weight)	width times √(mass) 0.1085 within 3.2 per cent over eleven weights	eleven weights	LANE_EDGES 20261017_164501
A runaway rider caught by the passing lane	expect more caught the faster it comes in	at weight 3 the passing lane catches 0.231 of runners coming in at rest, 0.423 at the pace and 0.538 at 1.6 times it; the slow lane takes the rest	four weights, seventeen speeds coming in	RAMP_SPEED 20261031_232517

Four rows of this table are told in no chapter. The halt is the toy column of chapter 11, held at a fixed metre: a 10 s halt passed back man to man stands the thirtieth man for 46.2 s, while the same halt ordered to everyone at once stands every man exactly 10 s and the gaps never close[284]. Bear’s other halt is called by a sergeant at the tail whenever any gap opens past a threshold, and it buys contact with time. Below 2.4 m the column halts so often that it never arrives; from 2.4 to 3.2 m every man stays in touch, but the tail takes 2.9 to 5.5 times as long as with no rule; touch is lost from 3.4 m, and at 3.8 and 4 m the rule still costs 214 to 219 s, 190 to 230 s out to 8 m. Up to 2.7 m the largest gap ever seen is the threshold plus the same 1.4205 m, the distance the column keeps opening after the halt is called[285]. The passing lane comes from Bear’s mountain road: heavy trucks in the slow lane, and a passing lane where the legs between switchbacks run long. Chapter 10’s needle that rides two waves, at weights from 0.5 to 3, plays the truck. Its slow lane is its first gear and its passing lane its second, and the stretch of push where it can pass narrows as one over the square root of its weight, which is its memory[286]. Where both lanes hold, a rider that runs away is caught by one or the other, and its weight is only half of what decides which: the other half is how fast it comes in. At every weight tried, from 1 to 3, the passing lane catches more of those arriving at 1.6 times the pace than of those starting at rest, though not at every step between. The heaviest catch only 0.423 of their runners coming in at the pace, but 0.538 of those coming in at 1.6 times the pace, and only 0.231 of those that start at rest[287].

Maintenance and power

Knob	Set it so	What it buys	Tried on	Run
A maintenance alarm	read the rested reaction, where a hiding clock cannot mask a wearing one	a fixed alarm wastes 0.031 to 0.051 of a machine’s life, where the calendar wastes 0.250 to 0.333	one fleet design, five seeds	RESTED_LAG 20261016_205504
A controlled machine’s headroom	meter it	its command sits at the limit on 0.988 of last flights; a fixed effort alarm wastes 0.190 of a life, one learned on the wobble 0.036, the calendar 0.344	one fleet	WORN_RIDER 20261016_065645
An alarm on power drawn	do not wait for more power: a worn machine may grow quieter, not hungrier	an alarm waiting for more power wastes 0.917 of a life	one fleet	RIDER_FLEET 20261016_044707
An inductor’s switch	switch at zero current	97.8 per cent arrives; mid-swing, 48.9 per cent is dumped	one circuit, two switch times	the first nights’ toy run 5

Bodies in the air

Knob	Set it so	What it buys	Tried on	Run
A long jumper’s arms and legs	two arm circles and two deep strides	hold off nearly all of the forward turn, 0.996; each circle adds 0.257 on a fixed 0.152	one to three circles with one to three strides, at two knee folds	HITCH_LEGS 20261016_065650; ARM_SHARE 20261016_182910; ARMS_LEGS 20261017_012927
Practice	rehearse a routine’s shapes on a mat	the shape part is 78 to 95 per cent of what it holds off, and no speed changes it	every pairing tried	ARMS_LEGS 20261017_012927
A dive’s kick-out	on the body’s angle, not the clock	0.420 of the clock’s miss; a clock-timed kick-out must land inside a window 28.8 ms wide	spin errors of 2 to 10 per cent, both signs	DIVE_CUE 20261017_082454; EXPLORER_ROUTINE_CHECK 20261017_164432
A twist-free cat	bend it to its sweet spot	78.837 degrees at radius 0.3 lands it within 3e-7 degrees, level	three radii	CAT_SWEET 20261018_034953

What this chapter found. Every machine’s settings in a builder’s order, led by the traders’ law carried into controllers. Each rule comes with its number, the cases it was tried on, and its run. They are toy numbers in toy units unless they say otherwise, and each wants its own trial before a real machine is built on it.

16. What is ours, and who was there first

> A flag went up today on the hill where the two cats meet, with a note pinned under it: as far as we walked. The note is the honest part.
>
> *— a no-physics read*

Someone was always there first

At the first sitting, on 2 October, looking back over the months of work before this book, Bear wrote this:

“The realization is that holonomy crosses domains. For me it shows how geometry and mathematics are tied together. We crossed a lot of domains, and someone was always there before us. Fifteen substrates or so, the same holonomy lens on each, and nothing magical: the experts were already there, again and again. That in itself is amazing, an arc of maybe twelve thousand years of history and tool-making.”

“Celebrate the plateaus, where the signposts left by those before us still stand.”

Every wall in the chapters you have just read has someone’s name on it. The swimmer that goes exactly nowhere when its drag is the same every way meets the first rule of slender swimming by drag, and the one that opens and closes the same way meets the scallop theorem. The needle that slips past its pace is Adler’s locking law, worked out for radios. The ear that flips at a rate you can predict from the soup’s rhythm is Rice’s, worked out for telephone noise in the 1940s. The leftover that equals the room inside the loop is Stokes, Gauss and Bonnet. We did not look any of them up first, on purpose, and we walked into every one of them from a direction their discoverers did not use.

That is the plateau Bear means, and it is worth more than it looks. When you arrive at a known wall by a road nobody else took, and it is the same wall, to the same number, then the road was real. A toy that disagreed with Adler would have told us our toy was broken. A toy that agreed with him, from a swimmer’s side of the world, told us the swimmer and the radio are one machine. We stood beside them, not on their shoulders.

There were 51 such meetings, and Appendix B sets out every one, each with its source, what the toy gave and the run that printed it. Two readers who had not seen our own count tagged the 44 the register then held by a rule written down before they began: 26 are pure numbers, fixed by mathematics alone and the same for every system their law covers, and 18 carry something the world sets, an inertia, a grip, a speed, a particular drive[288]. The pure ones are the ruler, and the rest are the floor the ruler was laid on.

The lens itself had a signpost. At the end of a long day Bear judged that nothing here was discovered, though everything was modelled, perhaps never before all through one lens. So we went looking for whoever had put the lens itself together, and found him. The philosopher Robert Batterman wrote a paper named for three of our toys, *Falling cats, parallel parking, and polarized light* (2003). His point, in his own words, is that structures of an abstract space of parameters can have “observable, physical, but obviously noncausal ‘effects’”, a line the philosopher Juha Saatsi quotes too (Saatsi, 2018): the geometry of a space of shapes or settings has effects you can observe and measure, effects no push causes. For the falling cat he puts the explanation in one place, “the mechanical connection on the bundle over the space of shapes”, because a Newtonian story told cat by cat “will be completely disjoint from the others”. Saatsi carries the argument on: such explanations are robust, because they do not depend on the dynamics underneath. That robustness is our clock test, the check that a loop’s answer does not change when the loop is run at another speed, which the cat, the diver’s shape part and the pumping coil all pass. The mathematician Robert Bryant taught the same lens to the public in 2013, with a trackball, a car parallel parking, a cat landing on its feet and a ball rolling without slipping. What this book can add to their argument is measurement: a bench of toys on which the part of each answer that is shape and the part that is clock can be counted, and a ledger of what the toys got wrong.

What is ours

Known, and confirmed here. 51 times the toys arrived at a number already in print, most of them blind, the rest flown on the discoverers’ own inputs as a check of the bench: Adler’s 866 slips against 866.0, Feigenbaum’s constant, reached to 4.6691 inside a needle’s chaos, the crisis exponent’s $-1/2$ as -0.498 ± 0.001, Pancharatnam’s phase at 1.000000, the cone’s missing wedge to 8.4e-15, the Sagnac delay to a part in a billion, Kane and Scher’s cat and Tong and Dullin’s diver rebuilt. Those are not small. They are why the rest of the book can be trusted. Two results we first thought were ours turned out, on the literature pass, to be Hatton and Choset’s height functions for the locomotion of shape-changing bodies; we arrived at them blind, and the book now credits them where they appear.

Known in kind, new in place or in number. These stand on ideas that are known, and say so; what is new is where the idea was carried or the number it was given. The traders’ correction is known in finance (Broadie, Glasserman and Kou, 1997); carried to a sampled switch’s dead band, with its reach measured, it is new in place. Act-and-wait control is known; sized taps that never tap blind, holding a delayed balancer to 1.75 and 2.00 fall times, past the $\sqrt{2}$ of a delayed tilt-and-rate push, are new in number. Condition-based maintenance is known; reading the rested reaction so that a hiding clock cannot mask a wearing one is new in place. String stability is known (Swaroop and Hedrick, 1996); carried to a marching column it is new in place. The split of a somersault into a shape part and a count part is Tong and Dullin’s; the long jumper’s 0.996 and the share a mat can rehearse are new in number. Leapfrogging smoke rings are known (Borisov, Kilin and Mamaev, 2013); the fair-fuel verdict is new in number.

Ours, as far as the reads go. Then there are the 16 results we have not yet seen in print. Every one has had a check of the literature, read as far as its pages could be reached, and Appendix B gives what each check found; the first, read against six sources in full, is claimed, and the others stay candidates until their checks are complete: most of the ideas are in print in kind and the numbers are ours, and a few came back empty in the pages reached. A found-here claim stands only where its check came back empty. So they are candidates, and one of them is claimed. The second already shows what the check can bring home. A swing pump that listens, when it has the wrong idea of the swing’s beat, loses push by exactly the share of power a mismatched cable loses, and for one reason: its misread beat is the map that turns an impedance ratio into a reflection. One of the fresh readers who checked this book saw the likeness; the reason was found on the second pass, and as far as the pages read go, the law is not in print. The two falling cats, a thick cat that twists at its waist and a twist-free cat half as thick, bent to mirror-image angles, turn alike: exactly in the algebra, at a steady circuit, and to ten decimal places of a degree in the runs. Both halves sit in Montgomery’s equation of 1993, which studies only the twist-free cat and never compares the two, and none of the six sources read in full says it. In brief, the second to the sixth. A pump that listens beats one that keeps time, and the arithmetic says how much it loses on the wrong swing. A bent swimmer reverses at a place we can name. Light carried round a loop of filters obeys a law registered before its tool was built. A published table of the human body most likely dropped one entry, at the knee. The first rare slips were the early warning of a noisy lock letting go. Appendix B sets out all 16, in the order a specialist would most want to see them, each with its number, its run and where its check stands.

What this chapter found. Arriving blind at a known wall, by a road nobody took, and at the same number, is what shows the road is real. Overclaiming and underclaiming are both errors. This chapter weighs the book in both directions.

17. What a loop leaves behind

> The best picture of all was a cat in mid-air, caught from tree to grass in one long smear of copies, each a little more turned over than the last, the last one standing on all four paws in the grass as if nothing had happened. That smear is the whole book in one image: a body with nothing to push against, coming home upright, having left a turn behind.
>
> *— a no-physics read*

Go back to the swivel chair of chapter 1. Arms out, swing them, arms in, swing them back, and round you go, a little at a time, with nothing pushing you. Every chapter since has been that chair in another coat. The falling cat is the chair with a waist instead of arms. The swimmer that steers by holding a curl is the chair in syrup. Light carried round a ring of filters is the chair with a beam for a body, and it comes home late instead of turned. The arrow carried round a paper mountain is the chair on a bent floor, and it reads the mountain back. The needle that locks to a turning field is the chair with a gear in it, and the needle with memory is the chair with a staircase of gears. The swing that listens is the chair with a refunder and a feeder, and the smoke rings are two chairs that keep passing through each other. One shape, met again and again in worlds that have nothing else in common, is the reason to believe it.

Two sentences hold across the book. The leftover is the room inside the loop times the grip the world gives it, for every loop in it, which is why doubling a small stroke goes about four times as far, and why a body touching nothing can turn and, on a flat floor, never travel. And what carries anything somewhere is a difference: a ship cannot tow itself, an even soup of waves gives a cork nothing, a push that is the same everywhere cannot be gripped from inside a loop. Under both sits the pair Bear named long before this book began, the gear and the rubber wheel, the count and the measure: a whole number that cannot drift, riding on a smooth fraction that can. The needle’s slips are whole numbers on a sliding staircase; the hidden mountains’ whole turns sit inside a leftover that is not whole; chaos in the remembering needle tumbles without end and still keeps five turns in every seven. Wherever we looked closely, a count was riding on a measure.

None of it is magic, and the ledger balances to the last lepton, as Bear said it would. But a balanced ledger turned out to have a great deal of room inside it, and the room is where the joy is. In that room the toys met the published record 51 times, to as many as fourteen figures, and found 16 results we have not yet seen in print. A cat turns over with nothing to push on. A diver who watches beats a diver who counts, missing 0.420 as much, a number predicted from the body’s inertias before the run. A compass needle given a memory walks Feigenbaum’s road into chaos and keeps an exact gear in the middle of it. A swing pumped by listening works on swings from half a metre to twelve and a half without retuning. A controller’s dead band is wider than the one you set, by an amount a trader worked out for barrier options. And two falling cats that should be strangers turn out to be mirror images of each other, which, as far as our reading goes, no one had said.

The toys are the part of this book most people will keep. A child who drops Holonomy out of a tree and watches it land on its feet has seen holonomy, whether or not the word ever comes. A coach who lays a dive along its clock has walked a loop. That is enough. The rest of the book is for whoever wants to know why it works, and it is all here, every number with the run that printed it, and every miss with the reason it missed. So climb on. Go round. Come home to the very spot where you got on, and look at what is in your hand.

Appendix A. How we worked, and how we went wrong

> A workshop where every toy on the bench wears a tag saying what it once got wrong. Most of the tags are in the same handwriting: the toy was fine and the ruler lied.
>
> *— a no-physics read*

A toy can lie, an instrument can lie, and the people reading them can lie to themselves. The bench’s way of working is a result of its own, and it is the one most likely to travel.

The machinery. Every prediction in this book was written down before the run that judged it: what we expected, the bar it had to clear, and, from the middle of the work on, a rival that would win if we were wrong, most often only the prediction’s opposite, and a long shot we did not expect to win. Every tool since the blind-run practice began ran once blind first (217 blind runs), printing only its planted checks, so that a broken tool could not be mistaken for a finding; the tools passed 555 of 572 planted checks, the checks that each tool repeats a known answer counted among them, and every one they failed was a fault or a limit of the tool, caught before its number was used. Then the run of record

judged the prediction. Misses were written up in the misses file, with their reasons, every one printed whole in the record's Part 3, and the question asked aloud: can this be redesigned, or is the redesign already covered somewhere else? 367 of the 656 rows name a redesign, and many of the book's best results are redesigns that landed.

The misses file is the field list of how a toy, a tool, an instrument or a guess can lie. Here are the lessons that earned a page of their own.

Run it backward. A drifting stepping rule showed itself when the first snake went 0.1508 forward and 0.1472 back[289]. Run backward, a slipping needle retraces itself to one part in ten million million while a locked one misses by half a turn[290], so running backward is also an instrument that detects a lock.

A test that cannot fail is not a test. The first bent-paper test started its arrow down the ridges and printed zero whatever the arithmetic did. It was rebuilt so that it had something to get wrong[291].

A question asked inside a loop. A quick orbit test hid a feedback loop that multiplied its reading by one over one minus the loop's gain, at 111 of 112 angles, to 1.9e-14: from 0.43 times the truth to 1,176 times, and past a gain of one reversed, up to 192 times, with phantom orbits where the gain reached one[292].

A sieve finer than the arithmetic reads its own grain as structure. An unwrapped phase of 31,600 radians is held only to about four trillionths. Carried in extended precision the median scatter fell about 7,700 times, and with the phase wrapped about 38,000 times, and the "long orbit" closed after 19 cycles[293]. Sixteen-figure arithmetic printed false step widths near 1e-13, and forty figures found true ones down to 9.99e-26.

A grid search can manufacture a trend. It read the long snake's steady climb as a slowdown, missing the best by more at each length. Five exact solves fixed it[294].

A trial of fixed length cannot place an edge past an inner fold. The ghost of a lock fooled a 400-cycle trial by 1.6e-8, and 34 trials of 200 answered "on" by chance[295]. And a lock detector's window must outlast the slips: one that reads in windows of 100 passes a rider that slips every 200 to 400 as locked.

The free lunch caught. The listening horse of chapter 3 had been given its charge for free; billed honestly to the ship, the pull is exactly 0.00000 at zero tether[296].

The unfair budget caught. One merged smoke ring seemed to beat two leapfrogging rings because it carried 1.41 to 1.73 times their energy[297]. On the same fuel the pair wins, as chapter 9 tells.

The model's own rule demoted. The bell toys' numbers came from an assumed kick rule and moved a tenth as far as a full fluid solver; the bells now stand on the solver alone[298].

The limit audit. The marching column's own collision limit braked walkers by mistake, often enough to move counts of stopping walkers by up to 100 per cent. Every column tool was rerun with only that repaired, and five verdicts in three tools changed[299].

A planted check must not ask chaos to repeat. A planted check asked smoke-ring trains to break at the times another computer had printed, and they did not: 4,494, 241 and 272 against 1,022, 1,011 and 303[300]. A nudge of 1e-15 of a radius moved the breaks by 28 to 84 per cent[301]. A planted check compares what chaos cannot scramble, the energy and impulse a flight keeps, or the statistics of many throws.

Put the error bar before the "within". Near the end, every number this book reads off a noisy run was given an error bar from its run's own units, seeds, soups, riders or noise paths, by script[302]. Of 35 such claims, 22 stood as written, 7 could not yet be given a bar, and 6 had to change their words. Later, four of the seven were given bars from their runs' own units: two stood with their bars, and two changed their words[303]. Five of the six were the same slip: a "within" printed tighter than its own error bar, such as Kramers' escape rate "within 5 per cent" where each count's own error is 5 to 9 per cent. The sixth was a lean read as scatter: a sweep of Kramers' rate printed as running "0.79 to 1.08" pools, all 6,549 of its slips together, to 0.909 ± 0.011, so his leading formula runs 9 per cent high there, and the old wording hid it; against the exact rate the same slips pool to 0.9885 ± 0.0122[304]. Chapter 11 tells the hardest case, where a finer ruler laid over the same noise repeated the same scatter and we twice named its owner too soon.

Count the wins too. For weeks each new result was weighed alone, against everything anyone had ever done, and sized "small but useful" or "medium"; no one added them up, and no record collected them. A benchmark matched to eight figures is an achievement whether or not someone was there first. The fix was a register that lists every result by kind, a benchmark as a benchmark, a rule as a rule, a candidate as a candidate, each with its run, read first at every step. Appendix B is that register's first page. The sparks of some milestones were getting smothered between one working session and the next, and catching each one in the moment, in the register, is what carried it through a long project.

Did the guesses grow safer? Late in the work Bear felt a twinge that the bench's guesses were growing safer. Read week by week, they had not: 648 of 934 held in the bench's first two weeks, 0.6938, and 35 of 58 in its last stretch, 0.6034, a difference of 1.37 of its standard error and no rise[305]. What had changed was that the guesses grew fewer

and their rivals stopped being ideas: of the 62 rivals written since 19 October, only five named a competing cause. From then on every rival had to name another cause, another direction or another mechanism, and every review of five exchanges reads the bench's own numbers again.

What the misses are worth. A miss with its reason is worth more than a hit without one. A hit says the bar was met; a miss says where the toy, the tool or the thinking broke, and the redesign check turns most of them into the next question. If a reader takes one thing from this appendix, take the order of work: write the prediction and its rival first, run blind, read the run whole, and print the miss beside the hit.

Appendix B. Every headline, with its numbers

The first page of the project's headline register, the record that lists every result by kind with its run, set here whole and rebuilt from the register at every build, so that no count in it is typed. The book's opening page draws its short list from this one. After it come the two tables that chapter 16 tells in words: the published laws the toys met, and the candidates that are ours.

Built blind from toys, one idea about loops meets the published record fifty-one times, to as many as fourteen figures, and then finds ground of its own.

An independent researcher and an AI asked one question of things that swim, swing, fly, pump, hum, lock, march and fall: what does a loop leave behind? They answered it with 500 small model tools and 584 runs of record, built without looking the answers up first; 479 of the tools are written in Python and 21, the explorer's, in JavaScript. Again and again the toys walked into the published record and arrived at its number, and then went past it into results this book has not yet seen in print.

What this achieved

1. **Two falling cats are one cat in a mirror** (chapter 12). A thick cat that twists at its waist, bent to b, and a twist-free cat bent to 180 − b turn at mirror-image rates at every shape (in the runs, with tubes of unit length and mass), their turning axes a right angle apart, so they turn alike on every path: 5e-13 degrees on a 1-degree grid, exact in algebra on a steady circuit, and within 2.842e-13 on nine paths whose bend swells and shrinks. Hinged at the ends the twist-free partner is half as thick; hinged nearer the middle it needs another thickness, which a faulty test had hidden. Both cats are Montgomery's 1993 ball-jointed cat, under his no-twist rule and under a second rule he does not study, halves spinning opposite ways about the spine; by that reading the mirror is a short step from his equation (9) that his paper does not take. We claim it as new, on a limited, noncommercial review of the citations and references[306].
2. **A pump that listens beats one that keeps time, and a mistuned listener loses push exactly as a mismatched cable loses power** (chapter 6). Three predictions worked by hand and registered before the tool all held: the growth was 1.26570 a swing against 1.26569. On any swing from 0.5 to 12.5 m the listener's push weakens only to $4r/(1+r)^2$ of full strength, within 0.14 per cent, where a clock rider never passes 12.67 degrees[307]. That factor is the share of power that crosses a joint between two cables whose impedances differ by the ratio r, and for one reason, not by likeness: the rider's misread beat is the very map that turns an impedance ratio into a reflection, the map behind a radio engineer's Smith chart, so a swing and a cable lose the same share, whichever way the mismatch runs. One of the book's fresh readers saw the likeness first; as far as our reading goes, the law is not in print.
3. **Where a bent swimmer reverses** (chapter 5). After a wrong first guess, a mechanism read off one lap size put the reversal at the lap's radius over √2, and it held at the two other sizes tried within half a per cent[308]. It was the first sign of something larger: as the water's sideways grip hardens, the swimmer's grip map gathers into a ridge along its C shapes, as a wheeled snake's does, and at a grip of 100,000 a lap swims forward exactly when its disk crosses that line; as far as our reading reached, the ridge a resistive swimmer reaches is not in print[309].
4. **A registered law for light carried round a loop** (chapter 7). $2\pi\, a\, J_1(2a)\, \sin(\text{gap})$, written down before its tool, held within 2.51e-7 over loop sizes of 1 to 40 degrees and gaps of 30 to 150[310]. The same bench meets Pancharatnam's phase at 1.000000 and the Sagnac delay to a part in a billion.
5. **Two published bodies rebuilt** (chapter 12). Kane and Scher's 1969 cat, flown along their own path, lands 0.10 degrees from straight down[311]. Tong and Dullin's 2019 diver comes back within 0.15 per cent, its best-dive gains 0.0189 and 0.2326 radians against their 0.0189 and 0.2327, once one entry their table most likely drops is restored[312].

6. **The universal numbers of chaos, inside a toy needle** (chapter 10). Feigenbaum's δ comes back to eight figures on the reference map, 4.66920158752, and as 4.6691 in the needle's own cascade, every split forecast within about 2.5 billionths of a push. Inside the needle's 5/7 lock it doubles again from 7 to 1,792 cycles on both sides of the step, and its bands shrink by 3.945, 4.482, 4.626, 4.660 then 4.667 from above and 3.994, 4.480, 4.629, 4.660 then 4.667 from below, closing on δ alike, on the road A Rider in Two Waves' rider takes in its own locked step, whose rule forecast the fourth ratio within 0.0015 and the fifth within 0.0001; past each end of the doubling its windows of order come in the logistic map's own order (Metropolis, Stein and Stein's), three of them found where that order forecast them, each still there when flown four times longer. Why each gap to δ is about a fifth of the last is read back too, from the logistic map's own values to period 262,144: the gap closes by 1/δ, 0.214169, the bend of the dial it is read on, and swings about that by the doubling operator's second eigenvalue, −0.123652 against the published −0.1236527; on a straightened dial the closing stops and the gaps change sign at every step. Each doubling turns a corner in the needle's time-averages, and the corners shrink by −1.966 a generation in all four of its lanes; the number is the cascade's own: the logistic and sine maps, solved exactly in 80-digit arithmetic, give −1.9644458, agreeing to the ninth figure, and a flatter, quartic top gives −0.603, its corners growing; on 1 − r|x|^z the corners stop shrinking near z = 2.89 (not found named in a limited pass of free records). The 9/4 window law holds within 0.7 per cent from memory 1 to 3, and the crisis exponent is −0.498 ± 0.001 against $-1/2$[313].
7. **Adler and Kramers met with numbers** (chapters 4 and 10). The needle slipped 866 times in a thousand cycles against Adler's 866.0, and escaped at Kramers' rate at the four barriers first read, each four noise units or more, within their counting error of 5 to 9 per cent; a later, wider sweep from 4 to 9 units, with six times the slips, pools to 0.909 ± 0.011 of Kramers' rate, his leading formula 9 per cent high there, a shortfall the counts resolve; against the exact rate, Stratonovich's, the same slips pool to 0.9885 ± 0.0122, so the shortfall is the formula's own[314].
8. **The geometry of a carried arrow, to fourteen figures** (chapter 7). The cone's missing wedge comes back to 8.4e-15, and bent paper reads zero to 3.1e-12. Hidden mountains, noiseless ones so far, are read back from the arrows alone. And a sampling rule came out of it: look again before the reading can move half a turn (3.018 radians kept the count; 3.364 lost a turn)[315].
9. **A traders' law carried into controllers** (chapter 11). Broadie, Glasserman and Kou's 1997 correction predicts a sampled switch's missed flips with nothing fitted, and read back out of the flips its constant's small-step limit is 0.57947 ± 0.00254 against their 0.5826[316]: for slow noise within two standard errors at six of seven steps and 2.2 at the coarsest, its error running from 14 per cent at the smallest step to 2 per cent at the largest, and for faster noise within 15 per cent while the sample step is at most a tenth of the noise's memory, a reach read after the registered bar missed at the fastest noises. Counted across a band, a slow sampler's count is step-free within 0.7 per cent over a 128-fold range of steps at a noise memory of 1. And read where the law lives, in the knee's extra size, its square root comes back: powers of 0.458 and 0.475 of the step against the law's own 0.481 and 0.454, where read from the flips it had seemed 0.332; and against a reference watcher four times finer the law holds at every step from 0.0003125 to 0.04 on every reader tried, within 1.6 standard errors[317].
10. **A maintenance alarm that reads the wearing clock where nothing hides it** (chapter 14). A fixed alarm on the rested reaction wastes 0.031 to 0.051 of a machine's life, where the calendar wastes 0.250 to 0.333[318]. Maintenance by condition is known to beat the calendar; what the toy adds is where to read the wear, which hides behind the rider's rest and shows only at taps where the rest has run out.
11. **Past the √2 limit of a delayed tilt-and-rate push** (chapter 4). Sized taps that never act blind hold a delayed balancer to 1.75 and 2.00 fall times in quiet, though the registered bar of 1.80 for both kicks missed[319]. Act-and-wait control and a smooth push fed the tilt's acceleration are known to pass that limit too, and in quiet the taps reach the second's bound, 2.00.
12. **Watch, don't count** (chapter 12). Kicking out of a dive on the body's angle cuts a nervous take-off's miss to 0.420 of a clock-timed one, a ratio of inertias predicted before the run and matched to 0.00002[320]. Somersaulters are known to time their opening by the body's angle and how fast it turns; what the toy adds is the size of the gain against a clock, and why.
13. **The early warning was the first rare slips** (chapter 4). With the slips left out, a noisy lock's gauge reads its softened hold within 0.034 where slips are rare, a little high as the noise grows and the pace nears the edge: about 0.03 high at the loudest noise, four to six times its standard error, and 0.007 to 0.026 at seven quieter settings near the edge; what warns of the cliff is hearing the first falls. It took two slip-finders that turned out blind, and a third tested on planted slips before it was trusted, to see it[321].

14. **A sign in Feigenbaum's table**. The standard table of Feigenbaum's constants for maps 1 − μ|x|^d (Briggs 1991 and his 1997 thesis; MathWorld; Finch 2003) is right in α at every order 2 to 12 and in δ at every even order, but its δ at every odd order is not the map's: 5.9679687038 printed at order 3, where the map gives 6.0846890743, as McCartney's values from superstable orbits, in 2020, also say: he notes a discrepancy with the earlier work at odd orders. Our own three readings agree with him, and so does a fresh helper's code written with no number given. One sign reproduces every printed odd entry: the doubling operator's change evaluated at the negative point x/α with signed powers where the map needs absolute ones; with it the solver meets the printed order-3 value to 29 places, without it the map's own to 30 (not found named in a limited pass; McCartney names no cause)[322]. It is written up as its own short note, *A Mirror in Feigenbaum's Table*, deposited beside this book.
15. **Chaos that keeps its gear when a lock loses its own** (chapter 10). Inside the needle's banded chaos, its exponent positive at nine of ten paces, the count keeps five turns in every seven within a quarter turn over 30,000 cycles (the bar asking for all ten paces missed); and jostled with a noise of 0.001, the true locks near their edges move their turn rates 0.013 and 0.014, the chaos 0.003 or less[323]. Phase locking of chaos is known (Rosenblum, Pikovsky and Kurths 1996), and so is a lock slipping near its edge; the pairing, the chaos shrugging off a noise that moves the near-edge locks more than four times as far, was not found in the pages reached.
16. **Smoke rings that leapfrog go further on the same fuel** (chapter 9). On equal energy a leapfrogging pair goes 14 to 19 per cent faster than one ring, and three packed half a radius apart 26 per cent; near a three-ring train's soft edge the squeeze it breaks through lies on one line with the core, a line that forecast two cores never flown, 0.045 and 0.055, every one of 15 breaks within 0.0025 of it[324]. Leapfrogging rings are known; no speed compared at equal energy was found in the pages reached.
17. **The needle needs memory to tumble** (chapter 10). Swept over twelve memories and 49 paces, the driven needle tumbles nowhere at memory 0.4 or below, and in thick fluid not even a second wave twelve times as strong tumbles it; inside the steps its circle folds before it tumbles, by 0.015 to 0.030 of memory at the three paces read finely, and inside a lock's window the lock's slope walks through 0 to −1 and the loop doubles, the order a single-humped map gives. The first test looked for the fold among the needles drifting between the steps and could not see it, because the needles that were folding had left that crowd; read where the needle tumbles, the fold was there[325]. The breakdown of a driven pendulum's circle into chaos is known; the sweep over memory and the reading of the fold through the lock were not found in the pages reached.

The count

Predictions are written down before the runs that judge them, and since the blind-run practice began every tool has run blind before its run of record (217 blind runs). The runs judged 1,104 registered predictions about the toys: 706 held, 369 missed and 29 were left undecided. Counted as distinct statements, a statement judged again in a later run counted once, that is 1,034, of which 666 held. Against them stood 154 distinct rivals, written to miss if the prediction was right, and 116 missed; but only 42 were competing ideas, of which 12 held, and the rest were the bare negation of their prediction (99, 25 held) or checks of a tool (13), so the rivals tested the predictions' bars more than they tested other ideas. The tools passed 555 of 572 planted checks, their fidelity and rig checks counted among them; the 17 they failed were faults or limits of the tool, caught. The 656 misses are printed beside the hits, each with its reason, and 367 of them name a redesign. The bench met the published record 51 times, and 16 results are candidates for found-here: 1 claimed as new on a limited, noncommercial review of the citations and references, and the other 15 each with a prior-art pass beside it in the register: most are known in kind, their numbers ours, and the rest were not found in the pages reached; none is claimed until its check is complete.

Limits. A preprint, no peer review, a limited literature pass. Toy models with assumed numbers, not experiments. An AI did the building with an independent researcher, in the open.

The published laws the toys met

All 51 meetings between a toy, built blind, and a law already in print, as the project's headline register counts them, each law with its source; a source marked † in the references has so far been checked only against a secondary record or a citation of it, not its publisher's own page. Chapter 16 tells what they are worth.

	The published law	What the toy gave	Run
1	Adler's injection-locking law (Adler 1946)	866 slips in 1,000 cycles at pace 2, against 866.0; past the edge at pace 1.2 the needle keeps 0.4475 of the drive and a conveyor 0.4464, against the closed form's 0.4472; the share kept matches the closed form within a few parts in ten thousand at every pace	NOVIKOV_NEEDLE 20261004_152937; HOW_IT_COUNTS 20261004_183911; NEEDLE_GEARS 20261003_210142

	The published law	What the toy gave	Run
2	Kramers' escape rate (Kramers 1940)	measured over predicted 1.000, 0.991, 0.999 and 0.949 at the four barriers first read, each four noise units high or more, each within its counting error of 0.05 to 0.09; a later sweep from 4 to 9 units read 0.79 to 1.08 row by row and, 6,549 slips in all, 0.909 ± 0.011 pooled, Kramers' leading formula 9 per cent high there, and against the exact rate (Stratonovich 1958) the same slips pool to 0.9885 ± 0.0122, the shortfall the leading formula's own (KRAMERS_EXACT 20261020_073520), each row's counting error, measured needle by needle, the square root of its slips within 3 per cent (SLIP_SPREAD 20261020_075635), and the 1.2 per cent left between the counts and the exact rate accounted for, within its errors, by the start, every needle set down at rest: on the seven rows counted for 500 time units, on 32,768 new needles, the slips from 50 time units on read 1.0027 ± 0.0036 of it (SLIP_RECHECK 20261020_083908, SLIP_START 20261020_090154); read backward, the slip rate gives the pace within 0.0064 in 15 rows	NOVIKOV_NEEDLE 20261004_152937 and _B 20261004_153129; SLIP_SENSE 20261011_060035; ERROR_BARS 20261019_162750
3	Feigenbaum's δ and α on the reference map, his own road and so a check of the bench (Feigenbaum 1978)	δ 4.66920158752 against 4.6692016091, α 2.502907898 against 2.5029078751: eight figures	DELTA_BY_TOP 20261008_034013
4	Feigenbaum's δ for flatter humps (van der Weele, Capel and Kluiving 1987)	7.2847 for a fourth-power top; order 6 lands 9.5e-6 from 9.2962468327; order 8 settles at 10.94862 on the reference 10.9486 after a second extrapolation (the first one's bar missed)	DELTA_BY_TOP 20261008_034013; DELTA_TAIL_2 20261011_051600
5	Feigenbaum's δ inside the chaos of a needle with memory 0.75 riding two waves (Feigenbaum 1978)	ratios 4.6155, 4.6570, 4.6667, 4.6687, 4.6691, closing on 4.6692; every split forecast in advance landed within about 2.5 billionths of a push, the largest miss 2.54e-9	SPOKES to CASCADE_5 20261006_225126
6	Feigenbaum's α and its square in the needle (Feigenbaum 1978)	lanes narrower by 2.8231, 2.3951 and 2.5491, closing on 2.5029 from alternate sides (a registered bar on the first two missed), and by about its square, 4.99 to 6.29 times, at half the orbit's points	CASCADE_4 20261006_221804, CASCADE_5
7	The 9/4 window law (Yorke, Grebogi, Ott and Tedeschini-Lalli 1985)	2.252 to 2.266 across memory 1 to 3, within about 0.7 per cent, while the window's width changes more than twentyfold	HEAVY_SHAPE 20261014_155553 to HEAVY_WINDOW_4 20261015_032553; STOP_WALL_3 20261015_000651
8	Crisis-induced intermittency (Grebogi, Ott, Romeiras and Yorke 1987)	slope −0.4982, standard error 0.0011, against −1/2; Claude's recalled −0.553 is not there	CRISIS_BURSTS 20261014_091405
9	Type-I intermittency (Pomeau and Manneville 1980)	254, 522, 1,058 and 2,130 drive cycles of lingering at distances a quarter apart: twice as long each time it is four times closer, within about 1 per cent at the closest of a prediction worked from the window edge's multipliers (the book's arithmetic, not printed in a run)	REINJECT 20261013_015743, REINJECT_CLOCK 20261013_024653
10	The saddle-node square root (Strogatz 2015)	slope 0.4999 at the half gear's fold, 0.499 at the lock's edge; the slip time's slope −0.5000 near the edge and −0.9820 far from it	HALF_TOP 20261013_183449; LOCK_EDGE_EXACT 20261013_063227; SLIP_BEND 20261006_071248
11	The one-gear theorem for a radio-driven junction without capacitance (Buchstaber, Karpov and Tertychnyi 2010)	the one-cycle map fits the Möbius form to 2e-10 radians; no lock at any fraction with denominator 2 to 40; the misfit falls sixteen times a halving, the integrator's own, to 2e-12	NEEDLE_GEARS_WHY 20261003_210451; QUANT_ZERO 20261013_081905; QUANT_STEP 20261013_082454
12	The devil's staircase (Jensen, Bak and Bohr 1983), ordered by mediants	2/3, 5/7, 3/4, 7/9, 11/14, 4/5, 13/16, 9/11, 14/17, 5/6, the widest n/(n+1) and the mediants next, to the whole gear at 1.14410, with five step-backs along the sweep	GEAR_TRAIN 20261013_170243; STAIR_WIDTHS 20261013_184033

	The published law	What the toy gave	Run
13	Stewart and McCumber’s hysteresis (Stewart 1968; McCumber 1968)	the lock’s edge splits in two; the climb-back line at memory 1 is 0.9643271245 by three integrators, one of them at two tolerances, and at memories 2 and 5 it is 0.790 and 0.541	CLIMB_BACK_LINE_CHECK 20261006_041633; SEAM_MAP_B 20261005_023746
14	The exact period of a large swing (the elliptic integral; Beléndez and colleagues 2007)	1.0019, 1.0313, 1.0732 and 1.1375 of the small-swing time at 10, 40, 60 and 80 degrees, against 1.00191, 1.03134, 1.07318 and 1.13749	SWING_TWENTY 20261004_075715
15	Parametric pumping at twice the swing (Landau and Lifshitz 1976)	10 to 40.14 degrees in six swings; once per swing barely moves it (11.06); with gravity off, nothing	SWING_PUMP 20261002
16	Pancharatnam’s phase (Pancharatnam 1956; Berry 1987)	the delay over half the enclosed solid angle 1.000000 at every loop from 1 to 40 degrees; reversed, exactly opposite; twice round, exactly twice	POLARISATION_LOOP 20261004_071425
17	Lossless geometric-phase optics (Bhandari 1997)	carried by wave plates the light keeps its intensity to 2.5e-14 and the same delay to 4.44e-15	POLARISATION_LOOP_2 20261007_074423
18	Gauss and Bonnet’s cone deficit, $2\pi(1-\cos s)$ (Gauss 1828; Bonnet 1848)	the arrow comes home turned by exactly the missing wedge, the largest difference 8.44e-15	CONE_TOMOGRAPHY 20261007_235426 (part A)
19	Gauss’s Theorema Egregium (Gauss 1828)	paper bent without stretching reads zero on every loop, to 3.1e-12	BENT_PAPER_2 20261008_164721
20	The Sagnac delay, $4A\Omega/c^2$ (Post 1967)	a 360-sided loop at 1.000000000355 of the law; a square, a triangle and a thin rectangle the same within a part in a billion; a 60-degree tilt gives exactly half	SAGNAC_LOOP 20261018_032138
21	The half-turn limit of phase unwrapping (Itoh’s condition; Itoh 1982)	looks 3.018 radians apart keep the count; 3.364 apart lose a whole turn without a sound	GROW_SLIP 20261008_195205
22	A rocket’s push per joule, 2/v (Sutton and Biblarz 2017)	0.2002 kg·m/s per joule at an exhaust of 10 m/s, against 0.2000; the price per joule times the price per kilogram is 2.002 to 2.003	PUSH_PRICE 20261004_144831; PUSH_TRADE_READER 20261007_024449
23	Push equals power over wave speed (the radiation-pressure form; Longuet-Higgins and Stewart 1964)	1.000000 to 1.000003 for five bodies riding water waves, 1.000006 for a motor-driven snake	WAVE_SNAKE_DRIFT 20261004_074634
24	Cattaneo and Vernotte’s heat with memory (1948, 1958)	a rod rings below $2\pi\sqrt{(\tau D)}$ and settles above it: dips 1.52e-2 at 0.80 of that length, 7.06e-5 at 0.95, none from 1.05	HEAT_RING 20261016_174342
25	The counter-flow exchanger’s effectiveness (ε-NTU; Kays and London 1984)	side by side never past a half; counter-flow a half at length 1, 10/11 at 10, 100/101 at 100	COUNTER_FLOW 20261009_002621
26	The drag-only helical swimmer’s best winding, $\tan\theta = \sqrt{(\text{grip ratio})}$ (Darbois Texier, Ibarra and Melo 2017)	54.7 degrees and 2.2214 radii a turn at grip 2 (closed form 54.74, 2.2214); 67.8 and 6.4127 at grip 6 (67.79); the closed form held to 6.2e-15	COIL_SNAKE 20261017_204030
27	Taylor’s swimming sheet (Taylor 1951)	a wave run along a coil swims it against the wave, 0.0046 radii a cycle, twice the wave’s height four times as far	COIL_WAVE 20261018_082544

	The published law	What the toy gave	Run
28	Broadie, Glasserman and Kou's correction for prices watched now and then (1997)	with nothing fitted, a sampled switch's missed flips from 2.5 per cent under to 7.7 per cent over for slow noise, and within 15 per cent for faster noise while the sample step is at most a tenth of the noise's memory, a reach read from the run after the registered bar (25 per cent at noise times 0.1, 0.03 and 0.01) missed at the two fastest (misses row 249); read in the knee itself, the square root's power 0.458 and 0.475 against the law's own 0.481 and 0.454, the law within 15 per cent at 11 of 13 settings, the two outside it at the slowest noise's smallest steps, where the error bar is wider than 15 per cent, about two bootstrap standard errors above the law on every reader tried; against a reference watcher four times finer every step from 0.0003125 to 0.04 within 1.6 bootstrap standard errors on every reader, the fitted one 1.081, 1.014, 0.997, 1.054, 1.043, 1.025, 0.992, 0.988 of the law	CREEP_LAW 20261016_035058; BAND_BRIDGE 20261016_072612; KNEE_SIZE 20261019_035236; KNEE_FINE 20261019_040109; KNEE_REF 20261019_042520; ERROR_BARS 20261019_162750; KNEE_READER 20261019_163948
29	Rice's count of level crossings (Rice 1944)	the push per firing 0.3710 to 0.3789 at every level from 0 to 2, against Rice's 0.3745 for those soups; the ear switches at 0.96 to 0.98 of Rice's zero-crossing rate	TWO_HANDS 20261017_203811; toy runs 5u and 5v
30	Crossings of one line by a rough path grow as √2 a halving (Feller 1968)	1.4145 and 1.4147 against 1.4142	NEVER_BLINK 20261016_090509
31	The two-capacitor half loss, and switching at zero current (Powell 1979; Mohan, Undeland and Robbins 2003)	through a resistor half is lost whatever its size; opened at zero current 97.8 per cent arrives, mid-swing 48.9 per cent is dumped	toy run 5
32	The area rule of loop locomotion (Stokes's theorem; Shapere and Wilczek 1989)	double the stroke and the travel goes 3.98 and then 3.92 times as far; four toys, four grips, one rule	AREA_RULE 20261002b, part 2a
33	Swimming on a curved surface (Wisdom 2003)	three beads on a ball travel by shape alone, travel over stroke squared 0.176 to 0.177 from a stroke of 0.05 to 0.5, and nothing on a flat table	toy runs 5j and 5l (AREA_RULE 20261002b, part 2b)
34	An upright balance's time goes as the square root of its length (Landau and Lifshitz 1976)	taps a second 11.98 on a 0.25 m board and 2.994 on a 4 m board, exactly 4 in the toy, since every rider's taps fall at the same moments in fall times at every length	BALANCE_BOARD 20261004_161712 and _B 20261004_162140
35	Roots cross one at a time through zero, in pairs elsewhere (D-subdivision; Neimark 1992)	161 single crossings at a proportional gain of 1 and 204 paired ones everywhere else	ROOT_JUMPS 20261015_031511
36	A count the shape cannot change (the index of a segment; Milnor 1965)	stable minus unstable seats, counting the wall when it holds the car, is 1 at every speed, a bump in the bank or not	BANKED_TRACK 20261004_154940; BANK_BUMP 20261015_064644
37	Kepler's third law (Kepler 1619; Stephenson 1994)	escape time against binding energy, slope −1.492 against −3/2, over 495 excursions	the triple-star side room's runs, as SEAM_MAP read them (not flown again here)
38	The three-body shape holonomy (Guichardet 1984; Shapere and Wilczek 1989; Montgomery 1993)	the triangle turns by half the area its loop of shapes encloses, to 0.00003 radians on average over 58 loops	triple_cat_holonomy 20260930_161702, read in TRIPLE_CARD_CHECK 20261005_055204
39	The twisting somersault's threshold (Yeadon 1993; Frohlich 1979)	predicted from the body's three inertias at 4.412 degrees of tilt; the toy makes 0.1373 turns at 4.25 and 1.3721 at 4.50 (a registered bar on how sharp the switch is missed)	DIVER_B 20261005_160338
40	Kane and Scher's falling cat (1969)	flown along their own path at their inertia ratio, the cat lands 0.10 degrees from straight down, turned 179.90 and level	CAT_SWEET 20261018_034953
41	Tong and Dullin's six-body diver (2019)	with one table entry restored and nothing fitted: inertias 21.0597 and 5.2858 against their 21.0647 and 5.2888, the field's peak 0.3058 against 0.3058, the best-dive gains 0.0189 and 0.2326 radians against their 0.0189 and 0.2327 (1.08 and 13.33 degrees both), the corner 193.37 against 193.65	TONG_DULLIN_KNEE 20261017_182629

	The published law	What the toy gave	Run
42	The height function of a gait: a gait’s net displacement is the curvature of the local connection integrated over the area it encloses in shape space, by Stokes’ theorem, made accurate by a good choice of body frame (Hatton and Choset 2011 and their papers before and after it, read through Wiezel, Ramasamy, Justus, Or and Hatton (2023), Geometric analysis of gaits and optimal control for three-link kinematic swimmers, arXiv:2109.06511; the 2011 paper checked against its records, the others owed)	summed over each loop, the grip landscape gives the three-link swimmer’s travel within 2 per cent to 60 degrees, measured over the summed hill 1.0016 to 1.0189, where the one-number square rule overshoots by more than a quarter	BIG_LOOP 20261003_175803
43	The rotational part of the same theorem: a gait’s net turn is the rotational curvature integrated over the area it encloses (Hatton and Choset 2011; Shapere and Wilczek 1989; that it is exact where the turning commutes is from recall)	a three-link swimmer held at a C of 45 degrees turns 2.2042 degrees a lap, the same to thirteen places for forty laps, against 2.19 from the curvature times the lap’s area	LANDSCAPE 20261003_201112
44	Rademaker and ter Braak’s two-cylinder cat (1935), its steady-circuit formula as Montgomery writes it for Kane and Scher’s cat (1993, his section 6.3)	the twist-free cat’s sweet spot, found blind on 18 October, is the formula’s flip bend within 1.43e-7 degrees at radius 0.3 and 9.03e-8 at 0.45; every steady circuit flown matches the formula within 2.2e-10 degrees at 45 bends, and its turn does not move with the hinge, his Fact 2; read after the paper, their own worked example, 157.771365 degrees by their formula and 160 as they rounded it, is flown within 2.73e-11 at two hinges	CAT_SWEET 20261018_034953; CAT_MONTGOMERY 20261020_035025; CAT_RTB 20261020_063617
45	Stratonovich’s exact mean speed of an overdamped needle with noise on a tilted washboard (Stratonovich 1958, as Risken 1989 and later writers give it)	the 15 rows of 4 to 9 noise units, 6,549 slips counted blind on 11 October, pool to 0.9885 ± 0.0122 of the exact rate, against 0.9093 ± 0.0112 of Kramers’ leading rate; the exact rate is 0.905 of Kramers’ at 4 noise units and 0.965 at 9; on new seeds, four times the needles and half the step, 0.9877 ± 0.0061, and the start accounts for the 1.2 per cent within its errors: a needle set down at rest is about 12 time units late to its first slip, and on the seven rows counted for 500 time units, on 32,768 new needles, the 78,593 slips from 50 time units on read 1.0027 ± 0.0036 of the exact rate	SLIP_SENSE 20261011_060035; KRAMERS_EXACT 20261020_073520; SLIP_RECHECK 20261020_083908; SLIP_START 20261020_090154
46	The exact long-run spread of an overdamped needle with noise on a tilted washboard (Reimann, Van den Broeck, Linke, Hänggi, Rubi and Pérez-Madrid 2001)	at 3 noise units the needle’s slips are 8.4 to 8.7 per cent more regular than independent events: the formula’s variance over mean, 0.9133, 0.9145 and 0.9160 at noise 0.1, 0.2 and 0.3, against the needle’s own waits between slips, read so that no long one is dropped, 0.9134 ± 0.0031, 0.9135 ± 0.0024 and 0.9182 ± 0.0020, and the counts’ long-run variance growth, 0.9061 ± 0.0183, 0.9121 ± 0.0181 and 0.9248 ± 0.0186; and at a fixed pace of 0.95, where noise makes the slips most regular at one level, the needle’s waits read 0.51695 ± 0.00248, 0.42357 ± 0.00126 and 0.50779 ± 0.00112 at noise 0.3, 0.7 and 1.5 against the formula’s 0.51429, 0.42341 and 0.50967	SLIP_RHYTHM 20261020_083426; SLIP_RHYTHM_2 20261020_084919; SLIP_RHYTHM_3 20261020_090717; RHYTHM_BAND 20261020_154009; SLIP_BAND 20261020_161039; SLIP_BAND_2 20261020_161539
47	The three-distance theorem (Sós 1958)	in a triple of stars whose outer orbit lasts 9.449 inner orbits ([9; 2, 4, 2, 2]), the gaps between the shape’s returns obey it on 10 starts of 10	triple_recurrence_plot 20261005_023346; triple_poincare_strobe 20261005_034156 (the companion room’s runs)

	The published law	What the toy gave	Run
48	The far interaction of two vortex rings has the form of two small magnets' and falls as the cube of their distance (Horn, Nicolis and Penco 2015)	a three-ring train's third ring, swinging out from its pair and pulled home: on the 48 swings that reached past 8 radii its time out grows as its reach to the power 2.587 (R squared 0.9998), which a pull falling as distance to the power 3.17 gives, and the reaches of 137 pairs of swings thin out beyond 6 radii as a power of -2.97 +- 0.40, which a pull falling as distance cubed gives to rings leaving near their escape speed; the step from the published power to these two readings is Claude's, from a picture of a ball loosely held in a dish, the second reading registered after a long shot on the first missed (misses row 487)	TRAIN_PARTING_EDGE 20261029_070800; TRAIN_PARTING_EDGE_2 20261029_211416
49	The universal order of the windows of order inside chaos past a period-doubling cascade (Metropolis, Stein and Stein 1973)	past each end of the needle's doubling inside its 5/7 lock, where it meets no repeat, the windows where 5/7 repeats again come, in units of its 56-cycle loop, as 10, 6, 8, 7, 5 with its doubling to 10, 3 with its doubling to 6, then 4 on the high side, and 6, 5, 3 with its doubling, 5, 8, then 7 with its doubling (in units of 28) on the low side: the logistic map's own order, scanned on its own road on the bench (12, 10, 6, 8 in its two-banded chaos; 7, 5, 3, 9, 5, 4 above its band merge, each with its doubling); a registered lean on the low side's order held, and two windows forecast from the map's order, a 7 between the 8 and the 5 and a 10 before the 6, were found where forecast on a grid ten times finer; the 3 window's near edge lies 0.4046 and 0.4041 of the 56 band's width from the 56-to-112 onset on the two sides, and the band ratios close on Feigenbaum's 4.669 as 4.626 and 4.629; of three windows the paper's own order names and no scan had met, forecast on the needle through a map calibrated on the windows already met, a 10 was found 0.0000000024 of pace from its forecast, holding at three paces when flown four times longer; an 11 met at one pace 0.0000000013 from its forecast proved a quiet stretch of chaos when flown longer and is withdrawn, and a second 10 was not found (misses rows 501 to 503)	the THIRD_RATIO, WINDOWS_ORDER, LOGISTIC_WINDOWS, MSS_FORECAST, MSS_FORECAST_2 and WINDOWS_LONG checks of 30 October
50	The corrections to Feigenbaum's scaling: the spectrum of the linearized period-doubling operator, its second eigenvalue -0.123652712553 (Christiansen, Cvitanovic and Rugh 1990, J. Phys. A 23, L713S-L717S, read at its authors' free copy), with a power of delta among the parameter's correction exponents (Briggs 1994, arXiv chao-dyn/9407019, read at its free text)	the logistic map's superstable r carried to period 262,144 in 80-digit arithmetic and its gaps to delta fitted four at a time by two shrink factors give 0.214169 against 1/delta = 0.214169 and -0.123652 against -0.1236527; the sine map gives 0.214169 and -0.123795; the first factor is the dial's bend, the 1/delta closing the needle's and A Rider in Two Waves' gaps show (0.21 to 0.24 a step): on a dial straightened from periods up to 1,024 the gaps change sign at each of fifteen steps, and on a second bent dial, mu = r(r-2)/4, the closing stays (misses row 504)	the FACTOR_FIT check of 31 October

	The published law	What the toy gave	Run
51	Feigenbaum’s constants for a hump of order z, the map 1 - mu abs(x)^z (Briggs 1991, Math. Comp. 57, 435-439, as MathWorld’s ‘Feigenbaum Constant’ tabulates them, read 31 October, and as Finch’s Mathematical Constants, 2003, section 1.9.1, Table 1.3, prints them, read in a copy Bear sent)	the doubling operator’s fixed point, solved here from Feigenbaum’s own equation with 24 terms in 50-digit arithmetic, gives alpha at orders 2, 3 and 4 and delta at orders 2 and 4 to every printed digit (-2.5029078751, -1.9276909638, -1.6903029714; 4.6692016091, 7.2846862171), the maps’ own superstable parameters to period 16,384 agree, and so do their doublings; the table’s rows 5 and 6 read at exchange 20: alpha at order 5 and both columns at order 6 agree to every printed digit (1.5557712501; 9.2962468327, 1.4677424503); at order 3 all three readings give delta 6.0846890743, where the table prints 5.9679687038, and at order 5 the equation and the map flown to period 16,384 give 8.3443847487, where it prints 8.3494991320: two entries the map does not bear out, both at odd orders, in opposite directions; one legible line of Briggs’s own Table 2, delta at order 12, 13.5350756661 7027649570, is met by the same solver at 40 coefficients to 2e-14; Briggs’s own thesis (1997, Table 2.3, read in a copy Bear sent) prints the same odd entries, and its deltas at orders 7, 9 and 11 are off too, while its alphas and its even orders’ deltas agree throughout; one slip reproduces every odd entry, the doubling operator’s perturbation evaluated at the negative argument x/alpha with signed powers in place of absolute ones (Briggs’s delta(3) met to 2e-29, orders 5 to 11 to the reading’s truncation); the map’s own delta(3) is 6.084689074262121896046121066259 3, and with the slip removed orders 2 to 12 lie on one smooth curve; known before: McCartney (2020), from superstable orbits to period 2^32, noted a discrepancy with the earlier values at odd orders and gave the map’s values, which ours match at every integer order 2 to 10 (within one unit in his last printed place) and at orders e and pi, naming no cause; a Wolfram Demonstration asked for an explanation; the cause, the sign, not found stated in a limited pass, a pointer for a specialist (Finch’s table, which prints the same order-3 entry, cites Briggs 1991 and Kuznetsov, Kuznetsov and Sataev 1997; Finch’s errata of 2015 and of 2024 do not correct it; McCartney’s accepted manuscript read); Briggs’s own 1989 direct-method table for the quadratic map (his thesis, Appendix A) met by the maps’ superstable ladder at every printed figure of its parameters, its last ratios drifting with its arithmetic while ours do not move at 100 digits; Delbourgo and Kenny’s rough values follow one stopping rule at five of six orders, and one box of Briggs’s Table 2.2 is two sources, three values the maps read part way and two delta cut short; Briggs’s 1989 direct method with its power changed to 3 gives the maps’ 6.084689 in its own 64-bit arithmetic; the swing of the order-2 ratios over delta is the operator’s eigenvalue -0.12365	the DELTA_Z3 and FEIGENBAUM_EQ_Z3 checks of 31 October; the CORNER_CLASS check of 31 October; the TABLE_ROWS_FIVE_SIX, ORDER5_MAP, DELTA_INVERSE and BRIGGS_OWN_Z12_C checks of 31 October; the SIGN_SLIP_ODD, SLIP_DEEP and ONE_CURVE_B checks of 31 October; the INDEPENDENT_READER_JUDGE, SLIP_SECTOR and ODD_TABLE checks of 31 October (a fresh helper’s code, given no number, agreeing); the MCCARTNEY_READ_JUDGE and MCCARTNEY_NONINTEGER checks of 31 October; the DK_CUTOFF, LOST_LABEL and BRIGGS_1989_TABLE (with _B and _C) checks of 31 October; the BRIGGS_1989_ORDER3 (with _B) and SWING checks of 31 October

The candidates that are ours

The 16 results not yet seen in print, in the order a specialist would most want to see them. Only the first is claimed as new; each of the others has had a check of the literature, and the last column says what it found and how far it reached.

	Result	The number	Where the check stands
1	Two models of a falling cat are mirror images: a thick cat that twists at its waist at bend b and a twist-free cat at 180 − b turn at mirror-image rates. In algebra this is exact at a steady circuit for equal tubes of any length, mass and thickness hinged at their ends, the partner half as thick; in the runs, with tubes of unit length and mass, the rates mirror at every shape, so the turns agree on every path, and with the hinge nearer the middle a thicker partner pairs them	5e-13 degrees on a 1-degree grid; exact in algebra; within 2.842e-13 on nine swelling paths; the rates within 9.3e-15 at 2,000 random shapes; 4.5e-13 with the hinge moved in (CAT_MIRROR 20261018_041123; CAT_MIRROR_PROOF 20261018_174959; CAT_MIRROR_PATHS 20261020_034021; CAT_MONTGOMERY 20261020_035025; CAT_MIRROR_RATES 20261020_041152)	known in its parts: both parts are in Montgomery's equation (1993), who studies only the twist-free cat and never compares the two; not stated there, nor in Rademaker and ter Braak (1935), whose cat is the twist-free cat, nor in Kane and Scher (1969), Gbur (2019), Batterman (2003) or Marsden's lectures (1992), all six read in full; the volume Montgomery's chapter is in, two preprints he cites and Iwai's book of 2021, behind a paywall, not yet reached
2	A pump that listens beats one that keeps time: its push weakens only to 4r/(1+r)² of full strength on any swing, exactly the share of power a mismatched cable passes, by the same map	growth 1.26570 a swing against 1.26569 worked by hand; the law within 0.14 per cent from 0.5 to 12.5 m (SWING_LISTEN 20261007_013201)	that the best pump listens is known (Bryde, Davenport and Mahadevan 2023); the mismatch law is not in the pages read, a second pass included, and it is a cable's mismatch loss for one reason
3	Where a bent swimmer reverses: at the lap's radius over √2 on a blade-hard grip	2.130 against 2.121 and 5.629 against 5.657 degrees at the two sizes it was not read from (S_REVERSAL_SPIKE 20261003_213523)	known in kind (Becker, Koehler and Stone 2003; Hatton and others 2022); that a swimmer in resistive water reaches the wheeled snake's ridge along the C shapes as its grip hardens (SWIM_RIDGE 20261105_145417) was not found in three passes
4	A registered law for light carried round a loop of filters, 2π a J1(2a) sin(gap)	within 2.51e-7 at 4,096 filters (POLARISATION_LOOP_2 20261007_074423)	not found in six searches; it is a two-line consequence of Pancharatnam's rule, the shift half the area the light's walk encloses on the ball, and its Bessel factor is everyday in polarimeters driven by a photoelastic modulator, where it sizes a signal rather than a loop's shift, so new at most in its form (chapter 7)
5	A published table of the human body most likely drops one entry: the thigh's 0.08 at the knee	restored with nothing fitted, every rebuilt number inside its registered bar (TONG_DULLIN_KNEE 20261017_182629)	not found noted in print: the same entry is missing from their open preprint (Tong and Dullin 2018), and every forward offset their table prints is the thigh's radius, 0.08 m, in Tong's thesis (2016); theirs to confirm
6	The early warning was the first rare slips	a noisy lock's gauge, with the slips left out, reaches 0.95 in the same step of a 0.05 pace grid as the slips pass one per 100 cycles (SLIP_PROOF_3 20261013_225419; ERROR_BARS 20261019_162750)	flickering as a warning is known (Dakos, van Nes and Scheffer 2013), and so are jumps that noise brings with no slowing down before them (Ditlevsen and Johnsen 2010); the falls heard one at a time and counted, and this place and number, were not found
7	Sized taps that never act blind hold a delayed balancer past the √2 limit of a delayed tilt-and-rate push	1.75 and 2.00 fall times in quiet (QUIET_TAP 20261013_195754)	act-and-wait is known (Stépán and Insperger 2006), and so is the √2, the bound for a push on the tilt and its rate read late; a push fed the tilt's acceleration reaches 2.00 too; the numbers are the toy's
8	Read a wearing clock where nothing hides it	a fixed alarm wastes 0.031 to 0.051 of a life, the calendar 0.250 to 0.333 (RESTED_LAG 20261016_205504)	that maintenance by condition beats maintenance by the calendar is known (de Jonge, Teunter and Tinga 2017 review the studies); wear hidden behind a rest, read only where the rest has run out, was not found; the numbers are the toy's
9	A slow sampler can count what it missed	step-free within 0.7 per cent over a 128-fold range of steps (BAND_BRIDGE 20261016_072612)	the bridge between looks is known for prices (Beaglehole, Dybvig and Zhou 1997) and in simulation; counting a two-level switch's flips with it was not found; the numbers are the toy's

	Result	The number	Where the check stands
10	Time a dive's kick-out by the body's angle, not the clock	the cued miss 0.420 of the clock's, a ratio of inertias predicted before the run (DIVE_CUE 20261017_082454)	somersaulters are known to time their opening by the body's angle and its rate (Lee, Young and Rewt 1992); the size of the gain against a clock, a ratio of inertias, is the toy's
11	On the same fuel, leapfrogging smoke rings beat one ring	a pair 14 to 19 per cent faster, three at half a radius 26 per cent (RING_BUDGET 20261004_080107; TRAIN_HOLD 20261015_171853)	leapfrogging is known, with two and three rings, regular and chaotic (Borisov, Kilin and Mamaev 2013); no speed at equal energy was found in the pages reached
12	Banded chaos can keep an exact gear while a true lock near its edge loses its gear to a whisper of noise	5/7 within a quarter turn over 30,000 cycles; the near-edge locks move 0.013 and 0.014 at noise 0.001, the chaos 0.003 or less (BANDED_FIVE 20261013_224945; CHAOS_ORDER_2 20261013_225021)	each half is known, phase locking of chaos (Rosenblum, Pikovsky and Kurths 1996) and slipping near an edge; the pairing was not found
13	A jump needs a latch: holding until the push turns	1.36, 1.51 and 1.82 times the push of a memoryless trigger for the same switches (LATCH_EVENT 20261015_232643)	the latch is a Schmitt trigger, a switch with two levels, known since Schmitt (1938); its push per switch against a memoryless switch was not found; the numbers are the toy's
14	Hidden mountains read back from carried arrows alone, with a ghost peak only a tiny circle unmasks	each peak's centre to about 1.3e-8; on noiseless mountains (CONE_TOMOGRAPHY 20261007_235444)	known in kind: the turn is the enclosed curvature (Gauss and Bonnet), and reading a field back from disc sums is tomography's (Kunyansky 2006); peaks read from carried frames alone were not found; a noisy test is owed
15	Where an arch of the chaos touches its floor a window opens	18 of 19 touches, none of 7 control paces (one of 8 was drawn twice) (WINDOW_TOUCH_5 20261014_092920)	known in kind: every window of a one-hump map holds a setting where the hump's tip lies on its own cycle (Metropolis, Stein and Stein 1973); the four strips' rule and the count are the toy's
16	A three-ring smoke train has a soft edge, and a single break time is one throw of a chaotic process	48 of 55 copies broke from 0.560 to 0.570, 3 of 5 at 0.560; a nudge of 1e-15 moved breaks by 28 to 84 per cent (TRAIN_EDGE 20261018_181641; TRAIN_NUDGE 20261018_192138)	chaotic leapfrogging is known, and a break time drawn from a spread of lifetimes is transient chaos's (Lai and Tél 2011); the soft edge and the kept pair were not found

Appendix C. The record, and how to read it

Every prediction the runs judged, and every other check they made, are printed in full in a companion volume, *The Loop Atlas: the record*, which is deposited beside this book as a record of its own and is rebuilt from the same runs at every build, so that it cannot fall behind them. It is a book of tables, made for checking rather than for reading through. If you want the story, it is in the chapters; if you want to see exactly what a run was asked and what it answered, open the record.

Every prediction gets a row, 1,104 in all, day by day in the order the runs were made: the run that judged it, named as the notes name runs, by its tool and its stamp; the prediction's label in that run; its words as the run printed them, bar and all; how it came out, with the run's own number beside the verdict where it printed one; and, for a prediction that missed or was left undecided, its row among the misses. Three of its rows, two from a late run of the smoke rings (chapter 9) and one from the needle's scissors (chapter 10), look like this:

Run	Label	The prediction, as the run prints it	Verdict	Misses row
TRAIN_PARTING_EDGE 20261029_070800	P4	the long shot: an even number of returns before every parting that had any	Held	
TRAIN_PARTING_EDGE 20261029_070800	P1	read to the parting for good, the lifetime under 1.10 times that to the first 5-radius spread	Missed	480
MEMORY_SWITCH 20261011_084432	S1	the switch between 0.70 and 0.71	Undecided	144

Every other check a run made has a table of its own in the same form: the rivals written to win if a prediction was wrong, the checks planted in each tool before its numbers were trusted, the toy pages' engine checked against the runs, and the book builder's own checks. This book only counts them, on Appendix B's first page.

Every miss is printed whole in the record's Part 3, 656 in all, numbered in the order they were found, and each says what happened, what kind of miss it was, what it taught and whether a redesign was named, and whether the lesson has been proved by a run. The fourth, from the listening bead of chapter 11, reads:

> **4.** Claude expected a listener as noisy as the soup to catch nothing.
>
> *Kind:* WRONG GUESS. *Lesson:* The catch fades gently (72% kept at equal noise) and being late hurts far more than being noisy. *Proved?* Proved by run.

Notes: the run behind each number

Every number of ours in this book was printed by a run of a small model, and each raised number in the text points here. A run is named by its tool in capitals and its date and time: CREEP_LAW 20261016_035058 is the tool CREEP_LAW, run on 16 October 2026 at 03:50:58. Runs of the first two days, 1 and 2 October, carry only a date, with a letter for a later version run the same day, as in AREA_RULE 20261002b; the first nights' toy runs are named by number, as in toy run 5c, and a misses row is one of the misses printed whole in the record's Part 3.

1. FUNNEL_CURLS 20261006_203001.
2. STRETCH_READS 20261013_060151, reading the orbit of STRETCH_WHERE 20261013_045856.
3. STAIR_WIDTHS 20261013_184033.
4. The first nights' toy run 5b.
5. Toy run 5d.
6. Toy run 5j.
7. SNAKE, toy run 5c.
8. Toy run 5g.
9. Toy run 5m.
10. Toy run 5m.
11. Toy run 5q.
12. Toy run 5p.
13. Toy run 5s.
14. Misses row 18.
15. AREA_RULE 20261002b.
16. Toy run 5l.
17. Toy run 5l.
18. LANDSCAPE 20261003_201112.
19. BIG_LOOP 20261003_175803.
20. HEAT_RING 20261016_174342.
21. BRICK_QUARTER 20261103_180126.
22. BRICK_LINE 20261016_205112; BRICK_FINE 20261103_180438.
23. HOW_IT_COUNTS 20261004_183911.
24. BELL_CLOCK 20261016_072241.
25. BELL_BALANCE 20261016_193628.
26. BELL_SOFT_MAP 20261021_000736.
27. BALANCE_BOARD 20261004_161712.
28. Misses row 2.
29. Misses row 23.
30. NOVIKOV_NEEDLE 20261004_152937.
31. NOVIKOV_NEEDLE_B 20261004_153129.
32. KRAMERS_EXACT 20261020_073520.
33. KRAMERS_EXACT 20261020_073520.
34. SLIP_SPREAD 20261020_075635.
35. SLIP_RHYTHM_3 20261020_090717.
36. SLIP_RHYTHM 20261020_083426.
37. SLIP_RECHECK 20261020_083908.
38. SLIP_START 20261020_090154.

39. RHYTHM_BAND 20261020_154009.
40. SLIP_BAND 20261020_161039.
41. FALL_CHAIN_2 20261012_041438.
42. FALL_SHARE 20261012_075804, SPEED_CHAIN 20261012_171949.
43. TOP_SECTION 20261013_005355.
44. SADDLE_ASYMPTOTE 20261013_025252.
45. SADDLE_LOG 20261013_012417.
46. SADDLE_NOISE 20261013_024850.
47. BANKED_TRACK 20261004_154940.
48. TYRE_KNEE 20261014_174309.
49. A turn’s climb and its own fall are tied, a correlation of +0.590 on spinning turns, while a fall and the next climb are nearly strangers, +0.068; KNEE_RECOUNT_E 20261005_003928.
50. SEAM_MAP 20261005_021349; SEAM_MAP_B 20261005_023746; the triple star’s row from the triple-star study’s own runs, kept with that study and not in this archive.
51. CLIFF_EDGE 20261004_062732.
52. CLIFF_EDGE_B 20261004_062930.
53. CLIFF_EDGE 20261004_062732.
54. SLIP_BEND 20261006_071248.
55. WOBBLE_SOFT 20261013_192713.
56. SLIP_PROOF_3 20261013_225419; OWED_BARS 20261107_023832; CUT_WIDE 20261107_033035.
57. NEVER_BLIND 20261006_045833.
58. SIZED_TAP 20261006_060952.
59. SIZED_TAP_2 20261006_061606.
60. SIZED_TAP_3 20261006_061940.
61. SIZED_TAP_4 20261013_161615.
62. QUIET_TAP 20261013_195754.
63. SIZED_TAP_2 20261006_061606.
64. LANDSCAPE 20261003_201112.
65. The first nights’ toy run 5k.
66. Toy run 4.
67. FACE_ALONG 20261017_070629; the broad face’s 6 is assumed.
68. COIL_SNAKE 20261017_204030.
69. COIL_CHEAP 20261107_060011.
70. COIL_PUMP 20261018_082424.
71. COIL_WAVE 20261018_082544.
72. BLOB_RING 20261104_170523.
73. BLOB_SMALL 20261104_170747.
74. LANDSCAPE 20261003_201112.
75. LANDSCAPE_FIG 20261003_201923.
76. S_REVERSAL 20261003_213109; S_REVERSAL_SPIKE 20261003_213523.
77. S_REVERSAL_SPIKE 20261003_213523.
78. SWIM_RIDGE 20261105_145417.
79. SWING_PUMP, run of 2 October.
80. SWING_TWENTY 20261004_075715.
81. SWING_TWENTY_B 20261004_075902.
82. SWING_LISTEN 20261007_013201.
83. SLING_RUNDOWN 20261017_232234.
84. The first nights’ toy run 5.
85. SKATER, run of 2 October.
86. POLARISATION_LOOP 20261004_071425.
87. POLARISATION_LOOP_2 20261007_074423.

88. POLARISATION_LOOP_2 20261007_074423.
89. POLARISATION_LOOP_2 20261007_074423, part W3.
90. POLARISATION_VORTEX 20261004_074503.
91. SAGNAC_LOOP 20261018_032138.
92. CONE_TOMOGRAPHY 20261007_235426.
93. CONE_TOMOGRAPHY 20261007_235444.
94. CONE_TOMOGRAPHY_2 20261008_002939.
95. CONE_TOMOGRAPHY_3 20261008_011150.
96. PEAK_RING_2 20261008_040854.
97. GROW_LOOP 20261008_133646.
98. GROW_SLIP 20261008_195205.
99. PENTA_SHARE 20261008_060200.
100. BENT_PAPER_2 20261008_164721.
101. CONE_TOMOGRAPHY 20261008_001220.
102. PUSH_PRICE 20261004_144831.
103. PUSH_TRADE_READER 20261007_024449.
104. WAVE_SNAKE_DRIFT 20261004_074634.
105. PUSH_PRICE 20261004_144831.
106. SNAKE_EXACT 20261015_154247.
107. SNAKE_LONGEST 20261017_155052.
108. SNAKE_RACE 20261101_175747.
109. SNAKE_BEAT 20261101_190003.
110. The CLOCK_RULE check of 1 November, reading SNAKE_RACE 20261101_175747 and SNAKE_BEAT 20261101_190003.
111. FREE_SWIM 20261101_222356.
112. FREE_SWIM_2 20261101_222740.
113. SAIL 20261018_202451.
114. SAIL_HULL 20261018_202650.
115. The first nights' toy run 5m.
116. RING_BUDGET 20261004_080107.
117. RING_KNOBS 20261015_165858.
118. RING_PATHS 20261102_013613.
119. TRAIN_HOLD 20261015_171853.
120. RING_AREA 20261105_032546.
121. RING_SPLIT 20261105_050121.
122. TRAIN_CLOSE 20261018_103739.
123. TRAIN_NUDGE 20261018_192138.
124. TRAIN_EDGE 20261018_181641.
125. ERROR_BARS 20261019_162750.
126. TRAIN_FLOOR_2 20261021_083613.
127. TRAIN_FLOOR 20261020_224239.
128. TRAIN_FUSE_CLOCK 20261028_101647.
129. TRAIN_EDGE_DOOR 20261022_070632; TRAIN_LONG_EDGE 20261022_175421; TRAIN_DOOR_LINE 20261022_222935; TRAIN_FUSE_CLOCK 20261028_101647.
130. TRAIN_DRIFT 20261019_034330.
131. TRAIN_FUSE_CLOCK 20261028_101647; TRAIN_BEAT 20261028_212034.
132. TRAIN_BEAT 20261028_212034.
133. TRAIN_PARTING 20261029_112955.
134. TRAIN_PARTING_EDGE 20261029_070800.
135. TRAIN_PARTING_EDGE 20261029_070800, its sixteen flights resampled.
136. TRAIN_PARTING_EDGE_2 20261029_211416.

137. The READ_267 check of 30 October, pooling TRAIN_PARTING_EDGE 20261029_070800 with TRAIN_PARTING_EDGE_2 20261029_211416.
138. The ROPE-1 runs among the files that travel with Bear’s jellyfish paper, 18 September.
139. DRAFT_TEST 20261008_134707.
140. Toy run 5h.
141. Toy run 5i.
142. NEEDLE_GEARS 20261003_210142.
143. NEEDLE_GEARS_WHY 20261003_210451.
144. Toy run 4.
145. Toy run 5ae.
146. SWING_GRID 20261016_203014.
147. SEAM_MAP_B 20261005_023746.
148. CLIMB_BACK_LINE_CHECK 20261006_041633.
149. MEMORY_SWITCH_2 20261011_085331.
150. BENCH_LOOP 20261020_081001.
151. Misses row 427.
152. NOISE_EDGE 20261014_232230.
153. AVALANCHE 20261004_064914.
154. ROUGH_FLOOR 20261017_013514.
155. Misses rows 286 to 288.
156. ROUGH_COUNT 20261102_173941.
157. ROUGH_LONG 20261102_200436.
158. SPOKES 20261006_204038 to CASCADE_5 20261006_225126.
159. The LOGISTIC_DEEP check of 30 October; its long shot, that the factor sits within half a per cent of one over Feigenbaum’s number, missed at 0.67 per cent.
160. CLOCK_SHIFT_30 20261109_034237.
161. DELTA_BY_TOP 20261008_034013.
162. DELTA_AREA 20261008_062647.
163. CHAOS_SETTLES 20261011_060320.
164. SETTLED_COPIES 20261011_212014.
165. SETTLED_GROOVES_2 20261012_041558.
166. REINJECT 20261013_015743; REINJECT_CLOCK 20261013_024653.
167. INNER_OUTER 20261013_005545.
168. QUARTER_TREE_READ 20261014_053757.
169. QUARTER_TREE_CURVES 20261014_055131.
170. WINDOW_TOUCH_3 20261014_080520; WINDOW_TOUCH_5 20261014_092920.
171. SEAM 20261014_090006.
172. CRISIS_BURSTS 20261014_091405.
173. HEAVY_SHAPE 20261014_155553.
174. HEAVY_MARKS 20261014_164806.
175. HEAVY_SHAPE_2 20261014_164913.
176. HEAVY_SHAPE_3 20261014_170837.
177. STOP_WALL_3 20261015_000651; HEAVY_WINDOW_4 20261015_032553.
178. SWITCH_END 20261015_074716.
179. HALVINGS_2 20261015_064427; HALVINGS_3 20261015_064938.
180. GAPS_READ 20261015_061859.
181. GEAR_TRAIN 20261013_170243.
182. HALF_BUBBLE 20261013_083609, BUBBLE_FAR 20261013_084037.
183. HALF_PARENT 20261013_163612.
184. BANDED_FIVE 20261013_224945.
185. CHAOS_ORDER_2 20261013_225021.

186. The HIDDEN_WINDOWS and HIDDEN_CHAOS checks of 30 October.
187. THIRD_AXIS 20261109_170732.
188. THIRD_AXIS_3 20261109_191727.
189. THIRD_AXIS_2 20261109_181451; THIRD_AXIS_3 20261109_191727.
190. ORBIT_19 20261013_041840.
191. STRETCH_READS 20261013_060151.
192. STAIR_WIDTHS 20261013_184033.
193. The BASIN_BOUNDARY and BASIN_FINE checks of 30 October.
194. Toy run 5m.
195. Toy run 5m.
196. Toy run 5n.
197. Toy run 5v.
198. EAR_PRICE_CHECK 20261003_220245.
199. EAR_KNEE 20261003_220719.
200. KNEE_CREEP 20261015_232920.
201. FIG_CREEP_LAW 20261016_035636; KNEE_SIZE 20261019_035236.
202. CREEP_LAW 20261016_035058; ERROR_BARS 20261019_162750.
203. ZETA_FIT 20261020_055322; ZETA_LIMIT 20261020_060256.
204. KNEE_SIZE 20261019_035236.
205. KNEE_FINE 20261019_040109.
206. KNEE_REF 20261019_042520; KNEE_READER 20261019_163948.
207. BRIDGE_COUNT 20261016_044439.
208. NEVER_BLINK 20261016_090509.
209. BAND_BRIDGE 20261016_072612.
210. NEVER_BLINK 20261016_090509; OWED_BARS 20261107_023832.
211. EAR_CURES 20261004_045151 and EAR_CURES_B 20261004_045721.
212. FLIP_MISTAKES 20261015_195739.
213. LATCH_EVENT 20261015_232643.
214. JUMP_STORE 20261015_223918; LATCH_EVENT 20261015_232643.
215. SPRING_FLOOR 20261017_160023.
216. The LATCH_WALL and LATCH_CEILING checks of 31 October.
217. SLOPE_EAR 20261015_190935; Åström and Hägglund 1995.
218. The first nights’ toy run 5r.
219. PILOT_BEAT 20261015_203855; BEAT_SWING 20261015_204207.
220. BEAT_SWING 20261015_204207; PILOT_PRICE 20261015_220321.
221. EAR_COST_2 20261013_183913.
222. EAR_COST_3 20261013_192738.
223. COLUMN_ACCORDION_2 20261013_222950.
224. MORTAR_SPACING 20261013_224843, as corrected in LIMIT_AUDIT 20261017_190938.
225. NIGHT_SPACING 20261013_231417, as corrected.
226. CATS_EYES 20261014_063138, as corrected.
227. LIMIT_AUDIT 20261017_190938.
228. WORD_BACK 20261014_065420; a toy with no limits.
229. HOUSE_IN_SPACE 20261004_150225; HOUSE_IN_SPACE_B 20261004_150847.
230. PUSH_PRICE 20261004_144831.
231. The companion study’s run triple_cat_holonomy 20260930_161702, read here in TRIPLE_CARD_CHECK 20261005_055204.
232. DIVER_B 20261005_160338; every body number assumed.
233. HITCH_KICK 20261006_011635.
234. HITCH_LEGS 20261016_065650.
235. ARM_SHARE 20261016_182910.

236. ARMS_LEGS 20261017_012927.
237. STRIDE_MAP 20261016_213858.
238. Misses row 279.
239. ARM_SHARE 20261016_182910.
240. ARMS_LEGS 20261017_012927.
241. DIVE_CUE 20261017_082454.
242. DIVE_CUE 20261017_082454; EXPLORER_ROUTINE_CHECK 20261017_164432.
243. TONG_DULLIN 20261017_173718.
244. TONG_DULLIN_SCAN 20261017_181953.
245. TONG_DULLIN_KNEE 20261017_182629.
246. TONG_DULLIN_SCAN 20261017_181953.
247. CAT_TWIST 20261017_231928.
248. CAT_SWEET 20261018_034953.
249. CAT_MIRROR 20261018_041123.
250. CAT_MIRROR_PROOF 20261018_174959.
251. CAT_MIRROR_PATHS 20261020_034021.
252. CAT_MIRROR_RATES 20261020_041152.
253. CAT_MONTGOMERY 20261020_035025.
254. CAT_MONTGOMERY 20261020_035025.
255. CAT_RTB 20261020_063617.
256. CAT_MONTGOMERY 20261020_035025.
257. CAT_MIRROR_RATES 20261020_041152.
258. CAT_MIRROR_PROOF 20261018_174959.
259. CAT_SWEET 20261018_034953.
260. Toy run 5j.
261. Toy run 5c.
262. Toy run 5p.
263. Toy run 5s.
264. Toy run 5m.
265. JUMP_STORE 20261015_223918.
266. LATCH_EVENT 20261015_232643.
267. The companion book’s run MEM-5, in its record.
268. SLIP_BEND 20261006_071248.
269. BAND_BRIDGE 20261016_072612.
270. BELL_CLOCK 20261016_072241.
271. BELL_RATIO 20261016_183422; BELL_BALANCE 20261016_193628.
272. MAINTENANCE_METER 20261005_012150 and _B 20261005_012355.
273. RIDER_FLEET 20261016_044707.
274. EQUAL_CATCH 20261016_192548.
275. LAG_METER 20261016_203014.
276. RESTED_LAG 20261016_205504, one fleet design over five seeds.
277. RIDER_FLEET 20261016_044707.
278. WORN_RIDER 20261016_065645.
279. SHUVIT_WALK 20261004_183058.
280. WADDLE_PRICE 20261011_071302.
281. WADDLE_FOLLOW 20261011_080808.
282. WOBBLY_SHUVIT 20261015_074302.
283. COMB_STRETCH 20261101_002907; COMB_STRETCH_B 20261101_003130.
284. HALT_ORDER 20261015_052122; LIMIT_AUDIT 20261017_190938.
285. HALT_TRADE 20261031_235932.
286. LANE_EDGES 20261017_164501.

287. RAMP_SPEED 20261031_232517.
288. BENCHMARK_TAGS 20261020_055158.
289. Misses row 2.
290. Misses row 23.
291. BENT_PAPER_2 20261008_164721.
292. GROOVES_16 20261007_182009.
293. ORBIT_919_4 20261013_040039, ORBIT_19 20261013_041840.
294. SNAKE_EXACT 20261015_154247.
295. GHOST_TRIAL 20261013_005637; GHOST_TRIAL_2 20261013_015743.
296. Toy run 5s.
297. RING_BUDGET 20261004_080107.
298. KICK_RULE_CHECK 20261005_031525.
299. LIMIT_AUDIT 20261017_190938.
300. TRAIN_CLOSE 20261018_103739.
301. TRAIN_NUDGE 20261018_192138.
302. ERROR_BARS 20261019_162750.
303. OWED_BARS 20261107_023832.
304. KRAMERS_EXACT 20261020_073520.
305. LEAN_DRIFT 20261105_025051.
306. CAT_MIRROR 20261018_041123; CAT_MIRROR_PROOF 20261018_174959; CAT_MIRROR_PATHS 20261020_034021; CAT_MONTGOMERY 20261020_035025; CAT_MIRROR_RATES 20261020_041152.
307. SWING_LISTEN 20261007_013201.
308. S_REVERSAL_SPIKE 20261003_213523.
309. SWIM_RIDGE 20261105_145417.
310. POLARISATION_LOOP_2 20261007_074423.
311. CAT_SWEET 20261018_034953.
312. TONG_DULLIN_KNEE 20261017_182629.
313. DELTA_BY_TOP 20261008_034013; CASCADE_5 20261006_225126; the CASCADE_57, THIRD_RATIO, FOURTH_RATIO, FIFTH_RATIO, WINDOWS_ORDER and MSS_FORECAST checks of 30 October and the FACTOR_FIT, LOBE_FLIP_FINE, CORNER_UNIVERSAL and CORNER_CLASS checks of 31 October; HEAVY_SHAPE 20261014_155553 and STOP_WALL_3 20261015_000651; CRISIS_BURSTS 20261014_091405.
314. KRAMERS_EXACT 20261020_073520; NOVIKOV_NEEDLE 20261004_152937 and _B 20261004_153129; SLIP_SENSE 20261011_060035; ERROR_BARS 20261019_162750.
315. CONE_TOMOGRAPHY 20261007_235426; BENT_PAPER_2 20261008_164721; GROW_SLIP 20261008_195205.
316. ZETA_LIMIT 20261020_060256.
317. CREEP_LAW 20261016_035058; BAND_BRIDGE 20261016_072612; KNEE_SIZE 20261019_035236; KNEE_REF 20261019_042520; KNEE_READER 20261019_163948.
318. RESTED_LAG 20261016_205504; EQUAL_CATCH 20261016_192548.
319. QUIET_TAP 20261013_195754.
320. DIVE_CUE 20261017_082454.
321. SLIP_PROOF_3 20261013_225419; OWED_BARS 20261107_023832.
322. The DELTA_Z3, FEIGENBAUM_EQ_Z3, SIGN_SLIP_ODD, SLIP_DEEP, ODD_TABLE, INDEPENDENT_READER_JUDGE and MCCARTNEY_READ_JUDGE checks of 31 October; benchmark 51.
323. BANDED_FIVE 20261013_224945; CHAOS_ORDER_2 20261013_225021.
324. RING_KNOBS 20261015_165858; TRAIN_HOLD 20261015_171853; TRAIN_DOOR_LINE 20261022_222935.
325. THIRD_AXIS 20261109_170732; THIRD_AXIS_2 20261109_181451; THIRD_AXIS_3 20261109_191727.

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What travels with this book

Two things are deposited with this book, and a third beside it. With it go the toy pages of chapter 2 and the explorer's front page, each a single file that opens in a web browser, and one crate, the archive of files, with a short map of it, README.md, at its top. In it are every tool, named LOOP_ATLAS_ and the tool's name, as in LOOP_ATLAS_CREEP_LAW.py, most with a one-click file that runs it; every run's record, named run_ and the tool's name and the run's stamp, as in run_LOOP_ATLAS_CREEP_LAW_20261016_035058.txt, often with a data file of its numbers beside it; the figures; and the templates the explorer's rooms were built from. Beside this book, as a record of its own, goes the companion volume, *The Loop Atlas: the record*, made by script from the runs at every build, of every prediction a run judged and every other check it made, each with its verdict (Appendix C shows its format); each of the two records names the other among its related works.

Declaration

What this is. A preprint, not peer reviewed. It is a toy-model study: every number in it was printed by a small computer model with values we chose, and it can teach about the real thing only as far as the model is faithful to it. Nothing here was measured on a real body, machine or animal by us; where a real measurement is quoted, it is someone else's and is cited.

How it was made. The Loop Atlas was made by a partnership that is part of its point. Matthew "Bear" Schulz asked Claude (Anthropic), an AI model, what it wondered about, and Claude's answer became the book's question; then he invited it to co-direct the work, and over four weeks, across several working sessions, the two chased the questions together. Every prediction was written down before the run that judged it, every run was kept, and the working record of how the work went was kept as it happened, sitting by sitting. Chapter 1 tells the story. Every quotation in italics marked as Bear's is his own, tidied at his word into clean sentences in his voice.

How far the literature was checked. Far enough to claim one result, not yet the rest. The 16 found-here candidates are results we have not yet found in print. One of them, the mirror between the two falling cats, is claimed as new on a limited, noncommercial review of the citations and references: eight sources from 1935 to 2019 were checked, six of them read whole, and none of them mentions a mirror. Four that may hold it were not reached: Iwai's book of 2021, which is behind a paywall, the volume Montgomery's chapter appears in, and two preprints he cites. So the claim is held until someone shows it in print. Each of the others has had a check of the literature, read as far as its pages could be reached: most of the ideas are in print in kind and the numbers are ours, a few came back empty in the pages reached, and two have parts of their check still owed; none of them is claimed as new. The benchmarks are published laws, and the toys met them; that the toys met them is ours, the laws are their discoverers'.

The limits. This is a preprint, with no peer review and a literature pass not yet finished. Every result is a toy model with assumed numbers, and none of it is an experiment; a builder who wants to use a rule here should try it on the real machine first. The toy wing is quasi-static, so its quarter-cycle optimum is not yet a result about real flight. The tomography was done on noiseless mountains. Kane and Scher's path was read by hand from their figure. And where a number was read off noise, appendix A says how far its error bar reaches. An AI did the building with an independent researcher, in the open, and both are named on the cover.

Dual-use statement. None of the toys bears on weapons, on harming people or on getting round safeguards. The work is about swimmers, swings, needles, switches, columns of walkers and falling cats.

Tool and computational resource disclosure

Tools used. Claude (Anthropic), an AI model, took part as a research partner and co-director: it wrote and ran every tool, kept the records, and drafted the text, including the no-physics reads. The long runs were flown on the author's own computer. All numerical results reported here were produced by the tools that travel with this book, written in Python, and every number of ours names its run in the notes. Typesetting used pandoc and wkhtmltopdf, through the crate's builder.

Responsibility. Responsibility for the correctness and adequacy of the arguments and results presented here, and for the completeness and accuracy of the citations, rests entirely with the named human author. No automated system is an author of this work. Claude is named on the cover, and the book speaks of two partners, because that is how the work was made; naming a partner is not making it an author, and answering for the work is the named human author's alone. This follows the Leiden Declaration on Artificial Intelligence and Mathematics (2026), which keeps credit and responsibility with people and asks that the use of automated tools be disclosed transparently; this section and chapter 1 are that disclosure.

Verification performed. Every prediction was written down, with the bar it had to clear, before the run that judged it, and from the middle of the work on every tool was first run blind on checks planted in it, 217 blind runs in all. Every number of ours names the run that printed it, and every count of our own records was made by script. Every PDF was read back against its source, letter by letter, and its pages looked at. The new prose was read against the record by a separate reader that had not written it. Each citation was checked against a record of it; an entry marked † has been checked only at second hand.

Attribution limits. Each of the 16 found-here candidates has a check of the literature beside it, read as far as its pages could be reached; one is claimed, on a limited review, and the others are not. Names the text gives as from recall are not listed until they are checked.

Computational resources. Small models on ordinary computers: a cloud workspace for most runs, and the author's own computer for the long ones, at most eight cores at a time.